SIM82XX_SIM83XX Series_ AT Command Manual

5G Module

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Version History

Version	Date	Chapter	What is new
V1.00	2020.4.24		New version
V1.01	2020.5.11	15 AT Commands for NTP 16 AT Commands for HTP	Add these chapters
V1.01	2020.5.11	20.2.3 AT+CVAUXS 20.2.4 AT+CVAUXV	Delete these commands
V1.01	2020.6.4	3.2.20 AT+SMEID	Delete this command
V1.01	2020.6.19	11.3.4 AT+CPING 11.3.5 AT+CPINGSTOP	Add these commands
V1.01	2020.6.19	3.3 Summary of CME ERROR codes	Remove MMS related error codes.
V1.01	2020.7.2	8.3 Summary of Unsolicited Result Codes	Add this chapter
V1.01	2020.7.14	4.2.19 AT+CNWINFO	Add this command
V1.01	2020.7.27	1.6.1 Software flow control (XON/XOFF flow control)	Delete this command
V1.01	2020.7.31	4.2.20 AT+CBANDCFG 4.2.21 AT+C5GREG	Add this command
V1.01	2020.8.10	4.2.22 AT+CSYSSEL 25 AT Commands for Bluetooth 26 AT Commands for Wifi	Add these commands
V1.01	2020.8.21	4.2.23 AT+CCELLCFGSet lte cell configuration 4.2.24 AT+C5GCELLCFG Set NR5G cell configuration	Add these commands
V1.01	2020.9.9	6.2.5 AT+CNUM Subscriber number	Add these commands
V1.01	2020.9.23	25.2.1 AT+BTINIT Init Bluetooth service	Remove the command
V1.01	2020.9.23	5.2.18 AT+MORING Enable or disable report MO ring URC 4.2.22.1 AT+CSYSSEL="nr5g_disable" 4.2.22.2 AT+CSYSSEL="nr5g_band" 4.2.22.3 AT+CSYSSEL="nsa_nr5g_band" 4.2.22.4 AT+CSYSSEL="lte_band" 4.2.22.5 AT+CSYSSEL="w_band"	Add these commands
V1.01	2020.9.30	5.2.19 AT+CLVL Loudspeaker	Add these commands

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V1.01	2020.11.11	5.2.22 AT+CRXVOL Adjust RX voice output speaker volume 5.2.23 AT+CTXVOL Adjust TX voice mic volume 5.2.24 AT+CTXMICGAIN Adjust TX voice mic gain 5.2.25 AT+CECH Inhibit far-end echo 5.2.26 AT+CECDT Inhibit echo during doubletalk 25.2.1 AT+BTINIT Init Bluetooth service 25.2.2 AT+BTTERM Stop Bluetooth service 19.2.9 AT+CPCIEMODE Get or set the mode of PCIE	Add these commands
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V1.02	2020.12.10	20.2.11 AT+CCUART	Add these commands
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V1.02	2021.3.2		Update version (Please refer to JIRA issue)
V1.02	2021.3.30	5.2.25 AT+CTXMICGAIN	Modify parameters information
V1.02	2021.4.14	1.4.4 Combining AT commands on the same Command line	Add note description
V1.02	2021.4.14	4.2.16 AT+CTZU 4.2.17 AT+CTZR 8.2.4 AT+CGDCONT 8.2.7 AT+CGQREQ 8.2.8 AT+CGEQREQ 8.2.10 AT+CGEQMIN 8.2.11 AT+CGDATA 12.2.10 AT+CFTPSTYPE 14.2.5 AT+CMQTTSSLCFG 25.2.3 AT+BTPOWER	Modify these commands

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Contents

THIS DOCUMENT IS A REFERENCE GUIDE TO ALL THE AT COMMANDS.

1 Introduction

1.1 Scope of the document

This document presents the AT Command Set for SIMCom SIM820X series, SIM821X series, SIM826X series and SIM83XX series.

1.2 Related documents

You can visit the SIMCom Website using the following link:
http://www.simcom.com

1.3 Conventions and abbreviations

In this document, the GSM engines are referred to as following term:

- ME (Mobile Equipment);
- MS (Mobile Station);
- TA (Terminal Adapter);
- DCE (Data Communication Equipment) or facsimile DCE (FAX modem, FAX board);

In application, controlling device controls the GSM engine by sending AT Command via its serial interface. The controlling device at the other end of the serial line is referred to as following term:

- TE (Terminal Equipment);
- DTE (Data Terminal Equipment) or plainly "the application" which is running on an embedded system;

1.4 AT Command syntax

The "AT" or "at" or "aT" or "At" prefix must be set at the beginning of each Command line. To terminate a Command line enter **<CR>**.
Commands are usually followed by a response that includes. **"<CR><LF><response><CR><LF>"**
Throughout this document, only the responses are presented, **<CR><LF>** are omitted intentionally.

The AT Command set implemented by SIM82XX_SIM83XX Series is a combination of 3GPP TS 27.005, 3GPP TS 27.007 and ITU-T recommendation V.25ter and the AT commands developed by SIMCom.

NOTE

Only enter AT Command through serial port after SIM82XX_SIM83XX Series is powered on and Unsolicited Result Code "RDY" is received from serial port. If auto-bauding is enabled, the Unsolicited Result Codes "RDY" and so on are not indicated when you start up the ME, and the "AT" prefix, or"at" prefix must be set at the beginning of each command line.

All these AT commands can be split into three categories syntactically: "**basic**", "**S parameter**", and "**extended**". These are as follows:

1.4.1 Basic syntax

These AT commands have the format of "**AT<*x*><*n*>**", or"**AT&<x><n>**", where "**<*x*>**"is the Command, and "**<*n*>**"is/are the argument(s) for that Command. An example of this is "**ATE<*n*>**", which tells the DCE whether received characters should be echoed back to the DTE according to the value of "**<*n*>**". "**<*n*>**" is optional and a default will be used if missing.

1.4.2 S Parameter syntax

These AT commands have the format of "**ATS<*n*>=<*m*>**", where "**<*n*>**" is the index of the **S** register to set, and *"**<m>**"*is the value to assign to it. "**<*m*>**" is optional; if it is missing, then a default value is assigned.

1.4.3 Extended Syntax

These commands can operate in several modes, as in the following table:

Table 1: Types of AT commands and responses

Test Command **AT+<*x*>=?**	The mobile equipment returns the list of parameters and value ranges set with the corresponding Write Command or by internal processes.
Read Command **AT+<*x*>?**	This command returns the currently set value of the parameter or parameters.
Write Command **AT+<x>=<…>**	This command sets the user-definable parameter values.
Execution Command **AT+<x>**	The execution command reads non-variable parameters affected by internal processes in the GSM engine.

1.4.4 Combining AT commands on the same Command line

You can enter several AT commands on the same line. In this case, you do not need to type the "**AT**" or "**at**" prefix before every command. Instead, you only need type "**AT**" or "**at**" the beginning of the command line. Please note to use a semicolon as the command delimiter after an extended command; in basic syntax or S parameter syntax, the semicolon need not enter, for example: ATE1Q0S0=1S3=13V1X4;+IFC=0,0;+IPR=115200.
Note: The AT Command line buffer of Audio, Filesystem, BT and WIFI can accept a maximum of 99 characters.If the characters entered exceeded this number then one of the command will be executed and TA will return "ERROR".

1.4.5 Entering successive AT commands on separate lines

When you need to enter a series of AT commands on separate lines, please Note that you need to wait the final response (for example **OK**, CME error, CMS error) of last AT Command you entered before you enter the next AT Command.

1.5 Supported character sets

The SIM82XX_SIM83XX Series AT Command interface defaults to the **IRA** character set. The SIM82XX_SIM83XX Series supports the following character sets:

- GSM format
- UCS2
- IRA

The character set can be set and interrogated using the "AT+CSCS" Command (3GPP TS 27.007). The character set is defined in GSM specification 3GPP TS 27.005.

The character set affects transmission and reception of SMS and SMS Cell Broadcast messages, the entry and display of phone book entries text field and SIM Application Toolkit alpha strings.

1.6 Flow control

Flow control is very important for correct communication between the GSM engine and DTE. For in the case such as a data or fax call, the sending device is transferring data faster than the receiving side is ready to accept. When the receiving buffer reaches its capacity, the receiving device should be capable to cause the sending device to pause until it catches up.

There are basically two approaches to achieve data flow control: software flow control and hardware flow control. SIM82XX_SIM83XX Series support both two kinds of flow control.
In Multiplex mode, it is recommended to use the hardware flow control.

1.6.1 Hardware flow control (RTS/CTS flow control)

Hardware flow control achieves the data flow control by controlling the RTS/CTS line. When the data transfer should be suspended, the CTS line is set inactive until the transfer from the receiving buffer has completed. When the receiving buffer is **OK** to receive more data, CTS goes active once again.

To achieve hardware flow control, ensure that the RTS/CTS lines are present on your application platform.

1.7 Definitions

1.7.1 Parameter Saving Mode

For the purposes of the present document, the following syntactical definitions apply:

- **NO_SAVE**: The parameter of the current AT command will be lost if module is rebooted or current AT command doesn't have parameter.
- **AUTO_SAVE**: The parameter of the current AT command will be kept in NVRAM automatically and take in effect immediately, and it won't be lost if module is rebooted.
- **AUTO_SAVE_REBOOT**: The parameter of the current AT command will be kept in NVRAM automatically and take in effect after reboot, and it won't be lost if module is rebooted.

1.7.2 Max Response Time

Max response time is estimated maximum time to get response, the unit is seconds.

2AT Commands According to V.25TER

2.1 Overview of AT Commands According to V.25TER

Command	Description
A/	Re-issues the last command given
ATD	Mobile originated call to dial a number
ATD><mem><n>	Originate call from specified memory
ATD><n>	Originate call from active memory
ATD><n>	Originate call from active memory
ATA	Call answer
ATH	Disconnect existing call
ATS0	Automatic answer incoming call
+++	Switch from data mode to command mode
ATO	Switch from command mode to data mode
ATI	Display product identification information
AT+IPR	Set local baud rate temporarily
AT+ICF	Set control character framing
AT+IFC	Set local data flow control
AT&C	Set DCD function mode
ATE	Enable command echo
AT&V	Display current configuration
AT&D	Set DTR function mode
ATV	Set result code format mode
AT&F	Set all current parameters to manufacturer defaults
ATQ	Set Result Code Presentation Mode
ATX	Set CONNECT Result Code Format
AT\V	Set CONNECT Result Code Format About Protocol
AT&E	Set CONNECT Result Code Format About Speed
AT&W	Save the user setting to ME
ATZ	Restore the user setting from ME
AT+CGMI	Request manufacturer identification
AT+CGMM	Request model identification
AT+CGMR	Request revision identification

AT+CGSN	Request product serial number identification
AT+CSCS	Select TE character set
AT+CIMI	Request international mobile subscriber identity
AT+GCAP	Request overall capabilities
AT+CVMOD	Select the type of voice calls

2.2 Detailed Description of AT Commands According to V.25TER

2.2.1 A/ Re-issues the Last Command Given

A/ Re-issues the Last Command Given	
Execution Command **A/**	Response Re-issues the previous Command
Parameter Saving Mode	NO_SAVE
Max Response Time	120000ms
Reference	

Example

```
A/
+GCAP: +CGSM,+FCLASS,+DS

OK
```

2.2.2 ATD Mobile Originated Call to Dial A Number

This command can be used to set up outgoing data calls. It also serves to control supplementary services.

ATD Mobile Originated Call to Dial A Number	
Execution Command **ATD<n>[<mgsm>][;]**	Response If originate a voice call successfully: **OK** **VOICE CALL: BEGIN**

	If error is related to ME functionality **+CME ERROR: <err>** If no dial tone and (parameter setting ATX2 or ATX4) **NO DIALTONE** If busy and (parameter setting ATX3 or ATX4) **BUSY** If a connection cannot be established **NO CARRIER** If the remote station does not answer **NO ANSWER** If connection successful and non-voice call. **CONNECT<text>**TA switches to data mode. Note: <text> output only if ATX<value> parameter setting with the <value>>0 When TA returns to command mode after call release **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	
Reference	

Defined Values

<n>	String of dialing digits and optionally V.25ter modifiers dialing digits: 0-9,*, #,+,A,B,C Following V.25ter modifiers are ignored: ,(comma),T,P,!,W,@
Emergency call:	
<n>	Standardized emergency number 112 (no SIM needed)
<mgsm>	String of GSM modifiers: I Actives CLIR (Disables presentation of own number to called party) i Deactivates CLIR (Enable presentation of own number to called party) G Activates Closed User Group invocation for this call only g Deactivates Closed User Group invocation for this call only

Example

ATD10086;
OK
VOICE CALL: BEGIN

NOTE

This command may be aborted generally by receiving an ATH Command or a character during execution. The aborting is not possible during some states of connection establishment such as handshaking.

2.2.3 ATD><mem><n> Originate call from specified memory

This command is used to originate a call using specified memory and index number.

ATD><mem><n> Originate call from specified memory	
Execution Command ATD><mem><n>[;]	Response a)If originate a voice call successfully: **OK** **VOICE CALL: BEGIN** b)If Originate a data call successfully: **CONNECT <text>** c)Originate a call unsuccessfully during command execution: **ERROR** d)Originate a call unsuccessfully for failed connection recovery: **NO CARRIER** e)Originate a call unsuccessfully for error related to the MT: **+CME ERROR: <err>**
Max Response Time	
Reference V.25ter	

Defined Values

<mem>	Phonebook storage: (For detailed description of storages see AT+CPBS) "DC" ME dialed calls list "MC" ME missed (unanswered received) calls list "RC" ME received calls list "SM" SIM phonebook "ME" UE phonebook "FD" SIM fixed dialing phonebook "ON" MSISDN list "LD" Last number dialed phonebook "EN" Emergency numbers
<n>	Integer type memory location in the range of locations available in the selected memory, i.e. the index returned by AT+CPBR.
<;>	The termination character ";" is mandatory to set up voice calls. It must not be used for data and fax calls.
<text>	CONNECT result code string; the string formats please refer ATX/AT\V/AT&E command.
<err>	Service failure result code string; the string formats please refer **+CME ERROR** result code and AT+CMEE command.

Example

```
ATD>SM3;                                   //Specify the <mem>.
OK

VOICE CALL: BEGIN
```

2.2.4 ATD><n> Originate call from active memory(1)

This command is used to originate a call to specified number.

ATD><n> Originate call from active memory	
Execution Command **ATD><n>[;]**	Response a)If originate a voice call successfully: **OK** **VOICE CALL: BEGIN**

	b)If Originate a data call successfully: **CONNECT <text>** c)Originate a call unsuccessfully during command execution: **ERROR** d)Originate a call unsuccessfully for failed connection recovery: **NO CARRIER** e)Originate a call unsuccessfully for error related to the MT: **+CME ERROR: <err>**
Reference V.25ter	

Defined Values

<n>	Integer type memory location in the range of locations available in the selected memory, i.e. the index returned by AT+CPBR.
<;>	The termination character ";" is mandatory to set up voice calls. It must not be used for data and fax calls.
<text>	CONNECT result code string; the string formats please refer ATX/AT\V/AT&E command.
<err>	Service failure result code string; the string formats please refer **+CME ERROR** result code and AT+CMEE command.

Example

```
ATD>2;
OK

VOICE CALL: BEGIN
```

2.2.5 ATD><str> Originate call from active memory(2)

This command is used to originate a call to specified number.

ATD><n> Originate call from active memory	
Execution Command **ATD><str>[;]**	Response a)If originate a voice call successfully:

	OK **VOICE CALL: BEGIN** b)If Originate a data call successfully: **CONNECT <text>** c)Originate a call unsuccessfully during command execution: **ERROR** d)Originate a call unsuccessfully for failed connection recovery: **NO CARRIER** e)Originate a call unsuccessfully for error related to the MT: **+CME ERROR: <err>**
Reference V.25ter	

Defined Values

<str>	String type value, which should equal to an alphanumeric field in at least one phone book entry in the searched memories. <str>formatted as current TE character set specified by AT+CSCS.<str> must be double quoted.
<;>	The termination character ";" is mandatory to set up voice calls. It must not be used for data and fax calls.
<text>	CONNECT result code string; the string formats please refer ATX/AT\V/AT&E command.
<err>	Service failure result code string; the string formats please refer **+CME ERROR** result code and AT+CMEE command.

Example

```
ATD>"kobe";
OK

VOICE CALL: BEGIN
```

2.2.6 ATA Call answer

This command is used to make remote station to go off-hook, e.g. answer an incoming call. If there is no an incoming call and entering this command to TA, it will be return "**NO CARRIER**" to TA.

ATA Call answer	
Execution Command **ATA**	Response a)If originate a voice call successfully: **OK** **VOICE CALL: BEGIN** b)For data call, and TA switches to data mode: **CONNECT** c)No connection or no incoming call: **NO CARRIER**
Reference V.25ter	

Example

```
ATA
VOICE CALL: BEGIN

OK
```

2.2.7 ATH Disconnect existing call

This command is used to disconnect existing call. Before using **ATH** command to hang up a voice call, it must set **AT+CVHU=0**. Otherwise, ATH command will be ignored and "**OK**" response is given only.
This command is also used to disconnect PS data call, and in this case it doesn't depend on the value of **AT+CVHU**.

ATH Disconnect existing call	
Execution Command **ATH**	Response **a)** If AT+CVHU=0: **OK**

	VOICE CALL: END: <time> b) If ATD*99# **OK** **+PPPD: DISCONNECTED** c) No any call **OK**
Reference V.25ter	

Defined Values

<time>	Voice call connection time: Format HHMMSS (HH: hour, MM: minute, SS: second)

Example

```
AT+CVHU=0
OK
ATH
OK

VOICE CALL: END: 000017
```

2.2.8 ATS0 Automatic answer incoming call

The S-parameter command controls the automatic answering feature of the Module. If set to 000, automatic answering is disabled, otherwise it causes the Module to answer when the incoming call indication (RING) has occurred the number of times indicated by the specified value; and the setting will not be stored upon power-off, i.e. the default value will be restored after restart.

ATS0 Automatic answer incoming call	
Read Command **ATS0?**	Response a) If success: **<n>** **OK** d) If failed

	ERROR
Write command **ATS0=<n>**	Response a) If success: **OK** b) If failed **ERROR**
Reference V.25ter	

Defined Values

<n>	000 Automatic answering mode is disable. (default value when power-on) 001–255 Enable automatic answering on the ring number specified.

Example

```
ATS0?
000

OK
ATS0=003
OK
```

NOTE

1.The S-parameter command is effective on voice call and data call.
2.If <n> is set too high, the remote party may hang up before the call can be answered automatically.

2.2.9 +++ Switch from data mode to command mode

This command is only available during a connecting PS data call. The **+++** character sequence causes the TA to cancel the data flow over the AT interface and switch to Command Mode. This allows to enter AT commands while maintaining the data connection to the remote device.

+++ Switch from data mode to command mode	
Execution Command **+++**	Response **OK**

Reference V.25ter	

NOTE

To prevent the +++ escape sequence from being misinterpreted as data, it must be preceded and followed by a pause of at least 1000 milliseconds, and the interval between two '+' character can't exceed 900 milliseconds.

2.2.10 ATO Switch from command mode to data mode

ATO is the corresponding command to the **+++** escape sequence. When there is a PS data call connected and the TA is in Command Mode, **ATO** causes the TA to resume the data and takes back to Data Mode.

ATO Switch from command mode to data mode	
Execution Command **ATO**	Response a) TA/DCE switches to Data Mode from Command Mode: **CONNECT [<baud rate>]** b) If connection is not successfully resumed: **NO CARRIER** or **ERROR**
Reference V.25ter	

Example

ATO
CONNECT 115200

2.2.11 ATI Display product identification information

This command is used to request the product information, which consists of manufacturer identification, model identification, revision identification, International Mobile station Equipment Identity (IMEI) and overall capabilities of the product.

ATI Display product identification information

Execution Command **ATI**	Response **Manufacturer: <manufacturer>** **Model: <model>** **Revision: <revision>** **IMEI: [<sn>]** **+GCAP: list of <name>s** **OK**
Reference V.25ter	

Defined Values

<manufacturer>	The identification of manufacturer.
<model>	The identification of model.
<revision>	The revision identification of firmware.
<sn>	Serial number identification, which consists of a single line containing IMEI (International Mobile station Equipment Identity) number.
<name>	List of additional capabilities: +CGSM GSM function is supported +FCLASS FAX function is supported +DS Data compression is supported +ES Synchronous data mode is supported. +CIS707-A CDMA data service command set +CIS-856 EVDO data service command set +MS Mobile Specific command set

Example

```
ATI
Manufacturer: SIMCOM INCORPORATED
Model: SIMCOM_SIM8200G
Revision: SIM8200G _V1.4.0
IMEI: 351602000330570
+GCAP: +CGSM,+FCLASS,+DS

OK
```

2.2.12 AT+IPR Set local baud rate temporarily

This command sets the baud rate of module's serial interface temporarily, after reboot the baud rate is set to value of **IPREX**.

AT+IPR Set local baud rate temporarily	
Test Command **AT+IPR=?**	Response **+IPR: (**list of supported **<speed>s)** **OK**
Read Command **AT+IPR?**	Response **+IPR: <speed>** **OK**
Write Command **AT+IPR=<speed>**	Response **OK** or **ERROR**
Execution Command **AT+IPR**	Set the value to boot value: **OK**

Defined Values

<speed>	Baud rate per second: 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 921600, 3000000, 3200000, 3686400

2.2.13 AT+ICF Set control character framing

This command sets character framing which contains data bit, stop bit and parity bit.

AT+ICF Set control character framing	
Test Command **AT+ICF=?**	Response **+ICF: (**list of supported **<format>s),(**list of supported **<parity>s)** **OK**
Read Command **AT+ICF?**	Response **+ICF: <format>,<parity>** **OK**
Write Command **AT+ICF=<format>[,<parity>]**	Response **OK** or **ERROR**

Execution Command **AT+ICF**	Set default value: **OK**
Reference V.25ter	

Defined Values

<format>	1 data bit 8, stop bit 2 2 data bit 8, parity bit 1,stop bit 1 3 data bit 8, stop bit 1 4 data bit 7, stop bit 2 5 data bit 7, parity bit 1,stop bit 1 6 data bit 7, stop bit 1
<parity>	0 Odd 1 Even 2 Space 3 none

Example

```
AT+ICF?
+ICF: 3,3

OK
AT+ICF=?
+ICF: (1-6),(0-3)

OK
AT+ICF=3,3
OK
```

2.2.14 AT+IFC Set local data flow control

The command sets the flow control mode of the module.

AT+IFC Set local data flow control	
Test Command **AT+IFC=?**	Response **+IFC: (**list of supported **<DCE>s),(**list of supported **<DTE>s)** **OK** or

	ERROR
Read Command **AT+IFC?**	Response **+IFC: <DCE>,<DTE>** **OK** or **ERROR**
Write Command **AT+IFC=<DCE>[,<DTE>]**	Response **OK** or **ERROR**
Execution Command **AT+IFC**	Set default value: **OK**
Reference V.25ter	

Defined Values

<DCE>	0 none (default) 2 RTS hardware flow control
<DTE>	0 none (default) 2 CTS hardware flow control

Example

```
AT+IFC?
+IFC: 0,0

OK
AT+IFC=?
+IFC: (0,2),(0,2)

OK
AT+IFC=2,2
OK
```

2.2.15 AT&C Set DCD function mode

This command determines how the state of DCD PIN relates to the detection of received line signal from the distant end.

AT&C Set DCD function mode	
Execution Command **AT&C[<value>]**	Response **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 DCD line shall always be on. 1 DCD line shall be on only when data carrier signal is present. 2 Setting winks(briefly transitions off,then back on)the DCD line when data calls end.

Example

```
AT&C1
OK
```

2.2.16 ATE Enable command echo

This command sets whether or not the TA echoes characters.

ATE Enable command echo	
Execution Command **ATE[<value>]**	Response **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 Echo mode off 1 Echo mode on

Example

```
ATE1
OK
```

2.2.17 AT&V Display current configuration

This command returns some of the base configuration parameters settings.

AT&V Display current configuration	
Execution Command **AT&V**	Response **<text>** **OK** or **ERROR**
Reference V.25ter	

Defined Values

<text>	All relative configuration information.

Example

AT&V
&C: 0; &D: 2; &F: 0; E: 1; L: 0; M: 0; Q:
0; V: 1; X: 0; Z: 0; S0: 0;
S3: 13; S4: 10; S5: 8; S6: 2; S7: 50; S8:
2; S9: 6; S10: 14; S11: 95;
+FCLASS: 0; +ICF: 3,3; +IFC: 2,2;
+IPR: 115200; +DR: 0; +DS: 0,0,2048,6;
+WS46: 12; +CBST: 0,0,1;
……
OK

2.2.18 AT&D Set DTR function mode

This command determines how the TA responds when DTR PIN is changed from the ON to the OFF condition during data mode.

AT&D Set DTR function mode	
Execution Command **AT&D[<value>]**	Response **OK** or

	ERROR
Reference V.25ter	

Defined Values

<value>	0 TA ignores status on DTR. 1 ON->OFF on DTR: Change to Command mode with remaining the connected call 2 ON->OFF on DTR: Disconnect call, change to Command mode.During state DTR = OFF is auto-answer off.

Example

AT&D1
OK

2.2.19 ATV Set result code format mode

This parameter setting determines the contents of the header and trailer transmitted with result codes and information responses.

ATV Set result code format mode	
Execution Command **ATV[<value>]**	Response If <value> =0 **0** If <value> =1 **OK**
Reference V.25ter	

Defined Values

<value>	0 Information response: <text><CR><LF> Short result code format: <numeric code><CR> 1 Information response: <CR><LF><text><CR><LF> Long result code format: <CR><LF><verbose code><CR><LF>

Example

ATV1
OK

2.2.20 AT&F Set all current parameters to manufacturer defaults

This command is used to set all current parameters to the manufacturer defined profile.

AT&F Set all current parameters to manufacturer defaults	
Execution Command **AT&F[<value>]**	Response **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 Set some temporary TA parameters to manufacturer defaults. The setting after power on or reset is same as value 0.

Default Values

TA Parameters	VALUES
AT+CNMP[1]	55
AT+CTZU[2]	0
AT+CGPSSSL[2]	0
AT+CGPSURL[2]	""

[1] takes effect immediately
[2] takes effect after a modem reboot

Example

```
AT&F
OK
```

2.2.21 ATQ Set Result Code Presentation Mode

Specify whether the TA transmits any result code to the TE or not. Text information transmitted in response is not affected by this setting.

ATQ Set Result Code Presentation Mode	
Execution Command **ATQ<n>**	Response If <n>=0: **OK** If <n>=1: No Responses
Execution Command **ATQ**	Set default value: 0 **OK**
Reference V.25ter	

Defined Values

<n>	0	DCE transmits result code
	1	DCE not transmits result code

Example

ATQ0
OK

2.2.22 ATX Set CONNECT Result Code Format

This parameter setting determines whether the TA transmits unsolicited result codes or not. The unsolicited result codes are
<CONNECT><SPEED><COMMUNICATION PROTOCOL>[<TEXT>]

ATX Set CONNECT Result Code Format	
Execution Command **ATX<n>**	Response **OK** or **ERROR**
Execution Command **ATX**	Set default value: 1 **OK** or **ERROR**
Reference V.25ter	

Defined Values

<n>	0 CONNECT result code returned 1,2,3,4 May be transmits extern result codes according to AT&E and AT\V settings. Refer to AT&E.

Example

ATX1
OK

2.2.23 AT\V Set CONNECT Result Code Format About Protocol

This parameter setting determines whether report the communication protocol. If PS call, it also determines whether report APN, uplink rate, downlink rate.

AT\V Set CONNECT Result Code Format About Protocol	
Execution Command **AT\V<value>**	Response **OK** or **ERROR**
Execution Command **AT\V**	Set default value: 0 **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 Don't report 1 Report communication protocol. And report APN, uplink rate, downlink rate if PS call. Refer to AT&E. The maybe communication protocol report include "NONE","PPPoverUD","AV32K","AV64K","PACKET". And APN in string format while uplink rate and downlink rate in integer format with kb unit.

Example

AT\V0
OK

2.2.24 AT&E Set CONNECT Result Code Format About Speed

This parameter setting determines to report Serial connection rate or Wireless connection speed. It is valid only ATX above 0.

AT&E Set CONNECT Result Code Format About Speed	
Execution Command **AT&E<value>**	Response **OK** or **ERROR**
Execution Command **AT&E**	Set default value: 1 **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0	Wireless connection speed in integer format.
	1	Serial connection rate in integer format. Such as: "115200"

Example

AT&E0
OK

2.2.25 AT&W Save the user setting to ME

This command will save the user settings to ME which set by ATE, ATQ, ATV, ATX, AT&C AT&D, AT\V, AT+IFC and ATS0.

AT&W Save the user setting to ME	
Execution Command **AT&W<value>**	Response **OK** or **ERROR**
Execution Command	Set default value: 0

AT&W	**OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 Save

Example

```
AT&W0
OK
```

2.2.26 ATZ Restore the user setting from ME

This command will restore the user setting from ME which set by ATE, ATQ, ATV, ATX, AT&C AT&D, AT&S, AT\Q, AT\V, and ATS0.

ATZ Restore the user setting from ME	
Execution Command ATZ<value>	Response **OK** or **ERROR**
Execution Command ATZ	Set default value: 0 **OK** or **ERROR**
Reference V.25ter	

Defined Values

<value>	0 Restore

Example

```
ATZ0
OK
```

2.2.27 AT+CGMI Request manufacturer identification

This command is used to request the manufacturer identification text, which is intended to permit the user of the Module to identify the manufacturer.

AT+CGMI Request manufacturer identification	
Test Command **AT+CGMI=?**	Response **OK**
Execution Command **AT+CGMI**	Response **<manufacturer>** **OK** or **ERROR**
Reference V.25ter	

Defined Values

<manufacturer>	The identification of manufacturer.

Example

```
AT+CGMI
SIMCOM INCORPORATED

OK
```

2.2.28 AT+CGMM Request model identification

This command is used to requests model identification text, which is intended to permit the user of the Module to identify the specific model.

AT+CGMM Request model identification	
Test Command **AT+CGMM=?**	Response **OK**
Execution Command **AT+CGMM**	Response **<model>**

	OK or **ERROR**
Reference V.25ter	

Defined Values

<model>	The identification of model.

Example

```
AT+CGMM
SIMCOM_SIM8200G

OK
```

2.2.29 AT+CGMR Request revision identification

This command is used to request product firmware revision identification text, which is intended to permit the user of the Module to identify the version.

AT+CGMR Request revision identification	
Test Command **AT+CGMR=?**	Response **OK**
Execution Command **AT+CGMR**	Response **+CGMR: <revision>** **OK** or **ERROR**
Reference V.25ter	

Defined Values

<revision>	The revision identification of firmware.

Example

```
AT+CGMR
+CGMR: LE14B03SIM8200M44A-LGA
OK
```

2.2.30 AT+CGSN Request product serial number identification

This command requests product serial number identification text, which is intended to permit the user of the Module to identify the individual ME to which it is connected to.

AT+CGSN Request product serial number identification	
Test Command **AT+CGSN=?**	Response **OK**
Execution Command **AT+CGSN**	Response **<sn>** **OK** or **+CME ERROR: memory failure**
Reference V.25ter	

Defined Values

<sn>	Serial number identification, which consists of a single line containing the IMEI (International Mobile station Equipment Identity) number of the MT. If in CDMA/EVDO mode ,it will show ESN(Electronic Serial Number)

Example

```
AT+CGSN
351602000330570

OK
```

2.2.31 AT+CSCS Select TE character set

Write command informs TA which character set <chest> is used by the TE. TA is then able to convert character strings correctly between TE and MT character sets.

Read command shows current setting and test command displays conversion schemes implemented in the TA.

AT+CSCS Select TE character set	
Test Command **AT+CSCS=?**	Response **+CSCS:** (list of supported **<chset>s)** **OK**
Read Command **AT+CSCS?**	Response **+CSCS: <chset>** **OK**
Write Command **AT+CSCS=<chset>**	Response **OK** or **ERROR**
Execution Command **AT+CSCS**	Set subparameters as default value: **OK**
Reference V.25ter	

Defined Values

<chset>	Character set, the definition as following: "IRA" International reference alphabet. "GSM" GSM default alphabet; this setting causes easily software flow control (XON /XOFF) problems. "UCS2" 16-bit universal multiple-octet coded character set; UCS2 character strings are converted to hexadecimal numbers from 0000 to FFFF.

Example

```
AT+CSCS="IRA"
OK
```

2.2.32 AT+CIMI Request international mobile subscriber identity

Execution command causes the TA to return <IMSI>, which is intended to permit the TE to identify the individual SIM card which is attached to MT.

NOTE: If USIM card contains two apps, like China Telecom 4G card, one RUIM/CSIM app, and another

USIM app; so there are two IMSI in it; AT+CIMI will return the RUIM/CSIM IMSI; AT+CIMIM will return the USIM IMSI.

AT+CIMI Request international mobile subscriber identity	
Test Command **AT+CIMI=?**	Response **OK**
Execution Command **AT+CIMI**	Response **<IMSI>** **OK** or **+CME ERROR: memory failure**
Reference V.25ter	

Defined Values

<IMSI>	International Mobile Subscriber Identity (string, without double quotes).

Example

```
AT+CIMI
460010222028133

OK
```

2.2.33 AT+GCAP Request overall capabilities

Execution command causes the TA reports a list of additional capabilities.

AT+GCAP Request overall capabilities	
Test Command **AT+GCAP=?**	Response **OK**
Execution Command **AT+GCAP**	Response **+GCAP:** (list of **<name>**s) **OK**
Reference V.25ter	

Defined Values

<name>	List of additional capabilities. +CGSM GSM function is supported +FCLASS FAX function is supported +DS Data compression is supported +ES Synchronous data mode is supported. +CIS707-A CDMA data service command set +CIS-856 EVDO data service command set +MS Mobile Specific command set

Example

```
AT+GCAP
+GCAP: +CGSM,+FCLASS,+DS

OK
```

2.2.34 AT+CVMOD Select the type of voice calls

Execution command to select the type of voice calls: CS_ONLY, VOIP_ONLY, CS_PREFERRED or VOIP_PREFERRED.

AT+CVMOD	Request overall capabilities
Test Command **AT+CVMOD=?**	Response **+CVMOD:** (list of **<mode>)** **OK**
Read Command **AT+CVMOD?**	Response **+CVMOD: (<mode>)** **OK**
Execution Command **AT+CVMOD=<mode>**	Response **OK**

Defined Values

<mode>	List of <mode>: 0 CS_ONLY 1 VOIP_ONLY

	2 CS_PREFERRED 3 VOIP_PREFERRED

Example

```
AT+CVMOD=?
+CVMOD: (0-3)
OK
```

3 AT Commands for Status Control

3.1 Overview of AT Commands for Status Control

Command	Description
AT+CFUN	Set phone functionality
AT+CPIN	Enter PIN
AT+CICCID	Read ICCID from SIM card
AT+CSIM	Generic SIM access
AT+CRSM	Restricted SIM access
AT+SPIC	Times remain to input SIM PIN/PUK
AT+CSPN	Get service provider name from SIM
AT+CSQ	Query signal quality
AT+AUTOCSQ	Set CSQ report
AT+CSQDELTA	Set RSSI delta change threshold
AT+CATR	Configure URC destination interface
AT+CPOF	Power down the module
AT+CRESET	Reset the module
AT+CACM	Accumulated call meter
AT+CAMM	Accumulated call meter maximum
AT+CPUC	Price per unit and currency table
AT+CCLK	Real time clock management
AT+CMEE	Report mobile equipment error

AT+CPAS	Phone activity status
AT+SIMEI	Set IMEI for the module
AT+CSVM	Voice Mail Subscriber number
AT+SMSIMCFG	Single/Dual SIM configuration

3.2 Detailed Description of AT Commands for Status Control

3.2.1 AT+CFUN Set phone functionality

Description

This command is used to select the level of functionality <fun> in the ME. Level "full functionality" is where the highest level of power is drawn. "Minimum functionality" is where minimum power is drawn. Level of functionality between these may also be specified by manufacturers. When supported by manufacturers, ME resetting with <rst> parameter may be utilized.

NOTE: AT+CFUN=6 must be used after setting AT+CFUN=7. If module in offline mode, must execute AT+CFUN=6 or restart module to online mode.

AT+CFUN Set phone functionality	
Test Command **AT+CFUN=?**	Response **+CFUN**: (list of supported **<fun>**s),(list of supported **<rst>**s) **OK** or **ERROR** or **+CME ERROR: <err>**
Read Command **AT+CFUN?**	Response **+CFUN: <fun>** **OK** or **ERROR** or **+CME ERROR: <err>**

Write Command	Response
AT+CFUN=<fun>,[<rst>]	**OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<fun>	0 minimum functionality 1 full functionality, online mode 4 disable phone both transmit and receive RF circuits 5 Factory Test Mode 6 Reset 7 Offline Mode
<rst>	0 do not reset the ME before setting it to <fun> power level 1 reset the ME before setting it to <fun> power level. This value only takes effect when <fun> equals 1.

Example

AT+CFUN?
+CFUN: 1

OK
AT+CFUN=0
OK

3.2.2 AT+CPIN Enter PIN

Description

This command is used to send the ME a password which is necessary before it can be operated (SIM PIN, SIM PUK, PH- SIM PIN, etc.). If the PIN is to be entered twice, the TA shall automatically repeat the PIN. If no PIN request is pending, no action is taken towards MT and an error message, **+CME ERROR**, is returned to TE.
If the PIN required is SIM PUK or SIM PUK2, the second pin is required. This second pin, <newpin>, is used to replace the old pin in the SIM.

AT+CPIN Enter PIN	
Test Command AT+CPIN=?	Response **OK**
Read Command AT+CPIN?	Response **+CPIN: <code>** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command AT+CPIN=<pin>[,<newpin>]	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<pin>	String type values.
<newpin>	String type values.
<code>	Values reserved by the present document: READY – ME is not pending for any password SIM PIN – ME is waiting SIM PIN to be given SIM PUK – ME is waiting SIM PUK to be given PH-SIM PIN – ME is waiting phone-to-SIM card password to be given SIM PIN2 – ME is waiting SIM PIN2 to be given SIM PUK2 – ME is waiting SIM PUK2 to be given PH-NET PIN – ME is waiting network personalization password to be given

Example

```
AT+CPIN?
+CPIN: SIM PUK2
```

```
OK
```

3.2.3 AT+CICCID Read ICCID from SIM card

Description

This command is used to Read the ICCID from SIM card

AT+CICCID Read ICCID from SIM card	
Test Command **AT+CICCID=?**	Response **OK**
Execution Command **AT+CICCID**	Response **+ICCID: <ICCID>** **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<ICCID>	Integrate circuit card identity, a standard ICCID is a 20-digit serial number of the SIM card, it presents the publish state, network code, publish area, publish date, publish manufacture and press serial number of the SIM card.

Example

```
AT+CICCID
+ICCID: 898600700907A6019125

OK
```

3.2.4 AT+CSIM Generic SIM access

Description

This command is used to control the SIM card directly.
Compared to restricted SIM access command AT+CRSM, AT+CSIM allows the ME to take more control over the SIM interface.
For SIM–ME interface please refer 3GPP TS 11.11.
NOTE: The SIM Application Toolkit functionality is not supported by AT+CSIM. Therefore the following SIM commands can not be used: TERMINAL PROFILE, ENVELOPE, FETCH and TEMINAL RESPONSE.

AT+CSIM Generic SIM access	
Test Command **AT+CSIM=?**	Response **OK**
Write Command **AT+CSIM=<length>,<command>**	Response **+CSIM: <length>,<response>** **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<length>	Integer type; length of characters that are sent to TE in <command>or<response>
<command>	Command passed from MT to SIM card.
<response>	Response to the command passed from SIM card to MT.

Example

```
AT+CSIM=?
OK
```

3.2.5 AT+CRSM Restricted SIM access

Description

By using AT+CRSM instead of Generic SIM Access AT+CSIM, TE application has easier but more limited access to the SIM database.
Write command transmits to the MT the SIM <command> and its required parameters. MT handles internally all SIM-MT interface locking and file selection routines. As response to the command, MT sends the actual SIM information parameters and response data. MT error result code **+CME ERROR** may be returned when the command cannot be passed to the SIM, but failure in the execution of the command in the SIM is reported in <sw1> and <sw2> parameters.

AT+CRSM Restricted SIM access	
Test Command **AT+CRSM=?**	Response **OK**
Write Command **AT+CRSM=<command>[,<fileID>[,<p1>,<p2>,<p3>[,<data>]]]**	Response **+CRSM: <sw1>,<sw2>[,<response>]** **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<command>	Command passed on by the MT to the SIM: 176 READ BINARY 178 READ RECORD 192 GET RESPONSE 214 UPDATE BINARY 220 UPDATE RECORD 242 STATUS 203 RETRIEVE DATA 219 SET DATA
<fileID>	Identifier for an elementary data file on SIM, if used by <command>. The following list the fileID hex value, user needs to convert them to decimal. EFs under MF 0x2FE2 ICCID 0x2F05 Extended Language Preferences

0x2F00	EF DIR
0x2F06	Access Rule Reference
EFs under USIM ADF	
0x6F05	Language Indication
0x6F07	IMSI
0x6F08	Ciphering and Integrity keys
0x6F09	C and I keys for pkt switched domain
0x6F60	User controlled PLMN selector w/Acc Tech
0x6F30	User controlled PLMN selector
0x6F31	HPLMN search period
0x6F37	ACM maximum value
0x6F38	USIM Service table
0x6F39	Accumulated Call meter
0x6F3E	Group Identifier Level
0x6F3F	Group Identifier Level 2
0x6F46	Service Provider Name
0x6F41	Price Per Unit and Currency table
0x6F45	Cell Bcast Msg identifier selection
0x6F78	Access control class
0x6F7B	Forbidden PLMNs
0x6F7E	Location information
0x6FAD	Administrative data
0x6F48	Cell Bcast msg id for data download
0x6FB7	Emergency call codes
0x6F50	Cell bcast msg id range selection
0x6F73	Packet switched location information
0x6F3B	Fixed dialling numbers
0x6F3C	Short messages
0x6F40	MSISDN
0x6F42	SMS parameters
0x6F43	SMS Status
0x6F49	Service dialling numbers
0x6F4B	Extension 2
0x6F4C	Extension 3
0x6F47	SMS reports
0x6F80	Incoming call information
0x6F81	Outgoing call information
0x6F82	Incoming call timer
0x6F83	Outgoing call timer
0x6F4E	Extension 5
0x6F4F	Capability Config Parameters 2
0x6FB5	Enh Multi Level Precedence and Pri
0x6FB6	Automatic answer for eMLPP service
0x6FC2	Group identity

	0x6FC3	Key for hidden phonebook entries
	0x6F4D	Barred dialling numbers
	0x6F55	Extension 4
	0x6F58	Comparison Method information
	0x6F56	Enabled services table
	0x6F57	Access Point Name Control List
	0x6F2C	De-personalization Control Keys
	0x6F32	Co-operative network list
	0x6F5B	Hyperframe number
	0x6F5C	Maximum value of Hyperframe number
	0x6F61	OPLMN selector with access tech
	0x6F5D	OPLMN selector
	0x6F62	HPLMN selector with access technology
	0x6F06	Access Rule reference
	0x6F65	RPLMN last used access tech
	0x6FC4	Network Parameters
	0x6F11	CPHS: Voice Mail Waiting Indicator
	0x6F12,	CPHS: Service String Table
	0x6F13	CPHS: Call Forwarding Flag
	0x6F14	CPHS: Operator Name String
	0x6F15	CPHS: Customer Service Profile
	0x6F16	CPHS: CPHS Information
	0x6F17	CPHS: Mailbox Number
	0x6FC5	PLMN Network Name
	0x6FC6	Operator PLMN List
	0x6F9F	Dynamic Flags Status
	0x6F92	Dynamic2 Flag Setting
	0x6F98	Customer Service Profile Line2
	0x6F9B	EF PARAMS - Welcome Message
	0x4F30	Phone book reference file
	0x4F22	Phone book synchronization center
	0x4F23	Change counter
	0x4F24	Previous Unique Identifier
	0x4F20	GSM ciphering key Kc
	0x4F52	GPRS ciphering key
	0x4F63	CPBCCH information
	0x4F64	Investigation scan
	0x4F40	MExE Service table
	0x4F41	Operator Root Public Key
	0x4F42	Administrator Root Public Key
	0x4F43	Third party Root public key
	0x6FC7	Mail Box Dialing Number
	0x6FC8	Extension 6
	0x6FC9	Mailbox Identifier

	0x6FCA Message Waiting Indication Status 0x6FCD Service Provider Display Information 0x6FD2 UIM_USIM_SPT_TABLE 0x6FD9 Equivalent HPLMN 0x6FCB Call Forwarding Indicator Status 0x6FD6 GBA Bootstrapping parameters 0x6FDA GBA NAF List 0x6FD7 MBMS Service Key 0x6FD8 MBMS User Key 0x6FCE MMS Notification 0x6FD0 MMS Issuer connectivity parameters 0x6FD1 MMS User Preferences 0x6FD2 MMS User connectivity parameters 0x6FCF Extension 8 0x5031 Object Directory File 0x5032 Token Information File 0x5033 Unused space Information File EFs under Telecom DF 0x6F3A Abbreviated Dialing Numbers 0x6F3B Fixed dialling numbers 0x6F3C Short messages 0x6F3D Capability Configuration Parameters 0x6F4F Extended CCP 0x6F40 MSISDN 0x6F42 SMS parameters 0x6F43 SMS Status 0x6F44 Last number dialled 0x6F49 Service Dialling numbers 0x6F4A Extension 1 0x6F4B Extension 2 0x6F4C Extension 3 0x6F4D Barred Dialing Numbers 0x6F4E Extension 4 0x6F47 SMS reports 0x6F58 Comparison Method Information 0x6F54 Setup Menu elements 0x6F06 Access Rule reference 0x4F20 Image 0x4F30 Phone book reference file 0x4F22 Phone book synchronization center 0x4F23 Change counter 0x4F24 Previous Unique Identifier
<p1><p2><p3>	Integer type; parameters to be passed on by the Module to the SIM.
<data>	Information which shall be written to the SIM (hexadecimal character format, refer

	AT+CSCS).
<sw1><sw2>	Status information from the SIM about the execution of the actual command. It is returned in both cases, on successful or failed execution of the command.
<response>	Response data in case of a successful completion of the previously issued command. "STATUS" and "GET RESPONSE" commands return data, which gives information about the currently selected elementary data field. This information includes the type of file and its size. After "READ BINARY" or "READ RECORD" commands the requested data will be returned. <response> is empty after "UPDATE BINARY" or "UPDATE RECORD" commands.

Example

```
AT+CRSM=?
OK
```

3.2.6 AT+SPIC Times remain to input SIM PIN/PUK

Description

This command is used to inquire times remain to input SIM PIN/PUK.

AT+SPIC Times remain to input SIM PIN/PUK	
Test Command **AT+SPIC=?**	Response **OK**
Execution Command **AT+SPIC**	Response **+SPIC: <pin1>,<puk1>,<pin2>,<puk2>** **OK**

Defined Values

<pin1>	Times remain to input PIN1 code.
<puk1>	Times remain to input PUK1 code.
<pin2>	Times remain to input PIN2 code.
<puk2>	Times remain to input PUK2 code.

Example

```
AT+SPIC=?
OK
AT+SPIC
+SPIC: 3,10,0,10

OK
```

3.2.7 AT+CSPN Get service provider name from SIM

Description

This command is used to get service provider name from SIM card.

AT+CSPN Get service provider name from SIM	
Test Command AT+CSPN=?	Response **OK** or **ERROR**
Read Command AT+CSPN?	Response **+CSPN: <spn>,<display mode>** **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<spn>	String type; service provider name on SIM
<display mode>	0 doesn't display PLMN. Already registered on PLMN. 1 display PLMN

Example

```
AT+CSPN=?
OK
AT+CSPN?
+CSPN: "CMCC",0

OK
```

3.2.8 AT+CSQ Query signal quality

Description

This command is used to return received signal strength indication <rssi> and channel bit error rate <ber> from the ME. Test command returns values supported by the TA as compound values.

AT+CSQ Query signal quality	
Test Command **AT+CSQ=?**	Response **+CSQ: (**list of supported **<rssi>s),(**list of supported **<ber>s)** **OK**
Execution Command **AT+CSQ**	Response **+CSQ: <rssi>,<ber>** **OK** or **ERROR**

Defined Values

<rssi>	0	-113 dBm or less
	1	-111 dBm
	2...30	-109... -53 dBm
	31	-51 dBm or greater
	99	not known or not detectable
	100	-116 dBm or less
	101	-115 dBm
	102…190	-114... -26dBm
	191	-25 dBm or greater

	199 not known or not detectable 100…199 expand to TDSCDMA, indicate RSCP received
<ber>	(in percent) 0 <0.01% 1 0.01% --- 0.1% 2 0.1% --- 0.5% 3 0.5% --- 1.0% 4 1.0% --- 2.0% 5 2.0% --- 4.0% 6 4.0% --- 8.0% 7 >=8.0% 99 not known or not detectable

Example

```
AT+CSQ
+CSQ: 22,0

OK
```

3.2.9 AT+AUTOCSQ Set CSQ report

Description

This command is used to enable or disable automatic report CSQ information, when automatic report enabled, the module reports CSQ information every five seconds or only after <rssi> or <ber> is changed, the format of automatic report is **"**+CSQ: <rssi>,<ber>**"**.

AT+AUTOCSQ Set CSQ report	
Test Command **AT+AUTOCSQ=?**	Response **+AUTOCSQ: (**list of supported **<auto>s),(**list of supported **<mode>s)** **OK**
Read Command **AT+AUTOCSQ?**	Response **+AUTOCSQ: <auto>,<mode>** **OK**

Write Command	Response
AT+AUTOCSQ=<auto>[,<mode>]	**OK** or **ERROR**

Defined Values

<auto>	0 disable automatic report 1 enable automatic report
<mode>	0 CSQ automatic report every five seconds 1 CSQ automatic report only after <rssi> or <ber> is changed **NOTE:**If the parameter of <mode> is omitted when executing write command, <mode> will be set to default value.

Example

```
AT+AUTOCSQ=?
+AUTOCSQ: (0-1),(0-1)

OK
AT+AUTOCSQ?
+AUTOCSQ: 1,1

OK
AT+AUTOCSQ=1,1
OK

+CSQ: 23,0 (when <rssi> or <ber> changing)
```

3.2.10 AT+CSQDELTA Set RSSI delta change threshold

Description

This command is used to set RSSI delta threshold for signal strength reporting.

AT+CSQDELTA Set RSSI delta change threshold

Test Command **AT+CSQDELTA=?**	Response **+CSQDELTA: (**list of supported **<delta>s)** **OK**
Read Command **AT+CSQDELTA?**	Response **+CSQDELTA: <delta>** **OK** or **ERROR**
Write Command **AT+CSQDELTA=<delta>**	Response **OK** or **ERROR**
Execution Command **AT+CSQDELTA**	Response Set default value（<delta>=5）: **OK**

Defined Values

<delta>	Range: from 0 to 5.

Example

```
AT+CSQDELTA?
+CSQDELTA: 5

OK
```

3.2.11 AT+CATR Configure URC destination interface

Description

This command is used to configure the serial port which will be used to output URCs. We recommend configure a destination port for receiving URC in the system initialization phase, in particular, in the case that transmitting large amounts of data, e.g. use TCP/UDP and MT SMS related AT command.

AT+CATR Configure URC destination interface	
Test Command **AT+CATR=?**	Response **+CATR:** (list of supported **<port>s)** **OK**
Read Command **AT+CATR?**	Response **+CATR: <port>** **OK**
Write Command **AT+CATR=<port>**	Response **OK** or **ERROR**
Execution Command **AT+CATR**	Response Set default value（<port>=0）: **OK**

Defined Values

<port>	0 all ports 1 use UART port to output URCs 2 use MODEM port to output URCs 3 use ATCOM port to output URCs 4 use cmux virtual port1 to output URCs 5 use cmux virtual port2 to output URCs 6 use cmux virtual port3 to output URCs 7 use cmux virtual port4 to output URCs

Example

AT+CATR=1
OK
AT+CATR?
+CATR: 1

OK

3.2.12 AT+CPOF Power down the module

Description

This command is used to power off the module. Once the AT+CPOF command is executed, The module will store user data and deactivate from network, and then shutdown.

AT+CPOF Power down the module	
Test Command **AT+CPOF=?**	Response **OK**
Execution Command **AT+CPOF**	Response **OK**

Example

```
AT+CPOF
OK
```

3.2.13 AT+CRESET Reset the module

Description

This command is used to reset the module.

AT+CRESET Reset the module	
Test Command **AT+CRESET=?**	Response **OK**
Execution Command **AT+CRESET**	Response **OK**

Example

```
AT+CRESET=?
OK
AT+CRESET
OK
```

3.2.14 AT+CACM Accumulated call meter

Description

This command is used to reset the Advice of Charge related accumulated call meter value in SIM file EF_{ACM}.

AT+CACM Accumulated call meter	
Test Command **AT+CACM=?**	Response **OK** or **ERROR**
Read Command **AT+CACM?**	Response **+CACM: <acm>** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+CACM=<passwd>**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+CACM**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<passwd>	String type, SIM PIN2.
<acm>	String type, accumulated call meter value similarly coded as <ccm> under +CAOC.

Example

```
AT+CACM?
+CACM: "000000"

OK
```

3.2.15 AT+CAMM Accumulated call meter maximum

Description

This command is used to set the Advice of Charge related accumulated call meter maximum value in SIM file EF_{ACMmax}.

AT+CAMM Accumulated call meter maximum	
Test Command **AT+CAMM=?**	Response **OK** or **ERROR**
Read Command **AT+CAMM?**	Response **+CAMM: <acmmax>** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+CAMM=<acmmax>[,<passwd>]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+CAMM**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<acmmax>	String type, accumulated call meter maximum value similarly coded as <ccm> under AT+CAOC, value zero disables ACMmax feature.
<passwd>	String type, SIM PIN2.

Example

```
AT+CAMM?
+CAMM: "000000"

OK
```

3.2.16 AT+CPUC Price per unit and currency table

Description

This command is used to set the parameters of Advice of Charge related price per unit and currency table in SIM file EF_{PUCT}..

AT+CPUC Price per unit and currency table	
Test Command **AT+CPUC=?**	Response **OK** or **ERROR**
Read Command **AT+CPUC?**	Response **+CPUC: [<currency>,<ppu>]** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+CPUC=<currency>,<ppu>[,<passwd>]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<currency>	String type, three-character currency code (e.g. "GBP", "DEM"), character set as specified by command Select TE Character Set AT+CSCS.
<ppu>	String type, price per unit, dot is used as a decimal separator. (e.g. "2.66").
<passwd>	String type, SIM PIN2.

Example

```
AT+CPUC?
+CPUC: "GBP","2.66"

OK
```

3.2.17 AT+CCLK Real time clock management

Description

This command is used to manage Real Time Clock of the module.

AT+CCLK Real time clock management	
Test Command **AT+CCLK=?**	Response **OK**
Read Command **AT+CCLK?**	Response **+CCLK: <time>** **OK**
Write Command **AT+CCLK=<time>**	Response **OK** or **ERROR**

Defined Values

<time>	String type value; format is "yy/MM/dd,hh:mm:ss±zz", where characters indicate year (two last digits), month, day, hour, minutes, seconds and time zone (indicates the difference, expressed in quarters of an hour, between the local time and GMT; three

	last digits are mandatory, range -96…+96). E.g. 6th of May 2008, 14:28:10 GMT+8 equals to "08/05/06,14:28:10+32". **NOTE:** 1. Time zone is nonvolatile, and the factory value is invalid time zone. 2. Command +CCLK? will return time zone when time zone is valid, and if time zone is 00, command +CCLK? will return "+00", but not "-00".

Example

```
AT+CCLK="08/11/28,12:30:33+32"
OK
AT+CCLK?
+CCLK: "08/11/28,12:30:35+32"

OK
AT+CCLK="08/11/26,10:15:00"
OK
AT+CCLK?
+CCLK: "08/11/26,10:15:02+32"

OK
```


3.2.18 AT+CMEE Report mobile equipment error

Description

This command is used to disable or enable the use of result code "**+CME ERROR: <err>**" or "**+CMS ERROR: <err>**" as an indication of an error relating to the functionality of ME; when enabled, the format of <err> can be set to numeric or verbose string.

AT+CMEE Report mobile equipment error	
Test Command **AT+CMEE=?**	Response **+CMEE:** (list of supported **<n>s)** **OK**
Read Command **AT+CMEE?**	Response **+CMEE: <n>** **OK**

Write Command **AT+CMEE=<n>**	Response **OK** or **ERROR**
Execution Command **AT+CMEE**	Response *Set default value:* **OK**

Defined Values

<n>	0 – Disable result code,i.e. only "**ERROR**" will be displayed. 1 – Enable error result code with numeric values. 2 – Enable error result code with string values.

Example

```
AT+CMEE?
+CMEE: 2

OK
AT+CPIN="1234","1234"
+CME ERROR: incorrect password
AT+CMEE=0
OK
AT+CPIN="1234","1234"
ERROR
AT+CMEE=1
OK
AT+CPIN="1234","1234"
+CME ERROR: 16
```

3.2.19 AT+CPAS Phone activity status

Description

This command is used to return the activity status <pas> of the ME. It can be used to interrogate the ME before requesting action from the phone.

NOTE: This command is same as AT+CLCC, but AT+CLCC is more commonly used. So

AT+CLCC is recommended to use.

AT+CPAS Phone activity status	
Test Command **AT+CPAS=?**	Response **+CPAS: (**list of supported **<pas>s)** **OK**
Execution Command **AT+CPAS**	Response **+CPAS: <pas>** **OK**

Defined Values

<pas>	0 ready (ME allows commands from TA/TE) 3 ringing (ME is ready for commands from TA/TE, but the ringer is active) 4 call in progress (ME is ready for commands from TA/TE, but a call is in progress)

Example

```
RING (with incoming call)
AT+CPAS
+CPAS: 3

OK
AT+CPAS=?
+CPAS: (0,3,4)

OK
```

3.2.20 AT+SIMEI Set IMEI for the module

Description

This command is used to set the module's IMEI value.

AT+SIMEI Set IMEI for the module

Test Command **AT+SIMEI=?**	Response **OK**
Read Command **AT+SIMEI?**	Response **+SIMEI: <imei>** **OK** or **ERROR**
Write Command **AT+SIMEI=<imei>**	Response **OK** or **ERROR**

Defined Values

<imei>	The 15-digit IMEI value.

Example

```
AT+SIMEI=357396012183170
OK
AT+SIMEI?
+SIMEI: 357396012183170

OK
AT+SIMEI=?
OK
```

3.2.21 AT+CSVM Voice Mail Subscriber number

Description

Execute the following command returns the voice mail number related to the subscriber.

AT+CSVM Voice Mail Subscriber number

Test Command **AT+CSVM=?**	Response **+CSVM: (0-1),"(0-9,+)",(128-255)** **OK** or **ERROR**
Read Command **AT+CSVM?**	Response **+CSVM: <valid>,"<number>",<type>** **OK** or **ERROR**
Write Command **AT+CSVM=<valid>,"<number>",<type>**	Response **OK** or **ERROR**

Defined Values

<valid>	Whether voice mail number is valid: 0 Voice mail number is invalid. 1 Voice mail number is valid.
<number>	String type phone number of format specified by <type>.
<type>	Type of address octet in integer format. see also AT+CPBR <type>

Example

```
AT+CSVM?
+CSVM: 1,"13697252277",129

OK
```

3.2.22 AT+SMSIMCFG Single/Dual SIM configuration

Description

This command is used to switch between two slots of the module in SSSA or DSSA. This command will take effect immediately.If the action has been completed,module returns the unsolicited result code **+SMSIMCFG: <mode>,<slot>**. Otherwise, switching another slot is not allowed.

AT+SMSIMCFG Single/Dual SIM configuration	
Read Command **AT+SMSIMCFG?**	Response If SSSA: **+SMSIMCFG: <mode>,<slot>,<stat>** **OK** or If DSSA: **+SMSIMCFG: <mode>,<slot>,<stat>** **+SMSIMCFG: <mode>,<slot>,<stat>** **OK**
Write Command **AT+SMSIMCFG=<mode>[,<slot>]** NOTE: Only one slot can be activated at a time	Response **OK** or **ERROR**

Defined Values

<mode>	Integer type, indicates the module's mode 0 – SSSA 1 – DSSA
<slot>	Integer type, indicates the slot number 1 – Slot 1 2 – Slot 2
<stat>	Integer type, indicates the status of the SIM card on each slot 0 – SIM card not inserted 1 – SIM card inserted and activated 2 – SIM card inserted and deactivated 3 – SIM error

Example

```
AT+SMSIMCFG?
+SMSIMCFG: 0,1,1 //in SSSA mode
```

```
OK

AT+SMSIMCFG=1,2
OK
+SIMCARD: NOT AVAILABLE

+CPIN: READY

AT+SMSIMCFG?
+SMSIMCFG: 1,1,2
+SMSIMCFG: 1,2,1 //in DSSA mode: slot 2 inserted and activated, slot 1 inserted and deactivated

OK
```

NOTE

All 5G module series may support SSSA/DSSA except 8300G 5G modules.
Multi-SIMs platforms,i.e. DSDS/DSDA do not support this AT command.

3.2.23 Indication of Voice Mail

This module supports voice mail function; the subscriber number is configured by AT+CSVM command, the following table shows the URC related Voice Mail.

Indication of Voice Mail	
Box Empty **+VOICEMAIL: EMPTY**	Description This indication means the voice mail box is empty
New Message **+VOICEMAIL: NEW MSG**	Description This indication means there is a new voice mail message notification received. This is for CPHS.
Voice Mail Status Updated **+VOICEMAIL: WAITING, <count>**	Description This indication means that there are <count> number of voice mail messages that needs to be got.

Defined Values

<count>	Count of voice mail message that waits to be got.

Example

```
+VOICEMAIL: WAITING,<count>
+VOICEMAIL: WAITING,5
```

3.3 Summary of CME ERROR codes

This result code is similar to the regular ERROR result code. The format of <err> can be ethier numeric or verbose string, by setting AT+CMEE command.

<err> of numeric format	<err> of verbose format
0	Phone failure
1	no connection to phone
2	phone adaptor link reserved
3	operation not allowed
4	operation not supported
5	PH-SIM PIN required
6	PH-FSIM PIN required
7	PH-FSIM PUK required
10	SIM not inserted
11	SIM PIN required
12	SIM PUK required
13	SIM failure
14	SIM busy
15	SIM wrong
16	incorrect password
17	SIM PIN2 required
18	SIM PUK2 required
20	memory full
21	invalid index
22	not found
23	memory failure
24	text string too long
25	invalid characters in text string
26	dial string too long
27	invalid characters in dial string
30	no network service

31	network timeout
32	network not allowed – emergency calls only
40	network personalization PIN required
41	network personalization PUK required
42	network subset personalization PIN required
43	network subset personalization PUK required
44	service provider personalization PIN required
45	service provider personalization PUK required
46	corporate personalization PIN required
47	corporate personalization PUK required
100	Unknown
103	Illegal message
106	Illegal ME
107	GPRS services not allowed
111	PLMN not allowed
112	Location area not allowed
113	Roaming not allowed in this location area
132	service option not supported
133	requested service option not subscribed
134	service option temporarily out of order
148	unspecified GPRS error
149	PDP authentication failure
150	invalid mobile class
257	network rejected request
258	retry operation
259	invalid deflected to number
260	deflected to own number
261	unknown subscriber
262	service not available
263	unknown class specified
264	unknown network message
273	minimum TFTS per PDP address violated
274	TFT precedence index not unique
275	invalid parameter combination
"CME ERROR" codes of FTP	
201	Unknown error for FTP
202	FTP task is busy
203	Failed to resolve server address
204	FTP timeout
205	Failed to read file

206	Failed to write file
207	It's not allowed in current state
208	Failed to login
209	Failed to logout
210	Failed to transfer data
211	FTP command rejected by server
212	Memory error
213	Invalid parameter
214	Network error

Example

```
AT+CPIN="1234","1234"
+CME ERROR: incorrect password
```

3.4 Summary of CMS ERROR codes

Final result code +CMS ERROR: <err> indicates an error related to mobile equipment or network. The operation is similar to ERROR result code. None of the following commands in the same command line is executed. Neither ERROR nor OK result code shall be returned. ERROR is returned normally when error is related to syntax or invalid parameters. The format of <err> can be either numeric or verbose. This is set with command AT+CMEE.

<err> of numeric format	<err> of verbose format
300	ME failure
301	SMS service of ME reserved
302	Operation not allowed
303	Operation not supported
304	Invalid PDU mode parameter
305	Invalid text mode parameter
310	SIM not inserted
311	SIM PIN required
312	PH-SIM PIN required
313	SIM failure
314	SIM busy
315	SIM wrong

316	SIM PUK required
317	SIM PIN2 required
318	SIM PUK2 required
320	Memory failure
321	Invalid memory index
322	Memory full
330	SMSC address unknown
331	No network service
332	Network timeout
340	NO +CNMA ACK EXPECTED
341	Buffer overflow
342	SMS size more than expected
500	Unknown error

Example

```
AT+CMGS=02112345678
+CMS ERROR: 304
```

4 AT Commands for Network

4.1 Overview of AT Commands for Network

Command	Description
AT+CREG	Network Registration
AT+COPS	Operator selection
AT+CLCK	Facility lock
AT+CPWD	Change password
AT+CCUG	Closed User Group
AT+CUSD	Unstructured supplementary service data
AT+CAOC	Advice of Charge
AT+CSSN	Supplementary service notifications
AT+CPOL	Preferred operator list
AT+COPN	Read operator names
AT+CNMP	Preferred mode selection
AT+CPSI	Inquiring UE system information
AT+CNSMOD	Show network system mode
AT+CEREG	EPS network registration status
AT+CTZU	Automatic time and time zone update
AT+CTZR	Time and time zone reporting
AT+CNWINFO	Inquiring extra network info
AT+C5GREG	NR5G network registration status
AT+CSYSSEL	Set system selection pref
AT+CCELLCFG	Set lte cell configuration
AT+C5GCELLCFG	Set NR5G cell configuration
AT+CRSSI	Getting RSSI for each RX

4.2 Detailed Description of AT Commands for Network

4.2.1 AT+CREG Network registration

This command is used to control the presentation of an unsolicited result code +CREG: <stat> when <n>=1 and there is a change in the ME network registration status, or code **+CREG: <stat>[,<lac>,<ci>]** when **<n>=2** and there is a change of the network cell.
Read command returns the status of result code presentation and an integer **<stat>** which shows whether the network has currently indicated the registration of the ME. Location information elements **<lac>** and **<ci>** are returned only when **<n>=2** and ME is registered in the network.

AT+CREG Network registration	
Test Command **AT+CREG=?**	Response **+CREG:** (list of supported <**n**>s) **OK**
Read Command **AT+CREG?**	Response **+CREG: <n>,<stat>[,<lac>,<ci>]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CREG=<n>**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+CREG**	Response (Set default value "<n>=0"): **OK**

Defined Values

<n>	0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +CREG: <stat> 2 enable network registration and location information unsolicited result code +CREG: <stat>[,<lac>,<ci>]
<stat>	0 not registered, ME is not currently searching a new operator to register to 1 registered, home network 2 not registered, but ME is currently searching a new operator to register to 3 registration denied

	4 unknown 5 registered, roaming
<lac>	Two byte location area code in hexadecimal format(e.g."00C3" equals 193 in decimal). NOTE: The <lac> not supported in CDMA/HDR mode
<ci>	Cell Identify in hexadecimal format. GSM : Maximum is two byte WCDMA : Maximum is four byte TDS-CDMA : Maximum is four byte NOTE: The <ci> not supported in CDMA/HDR mode

Example

```
AT+CREG?
+CREG: 0,1

OK
```

NOTE

Location information elements <lac> and <ci> are returned only when <n>=2 and ME is registered in the network

4.2.2 AT+COPS Operator selection

Write command forces an attempt to select and register the GSM/UMTS network operator. <mode> is used to select whether the selection is done automatically by the ME or is forced by this command to operator **<oper>** (it shall be given in format **<format>**). If the selected operator is not available, no other operator shall be selected (except **<mode>=4**). The selected operator name format shall apply to further read commands (**AT+COPS?**) also. **<mode>=2** forces an attempt to deregister from the network. The selected mode affects to all further network registration (e.g. after **<mode>=2**, ME shall be unregistered until **<mode>=0**or**1** is selected).
Read command returns the current mode and the currently selected operator. If no operator is selected, **<format>** and **<oper>** are omitted.
Test command returns a list of quadruplets, each representing an operator present in the network. Quadruplet consists of an integer indicating the availability of the operator **<stat>**, long and short alphanumeric format of the name of the operator, and numeric format representation of the operator. Any of the formats may be unavailable and should then be an empty field. The list of operators shall be in order: home network, networks referenced in SIM, and other networks.

It is recommended (although optional) that after the operator list TA returns lists of supported **<mode>**s and **<format>**s. These lists shall be delimited from the operator list by two commas.

AT+COPS Operator selection	
Test Command **AT+COPS=?**	Response **[+COPS: [**list of supported **(<stat>,long alphanumeric <oper> ,short alphanumeric <oper>,numeric <oper>[,<AcT>])**s**]** **[,,(**list of supported **<mode>**s**),(**list of supported **<format>**s**)]]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Read Command **AT+COPS?**	Response **+COPS: <mode>[,<format>,<oper>[,<AcT>]]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+COPS=<mode>[,<format>[,<oper>[,<AcT>]]]**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+COPS**	Response **OK**

Defined Values

<mode>	<u>0</u> automatic 1 manual 2 force deregister 3 set only <format> 4 manual/automatic NOTE: if <mode> is set to 1, 4 in write command, the <oper> is needed.
<format>	0 long format alphanumeric <oper> 1 short format alphanumeric <oper> 2 numeric <oper>

<oper>	string type, <format> indicates if the format is alphanumeric or numeric.
<stat>	0 unknown 1 available 2 current 3 forbidden
<AcT>	Access technology selected 0 GSM 1 GSM Compact 2 UTRAN 6 UTRAN_HSDPA_HSUPA 7 EUTRAN 8 EC_GSM_IOT 9 EUTRAN_NB_S1 11 NR_5GCN (NR connected to 5G core Network) 12 NGRAN (NG-RAN access technology) 13 EUTRA_NR (Dual connectivity of LTE with NR) NOTE: the value 8 do not follow the 3gpp spec, we add this value to distinguish cdma/hdr.

Example

```
AT+COPS?
+COPS: 0,0,"CHINA MOBILE",11

OK

AT+COPS=?
+COPS:(1,"CHN-CT","CT","46011",12),(1,"CHN-CT","CT","46011",7),(3,"CHN-UNICOM"
,"UNICOM","46001",12),(3,"CHN-UNICOM","UNICOM","46001",7),(3,"CHINA MOBILE",
"CMCC","46000",12),(3,"CHINA MOBILE","CMCC","46000",7),(1,"CHINA BROADNET",
"CBN","46015",7),(1,"CHINA BROADNET","CBN","46015",12),,(0,1,2,3,4),(0,1,2)

OK
```

4.2.3 AT+CLCK Facility lock

This command is used to lock, unlock or interrogate a ME or a network facility **<fac>**. Password is normally needed to do such actions. When querying the status of a network service (**<mode>=2**) the response line for 'not active' case (**<status>=0**) should be returned only if service is not active for any **<class>**.

AT+CLCK Facility lock	
Test Command **AT+CLCK=?**	Response **+CLCK: (**list of supported **<fac>**s) **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CLCK=<fac>,<mode>[,<passwd>[,<class>]]**	Response (When <mode>=2 and command successful:) **[+CLCK: <status>[,<class1>[<CR><LF>** **+CLCK: <status>,<class2>** **[...]]]]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<fac>	"PF" lock Phone to the First inserted SIM card or USIM card "SC" lock SIM card or USIM card "AO" Barr All Outgoing Calls "OI" Barr Outgoing International Calls "OX" Barr Outgoing International Calls except to Home Country "AI" Barr All Incoming Calls "IR" Barr Incoming Calls when roaming outside the home country "AB" All Barring services (only for <mode>=0) "AG" All outGoing barring services (only for <mode>=0) "AC" All inComing barring services (only for <mode>=0) "FD" SIM fixed dialing memory feature "PN" Network Personalization "PU" network subset Personalization "PP" service Provider Personalization "PC" Corporate Personalization
<mode>	0 unlock 1 lock 2 query status
<status>	0 not active 1 active
<passwd>	Password.

	string type; shall be the same as password specified for the facility from the ME user interface or with command Change Password +CPWD
<classX>	It is a sum of integers each representing a class of information: 1 voice (telephony) 2 data (refers to all bearer services) 4 fax (facsimile services) 8 short message service 16 data circuit sync 32 data circuit async 64 dedicated packet access 128 dedicated PAD access 255 The value 255 covers all classes

Example

```
AT+CLCK="SC",2
+CLCK: 0

OK
```

NOTE

When querying the status of a network service (<mode>=2) the response line for 'not active' case (<status>=0) should be returned only if service is not active for any <class>.

4.2.4 AT+CPWD Change password

Write command sets a new password for the facility lock function defined by command Facility Lock **AT+CLCK**.
Test command returns a list of pairs which present the available facilities and the maximum length of their password.

AT+CPWD Change password	
Test Command **AT+CPWD=?**	Response **+CPWD:** (list of supported **(<fac>,<pwdlength>**)s) **OK** or **ERROR** If error is related to ME functionality:

	+CME ERROR: <err>
Write Command **AT+CPWD=<fac>,<oldpwd>,<newpwd>**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<fac>	Refer Facility Lock +CLCK for other values: "SC" SIM or USIM PIN1 "P2" SIM or USIM PIN2 "AB" All Barring services "AC" All inComing barring services (only for <mode>=0) "AG" All outGoing barring services (only for <mode>=0) "AI" Barr All Incoming Calls "AO" Barr All Outgoing Calls "IR" Barr Incoming Calls when roaming outside the home country "OI" Barr Outgoing International Calls "OX" Barr Outgoing International Calls except to Home Country
<oldpwd>	String type, it shall be the same as password specified for the facility from the ME user interface or with command Change Password AT+CPWD.
<newpwd>	String type, it is the new password; maximum length of password can be determined with <pwdlength>.
<pwdlength>	Integer type, max length of password.

Example

AT+CPWD=?
+CPWD: ("AB",4),("AC",4),("AG",4),("AI",4),("AO",4),("IR",4),("OI",4),("OX",4),("SC",8),("P2",8)

OK

4.2.5 AT+CCUG Closed user group

This command allows control of the Closed User Group supplementary service. Set command enables the served subscriber to select a CUG index, to suppress the Outgoing Access (OA), and to suppress the preferential CUG.

AT+CCUG Closed user group

Test Command **AT+CCUG=?**	Response **OK** or **ERROR**
Read Command **AT+CCUG?**	Response **+CCUG: <n>,<index>,<info>** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CCUG=<n>[,<index>[,<info>]]**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+CCUG**	Response (Set default value): **OK**

Defined Values

<n>	0 disable CUG temporary mode 1 enable CUG temporary mode
<index>	0...9 CUG index 10 no index (preferred CUG taken from subscriber data)
<info>	0 no information 1 suppress OA 2 suppress preferential CUG 3 suppress OA and preferential CUG

Example

```
AT+CCUG?
+CCUG: 0,0,0

OK
```

NOTE

This command not supported in CDMA/HDR mode

4.2.6 AT+CUSD Unstructured supplementary service data

This command allows control of the Unstructured Supplementary Service Data (USSD). Both network and mobile initiated operations are supported. Parameter **<n>** is used to disable/enable the presentation of an unsolicited result code (USSD response from the network, or network initiated operation) **+CUSD: <m>[,<str>,<dcs>]** to the TE. In addition, value **<n>=2** is used to cancel an ongoing USSD session.

AT+CUSD Unstructured supplementary service data	
Test Command **AT+CUSD=?**	Response **+CUSD:** (list of supported **<n>**s) **OK**
Read Command **AT+CUSD?**	Response **+CUSD: <n>** **OK**
Write Command **AT+CUSD=<n>[,<str>[,<dcs>]]**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+CUSD**	Response (Set default value): **OK**

Defined Values

<n>	0 disable the result code presentation in the TA 1 enable the result code presentation in the TA 2 cancel session (not applicable to read command response)
<str>	String type USSD string.
<dcs>	Cell Broadcast Data Coding Scheme in integer format (default 0).
<m>	0 no further user action required (network initiated USSD Notify, or no further information needed after mobile initiated operation) 1 further user action required (network initiated USSD Request, or further information needed after mobile initiated operation) 2 USSD terminated by network 4 operation not supported 5 network time out

Example

```
AT+CUSD?
+CUSD: 1

OK

AT+CUSD=0
OK
```

NOTE

This command not supported in CDMA/HDR mode

4.2.7 AT+CAOC Advice of Charge

This command refers to Advice of Charge supplementary service that enables subscriber to get information about the cost of calls. With **<mode>=0**, the execute command returns the current call meter value from the ME.
This command also includes the possibility to enable an unsolicited event reporting of the CCM information. The unsolicited result code **+CCCM: <ccm>** is sent when the CCM value changes, but not more that every 10 seconds. Deactivation of the unsolicited event reporting is made with the same command.

AT+CAOC Advice of Charge	
Test Command **AT+CAOC=?**	Response **+CAOC:** (list of supported **<mode>**s) **OK**
Read Command **AT+CAOC?**	Response **+CAOC: <mode>** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CAOC=<mode>**	Response **+CAOC: <ccm>** **OK**

	or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+CAOC**	Response (Set default value): **OK** or **ERROR**

Defined Values

<mode>	0 query CCM value 1 deactivate the unsolicited reporting of CCM value 2 activate the unsolicited reporting of CCM value
<ccm>	String type, three bytes of the current call meter value in hexadecimal format (e.g. "00001E" indicates decimal value 30), value is in home units and bytes are similarly coded as ACMmax value in the SIM.

Example

```
AT+CAOC=0
+CAOC: "000000"

OK
```

NOTE

This command not supported in CDMA/HDR mode

4.2.8 AT+CSSN Supplementary service notifications

This command refers to supplementary service related network initiated notifications. The set command enables/disables the presentation of notification result codes from TA to TE.

When **<n>=1** and a supplementary service notification is received after a mobile originated call setup, intermediate result code **+CSSI: <code1>[,<index>]** is sent to TE before any other MO call setup result codes presented in the present document. When several different **<code1>**s are received from the network, each of them shall have its own **+CSSI** result code.

When**<m>=1** and a supplementary service notification is received during a mobile terminated call setup or during a call, or when a forward check supplementary service notification is received, unsolicited result code **+CSSU: <code2>[,<index>[,<number>,<type>[,<subaddr>,<satype>]]]** is sent to TE. In case of MT call setup, result code is sent after every **+CLIP** result code (refer command "Calling line identification presentation +CLIP") and when several different **<code2>**s are received from the network, each of them shall have its own **+CSSU** result code.

AT+CSSN Supplementary service notifications	
Test Command **AT+CSSN=?**	Response **+CSSN:** (list of supported **<n>**s),(list of supported **<m>**s) **OK**
Read Command **AT+CSSN?**	Response **+CSSN: <n>,<m>** **OK** or **ERROR**
Write Command **AT+CSSN=<n>[,<m>]**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<n>	Parameter sets/shows the +CSSI result code presentation status in the TA: 0 disable 1 enable
<m>	Parameter sets/shows the +CSSU result code presentation status in the TA: 0 disable 1 enable
<code1>	0 unconditional call forwarding is active 1 some of the conditional call forwarding are active 2 call has been forwarded 3 call is waiting 5 outgoing calls are barred
<index>	Refer "Closed user group +CCUG".
<code2>	0 this is a forwarded call (MT call setup) 2 call has been put on hold (during a voice call) 3 call has been retrieved (during a voice call) 5 call on hold has been released (this is not a SS notification) (during a voice call)
<number>	String type phone number of format specified by <type>.

<type>	Type of address octet in integer format; default 145 when dialing string includes international access code character "+", otherwise 129.
<subaddr>	String type sub address of format specified by <satype>.
<satype>	Type of sub address octet in integer format, default 128.

Example

```
AT+CSSN=1
OK

AT+CSSN?
+CSSN: 1,1

OK
```

NOTE

This command not supported in CDMA/HDR mode

4.2.9 AT+CPOL Preferred operator list

This command is used to edit the SIM preferred list of networks.

AT+CPOL Preferred operator list	
Test Command **AT+CPOL=?**	Response **+CPOL: (**list of supported **<index>**s**),(**list of supported **<format>**s**)** **OK**
Read Command **AT+CPOL?**	Response **[+CPOL: <index1>,<format>,<oper1>[,<GSM_AcT1>,<GSM_Compact_AcT1>,<UTRAN_AcT1>,<LTE_AcT1>][<CR><LF>** **+CPOL: <index2>,<format>,<oper2>[,<GSM_AcT1>,<GSM_Compact_AcT1>,<UTRAN_AcT1>,<LTE_AcT1>]** **[...]]]** **OK** or **ERROR**
Write Command	Response

AT+CPOL=<index>[,<format>[,<oper>][,<GSM_AcT1>,<GSM_Compact_AcT1>,<UTRAN_AcT1>,<LTE_AcT1>]] NOTE: If using USIM card, the last four parameters must set.	**OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<index>	Integer type, the order number of operator in the SIM preferred operator list. If only input <index>, command will delete the value indicate by <index>.
<format>	0 long format alphanumeric <oper> 1 short format alphanumeric <oper> 2 numeric <oper>
<operX>	String type.
<GSM_AcTn>	GSM access technology: 0 access technology not selected 1 access technology selected
<GSM_Compact_AcTn>	GSM access technology: 0 access technology not selected 1 access technology selected
<UTRA_AcTn>	UTRA access technology: 0 access technology not selected 1 access technology selected
<LTE_AcTn>	LTE access technology: 0 access technology not selected 1 access technology selected

Example

```
AT+CPOL?
+CPOL: 1,2,"46001",0,0,1,0

OK

AT+CPOL=?
+CPOL: (1-8),(0-2)

OK
```

4.2.10 AT+COPN Read operator names

This command is used to return the list of operator names from the ME. Each operator code **<numericX>** that has an alphanumeric equivalent **<alphaX>** in the ME memory shall be returned.

AT+COPN Read operator names	
Test Command AT+COPN=?	Response **OK** or **ERROR**
Execution Command AT+COPN	Response **[+COPN: <numeric1>,<alpha1>[<CR><LF>** **+COPN: <numeric2>,<alpha2>** **[...]]]** **OK** or If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<numericX>	String type, operator in numeric format (see AT+COPS).
<alphaX>	String type, operator in long alphanumeric format (see AT+COPS).

Example

```
AT+COPN
+COPN: "46000","China Mobile Com"
+COPN: "46001","China Unicom"
......

OK
```

4.2.11 AT+CNMP Preferred mode selection

This command is used to select or set the state of the mode preference.

AT+CNMP Preferred mode selection	
Test Command	Response

AT+CNMP=?	**+CNMP:** (list of supported **<mode>**s) **OK**
Read Command **AT+CNMP?**	Response **+CNMP: <mode>** **OK**
Write Command **AT+CNMP=<mode>**	Response **OK** or (If <mode> not supported by module, this command will return ERROR.) **ERROR**

Defined Values

<mode>	2 Automatic 14 WCDMA Only 38 LTE Only 71 NR5G 54 WCDMA+LTE Only 55 WCDMA+LTE+NR5G 109 LTE+NR5G

Example

```
AT+CNMP=109
OK

AT+CNMP?
+CNMP: 109

OK
```

NOTE

1. The set value in Write Command will take efficient immediately; The set value will retain after module reset
2. The response will be returned immediately for Test Command and Read Command; The Max Response Time for Write Command is 10 seconds
3. SIM8210 Series support **<mode>**:
   2 Automatic
   38 LTE
   71 NR5G

4.2.12 AT+CPSI Inquiring UE system information

AT+CPSI Inquiring UE system information	
Test Command **AT+CPSI=?**	Response **+CPSI:** (scope of**<time>)** **OK**
Read Command **AT+CPSI?**	Response If camping on a wcdma cell: **+CPSI: <System Mode>,<Operation Mode>,<MCC>-<MNC>,<LAC>,<Cell ID>,<Frequency Band>,<PSC>,<Freq>,<SSC>,<EC/IO>,<RSCP>,<Qual>,<RxLev>,<TXPWR>** **OK** If camping on a lte cell: **+CPSI: <System Mode>,<Operation Mode>[,<MCC>-<MNC>,<TAC>,<SCellID>,<PCellID>,<Frequency Band>,<earfcn>,<dlbw>,<ulbw>,<RSRQ>,<RSRP>,<RSSI>,<RSSNR>]** **OK** If no service: **+CPSI: NO SERVICE,<Operation mode>** **OK** If camping on EN-DC connected mode: **+CPSI: LTE,<Operation Mode>[,<MCC>-<MNC>,<TAC>,<SCellID>,<PCellID>,<Frequency Band>,<earfcn>,<dlbw>,<ulbw>,<RSRQ>,<RSRP>,<RSSI>,<RSSNR>]** **+CPSI: NR5G_NSA, [<PCellID>,<Frequency Band>,<earfcn/ssb>,<RSRP>,<RSRQ>,<SNR>,<scs>,<NR_dl_bw>]** **OK** If camping on NR5G only mode: **+CPSI: NR5G_SA,<Operation Mode>[,<MCC>-<MNC>,<TAC>,<SCellID>,<PCellID>,<Frequency Band>,<earfcn>,<RSRP>,<RSRQ>,<SNR>]**

	OK **ERROR**
Write Command **AT+CPSI=<time>**	Response **OK** or **ERROR**

Defined Values

<time>	The range is 0-255, unit is second, after set <time> will report the system information every the seconds.
<System mode>	System mode, values: "NO SERVICE", "WCDMA", "LTE "… If module in LIMITED SERVICE state and +CNLSA command is set to 1, the system mode will display as "WCDMA-LIMITED"…
<Operation mode>	UE operation mode, values: "Unknown", "Online", "Offline", "Factory Test Mode", "Reset", "Low Power Mode".
<MCC>	Mobile Country Code (first part of the PLMN code)
<MNC>	Mobile Network Code (second part of the PLMN code)
<LAC>	Location Area Code (hexadecimal digits with leading character)
<Cell ID>	Service-cell Identify.
<Frequency Band>	Frequency Band of active set
<PSC>	Primary synchronization code of active set.
<Freq>	Downlink frequency of active set.
<SSC>	Secondary synchronization code of active set
<EC/IO>	Ec/Io valueReceived Signal Code Power
<RSCP>	Received Signal Code Power
<Qual>	Quality value for base station selection
<RxLev>	RX level value for base station selection
<TXPWR>	UE TX power in dBm. If no TX, the value is 500.
<TAC>	Tracing Area Code
<PCellID>	Physical Cell ID
<SCellID>	Serving Cell ID
<earfcn>	E-UTRA absolute radio frequency channel number for searching LTE cells
<ssb>	The SSB combines Synchronization Signal and PBCH block. It consists of Primary Synchronization Signals (PSS), Secondary Synchronization Signals (SSS) and PBCH.
<dlbw>	Transmission bandwidth configuration of the serving cell on the downlink.Range: 0 to 5. Values:

	0 – BW_1.4MHz 1 – BW_3MHz 2 – BW_5MHz 3 – BW_10MHz 4 – BW_15MHz 5 – BW_20MHz
<ulbw>	Transmission bandwidth configuration of the serving cell on the uplink.Range: 0 to 5. Values: 0 – BW_1.4MHz 1 – BW_3MHz 2 – BW_5MHz 3 – BW_10MHz 4 – BW_15MHz 5 – BW_20MHz
<RSRP>	Current reference signal receive power in dBm x10 as measured by L1.Range: -44 to -140.
<RSRQ>	Current reference signal receive quality as measured by L1.The quantities are in dB x10. Range: -20.0 to -3.0 dB.
<RSSI>	Current received signal strength indicator as measured by L1.Values are in dBm x10. Range: -120.0 to 0.
<RSSNR>	SINR(Signal to Interference Noise Ratio). Range of values: (0-30). The higher the value, the better.
<SNR>	SIGNAL-NOISE RATIO. Range of values: (-23-40), which represents -23.0 dB to 40.0 dB. The higher the value, the better.
<BID>	Base ID
<scs>	Values: 0 - SCSC15 1 - SCSC30 2 - SCSC60 3 - SCSC120
<NR_dl_bw>	Carrier bandwidth Values: 0 – BW_5MHz 1 – BW_10MHz 2 – BW_15MHz 3 – BW_20MHz 4 – BW_25MHz 5 – BW_30MHz 6 – BW_35MHz 7 – BW_40MHz 8 – BW_45MHz 9 – BW_50MHz 10 – BW_60MHz

	11 – BW_70MHz 12 – BW_80MHz 13 – BW_90MHz 14 – BW_100MHz 15 – BW_200MHz 16 – BW_400MHz

Example

```
AT+CPSI?
+CPSI: WCDMA,Online,460-01,0xA809,11122855,WCDMA IMT 2000,279,10663,0,1.5,62,33,
52,500

OK

AT+CPSI=?
+CPSI: (0-255)

OK
AT+CPSI?
+CPSI: LTE,Online,460-11,0x5A1E,187214780,257,EUTRAN-BAND3,1850,5,5,-94,-850,-545,15
+CPSI: NR5G_NSA,644,NR5G_BAND78,627264,-960,-120,95,1,12

OK
```

4.2.13 AT+CNSMOD Show network system mode

This command is used to return the current network system mode.

AT+CNSMOD Show network system mode	
Test Command **AT+CNSMOD=?**	Response **+CNSMOD:** (list of supported **<n>s)** **OK**
Read Command **AT+CNSMOD?**	Response **+CNSMOD: <n>,<stat>** **OK** or **ERROR** If error is related to ME functionality:

	+CME ERROR: <err>
Write Command **AT+CNSMOD=<n>**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<n>	0 disable auto report the network system mode information 1 auto report the network system mode information, command: +CNSMOD: <stat>
<stat>	0 no service 1 GSM 2 GPRS 3 EGPRS (EDGE) 4 WCDMA 5 HSDPA only(WCDMA) 6 HSUPA only(WCDMA) 7 HSPA (HSDPA and HSUPA, WCDMA) 8 LTE 9 TDS-CDMA 10 TDS-HSDPA only 11 TDS- HSUPA only 12 TDS- HSPA (HSDPA and HSUPA) 13 CDMA 14 EVDO 15 HYBRID (CDMA and EVDO) 16 1XLTE(CDMA and LTE) 23 eHRPD 24 HYBRID(CDMA and eHRPD) 36 NR5G

Example

AT+CNSMOD?
+CNSMOD: 0,2

OK

4.2.14 AT+CEREG EPS network registration status

The set command controls the presentation of an unsolicited result code **+CEREG: <stat>** when **<n>=1** and there is a change in the MT's EPS network registration status in E-UTRAN, or unsolicited result code **+CEREG: <stat>[,<tac>,<ci>[,<AcT>]]** when **<n>=2** and there is a change of the network cell in E-UTRAN; in this latest case **<AcT>,<tac> and <ci>** are sent only if available.
NOTE 1: If the EPS MT in GERAN/UTRAN/E-UTRAN also supports circuit mode services and/or GPRS services, the **+CREG** command and **+CREG:** result codes and/or the **+CGREG** command and **+CGREG:** result codes apply to the registration status and location information for those services.
The read command returns the status of result code presentation and an integer <stat> which shows whether the network has currently indicated the registration of the MT. Location information elements **<tac>,<ci>** and **<AcT>**, if available, are returned only when **<n>=2** and MT is registered in the network.

AT+CEREG EPS network registration status	
Test Command **AT+CEREG=?**	Response **+CEREG: (**list of supported **<n>**s) **OK** or **ERROR**
Read Command **AT+CEREG?**	Response **+CEREG: <n>,<stat>[,<tac>,<ci>[,<AcT>]]** **OK** or **ERROR**
Write Command **AT+CEREG[=<n>]**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+CEREG**	Response (Set default value"<n>=0"): **OK** or **ERROR**

Defined Values

<n>	0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +CEREG: <stat> 2 enable network registration and location information unsolicited

	result code +CEREG: <stat>[,<tac>,<ci>[,<AcT>]]
<stat>	0 not registered, MT is not currently searching an operator to register to 1 registered, home network 2 not registered, but MT is currently trying to attach or searching an operator to register to 3 registration denied 4 unknown (e.g. out of E-UTRAN coverage) 5 registered, roaming 6 registered for "SMS only", home network (not applicable) 7 registered for "SMS only", roaming (not applicable) 8 attached for emergency bearer services only (See NOTE 2)
<tac>	string type; two byte tracking area code in hexadecimal format without leading character (e.g. "9191" equals 37265 in decimal)
<ci>	string type; four byte E-UTRAN cell identify in hexadecimal format without leading character
<AcT>	A numberic parameter that indicates the access technology of serving cell 0 GSM (not applicable) 1 GSM Compact (not applicable) 2 UTRAN (not applicable) 3 GSM w/EGPRS (see NOTE 3) (not applicable) 4 UTRAN w/HSDPA (see NOTE 4) (not applicable) 5 UTRAN w/HSUPA (see NOTE 4) (not applicable) 6 UTRAN w/HSDPA and HSUPA (see NOTE 4) (not applicable) 7 E-UTRAN 10 EUTRA_5GCN 11 NR_5GCN 12 NG-RAN 13 EUTRA_NR

Example

```
AT+CEREG?
+CEREG: 0,1

OK
```

NOTE

If the EPS MT in GERAN/UTRAN/E-UTRAN/NR also supports circuit mode services and/or GPRS services, the +CREG command and +CREG: result codes and/or the +CGREG command and +CGREG: result codes apply to the registration status and location information for those services.

4.2.15 AT+CTZU Automatic time and time zone update

This command is used to enable and disable automatic time and time zone update via NITZ.

AT+CTZU Automatic time and time zone update	
Test Command **AT+CTZU=?**	Response **+CTZU: (**list of supported **<on/off>s)** **OK**
Read Command **AT+CTZU?**	Response **+CTZU: <on/off>** **OK** or If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CTZU=<on/off>**	Response **OK** or **ERROR**

Defined Values

<on/off>	Integer type value indicating: 0 Disable automatic time zone update via NITZ. 1 Enable automatic time zone update via NITZ. NOTE: 1. The value of <on/off> is nonvolatile, and factory value is 1. 2. For automatic time and time zone update is enabled (+CTZU=1): If time zone is only received from network and it isn't equal to local time zone (AT+CCLK), time zone is updated automatically, and real time clock is updated based on local time and the difference between time zone from network and local time zone (Local time zone must be valid). If Universal Time and time zone are received from network, both time zone and real time clock is updated automatically, and real time clock is based on Universal Time and time zone from network.

Example

```
AT+CTZU?
+CTZU: 1
```

OK

AT+CTZU=0
OK

4.2.16 AT+CTZR Time and time zone reporting

This command is used to enable and disable the time zone change event reporting. If the **AT+CTZR=1**, the MT returns the unsolicited result code**+CTZV: <tz>** whenever time zone received from network isn't equal to local time zone; If **AT+CTZR=2**, report **+CTZE: <tz>,<dst>,<time>** whenever the time zone and time is changed.

AT+CTZR Time and time zone reporting	
Test Command **AT+CTZR=?**	Response **+CTZR: (**list of supported **<on/off>s)** **OK**
Read Command **AT+CTZR?**	Response **+CTZR: <on/off>** **OK**
Write Command **AT+CTZR=<on/off>**	Response **OK** or **ERROR**
Execution Command **AT+CTZR**	Response (Set default value) **OK**

Defined Values

<on/off>	Integer type value indicating: 0 Disable time zone change event reporting (default). 1 Enable time zone change event reporting. 2 Display <tz>,<dst>,<time>
<tz>	Local time zone received from network, it's an integer, and the format is "±tz".
<dst>	Network daylight saving time, and if it is received from network, it indicates the value that has been used to adjust the local time zone. The values as following: 0 No adjustment for Daylight Saving Time.

	1 +1 hour adjustment for Daylight Saving Time. 2 +2 hours adjustment for Daylight Saving Time.
<time>	Universal time received from network, and the format is "yyyy/MM/dd,hh:mm:ss", where characters indicate year (two last digits), month, day, hour, minutes and seconds. **NOTE:**Here in,**<time>** is Universal Time or NITZ time.

Example

```
AT+CTZR?
+CTZR: 0

OK

AT+CTZR=1
OK

+CTZV: "+32"
AT+CTZR=2
OK

+CTZE: "+32",0,"2020/03/19,15:19:46"
```

NOTE

The time zone reporting is not affected by the Automatic Time and Time Zone command AT+CTZU.

4.2.17 AT+CNWINFO Inquiring extra network info

AT+CNWINFO Inquiring extra network info	
Read Command **AT+CNWINFO?**	Response If camping a LTE cell: **+CNWINFO:** **<SYS_MODE>,<ECGI>,<eNBID>,<MCS>,<DL_Layers>,<DL_MOD>,<UL_MOD>,<CQI>,<TRANS_MODE>,<TX_POWER>** **OK** if EN-DC connnect: **+CNWINFO:** **<SYS_MODE>,<ECGI>,<eNBID>,<MCS>,<DL_Layers>,<DL_MOD**

	>,<UL_MOD>,<CQI>,<TRANS_MODE>,<TX_POWER> **+CNWINFO: <SYS_MODE>,<Serving_SSB_ID>,<DL_MCS>,<UL_MCS>,<DL_Layers>,<DL_MOD>,<UL_MOD>,<CQI>** If camping a NR5G cell: **+CNWINFO: <SYS_MODE>,<NCGI>,<gNBID>,<Serving_SSB_ID>,<DL_MCS>,<UL_MCS>,<DL_Layers>,<DL_MOD>,<UL_MOD>,<CQI>,<BANDWIDTH>,<TX_POWER>,<RSSI>,<NR_QCI>** **OK** or **ERROR**
Write Command **AT+CNWINFO=<on>**	Response **OK** or **ERROR**

Defined Values

<SYS_MODE>	System mode: LTE NR5G
<ECGI>	A decimal value, E-UTRAN Cell Global Identifier.
<eNBID>	eNodeB ID, it's a hex value.
<NGCI>	A decimal value, NR Cell Global Identifier.
<gNBID>	gNodeB ID, it's a hex value.
<MCS>/<DL_MCS>/<UL_MCS>	Modulation and Coding Scheme, a decimal value(0-31).
<DL_Layers>	The num of DL Layers
<DL_MOD>/<UL_MOD>	Modulation type: "BPSK" "QPSK" "16QAM" "64QAM" "256QAM" "1024QAM" "UNKOWN"
<CQI>	Channel quality indication, a decimal value(0-15).
<TRANS_MODE>	Transmission Scheme.

<TX_POWER>	UL Tx power, a decimal value(-99-30)
<BANDWIDTH>	The bandwidth of attach cell: 5M 10M 15M 20M 25M 30M 40M 50M 60M 70M 80M 90M 100M 200M 400M
<RSSI>	0 -113 dBm or less 1 -111 dBm 2...30 -109... -53 dBm 31 -51 dBm or greater 99 not known or not detectable 100 -116 dBm or less 101 -115 dBm 102…191 -114... -26dBm 191 -25 dBm or greater 199 not known or not detectable 100…199 expand to TDSCDMA, indicate RSCP received
<on>	On or Off **<DL_MOD>/<UL_MOD>** display: 0 Close 1 Open
<Serving_SSB_ID>	Serving Cell SSB ID, Integer Type.
<NR_QCI>	QoS Class Identifier.

Example

```
AT+CNWINFO?
+CNWINFO: NR5G,4601141341260034,0x9A02191,0,0,0,0,UNKOWN,BPSK,0,100M,-99,-100,6

OK

AT+CNWINFO?
+CNWINFO: LTE,4600039589685,0x25C17,0,0,UNKOWN,16QAM,10,0,11
```

OK

4.2.18 AT+C5GREG NR5G network registration status

The set command controls the presentation of an unsolicited result code **+C5GREG: <stat>** when **<n>=1**, or unsolicited result code **+C5GREG: <stat>[,<tac>,<ci>,<AcT>,<octec_len>,<nssai>]** when **<n>=2**.

AT+C5GREG NR5G network registration status	
Test Command **AT+C5GREG=?**	Response **+C5GREG: (**list of supported **<n>s)** **OK** or **ERROR**
Read Command **AT+C5GREG?**	Response **+C5GREG: <n>,<stat>[,<tac>,<ci>,<AcT>,<octec_len>,<nssai>]** **OK** or **ERROR**
Write Command **AT+C5GREG=<n>**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command **AT+C5GREG**	Response (Set default value(<n>=0) **OK** or **ERROR**

Defined Values

<n>	0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +C5GREG: <stat> 2 enable network registration and location information unsolicited result code +C5GREG: <stat>[,<tac>,<ci>,<AcT>,<octec_len>,<nssai>]
<stat>	0 not registered, MT is not currently searching an operator to register to

	1 registered, home network 2 not registered, but MT is currently trying to attach or searching an operator to register to 3 registration denied 4 unknown (e.g. out of 5GC coverage) 5 registered, roaming 6 registered for "SMS only", home network (not applicable) 7 registered for "SMS only", roaming (not applicable) 8 attached for emergency bearer services only
<tac>	string type; three byte tracking area code in hexadecimal format without leading character (e.g. "720013" equals 7471123 in decimal)
<ci>	string type; five byte NR cell identify in hexadecimal format without leading character.
<AcT>	A numberic parameter that indicates the access technology of serving cell 0 GSM (not applicable) 1 GSM Compact (not applicable) 2 UTRAN (not applicable) 3 GSM w/EGPRS (not applicable) 4 UTRAN w/HSDPA (not applicable) 5 UTRAN w/HSUPA (not applicable) 6 UTRAN w/HSDPA and HSUPA (not applicable) 7 E-UTRAN 10 EUTRA_5GCN 11 NR_5GCN 12 NGRAN 13 EUTRA_NR
<octec_len>	The length of **<nssai>.**
<nssai>	Network slice selection assistance information, a string value.

Example

```
AT+C5GREG?
+C5GREG: 0,1

OK

AT+C5GREG=2
OK

AT+C5GREG?
+C5GREG: 2,1,"161816","3121F9002",11,1,"01"

OK
```

NOTE

The <nssai> return value varies depending on the SIM card configuration of the carrier.

4.2.19 AT+CSYSSEL Set system selection pref

AT+CSYSSEL Set system selection pref	
Test Command **AT+CSYSSEL=?**	Response **+CSYSSEL: "nr5g_disable",**(list of supported **<nr5g_disable_mode>)** **+CSYSSEL: "nr5g_band",**(list of supported **<sa_nr5gband>)** **+CSYSSEL: "nsa_nr5g_band",**(list of supported **<nsa_nr5gband>)** **[+CSYSSEL: "nrdc_nr5g_band",**(list of supported **<nrdc_nr5gband>)** **]** **+CSYSSEL: "lte_band",**(list of supported **<lte_band>)** **+CSYSSEL: "w_band",**(list of supported **<w_band>)** **+CSYSSEL: "rat_acq_order",**(list of supported **<rat_acq_order>)** **OK**
Write Command **AT+CSYSSEL="nr5g_disable"[,<nr5g_disable_mode>]**	Response If the parameter**<nr5g_disable_mode>** is omitted, return current configuration: **+CSYSSEL: "nr5g_disable",<nr5g_disable_mode>** **OK** or **ERROR** This command was applied to disable NSA or SA mode.
Write Command **AT+CSYSSEL="nr5g_band"[,<sa_nr5gband>]**	Response If the parameter **<sa_nr5gband>** is omitted, return current configuration: **+CSYSSEL: "nr5g_band",<sa_nr5gband>** **OK** If the parameter **<sa_nr5gband>** is specified, configure the preferred SA NR5G bands to be searched: **OK** or **ERROR**

Write Command **AT+CSYSSEL="nsa_nr5g_band"[,<nsa_nr5gband>]**	Response If the parameter **<nsa_nr5gband>** is omitted, return current configuration: **+CSYSSEL: "nsa_nr5g_band",<nsa_nr5gband>** **OK** If the parameter **<nsa_nr5gband>** is specified, configure the preferred NSA NR5G bands to be searched: **OK** or **ERROR**
Write Command **AT+CSYSSEL="nrdc_nr5g_band"[,<nrdc_nr5gband>]**	Response If the parameter **<nrdc_nr5gband>** is omitted, return current configuration: **+CSYSSEL: "nrdc_nr5g_band",<nrdc_nr5gband>** **OK** If the parameter **<nrdc_nr5gband>** is specified, configure the preferred NRDC NR5G bands to be searched: **OK** or **ERROR**
Write Command **AT+CSYSSEL="lte_band"[,<lte_band>]**	Response If the parameter **<lte_band>** is omitted, return current configuration: **+CSYSSEL: "lte_band",<lte_band>** **OK** If the parameter **<lte_band>** is specified, configure the preferred LTE bands to be searched: **OK** or **ERROR** This command was applied to configure LTE bands.
Write Command **AT+CSYSSEL="w_band"[,<w_band>]**	Response If the parameter **<w_band>** is omitted, return current configuration: **+CSYSSEL: "w_band",<w_band>** **OK** If the parameter **<w_band>** is specified, configure the preferred WCDMA bands to be searched: **OK** or **ERROR** This command was applied to configure WCDMA bands.

Write Command **AT+CSYSSEL="rat_acq_order"[,<rat_acq_order>]**	Response If the parameter **<rat_acq_order>** is omitted, return current configuration: **+CSYSSEL: "rat_acq_order",<rat_acq_order>** **OK** If the parameter **<rat_acq_order>** is specified, configure the RAT acquisition order. **OK** or **ERROR** This command will take effect after reboot.
Execution Command **AT+CSYSSEL**	Restore all band to default band capability. **OK**

Defined Values

<nr5g_disable_mode>	NR5G SA/NSA disable setting: 0 None is disabled 1 Disable SA 2 Disable NSA
<sa_nr5gband>	String type. Use the colon as a separator to list NR5G bands to be configured. The parameter format is B1:B2:B3:…:BN.
<nsa_nr5gband>	String type. Use the colon as a separator to list NSA NR5G bands to be configured. The parameter format is B1:B2:B3:…:BN.
<nrdc_nr5gband>	String type. Use the colon as a separator to list NRDC NR5G bands to be configured. The parameter format is B1:B2:B3:…:BN.
<lte_band>	String type. Use the colon as a separator to list LTE bands to be configured. The parameter format is B1:B2:B3:…:BN.
<w_band>	String type. Use the colon as a separator to list WCDMA bands to be configured. The parameter format is B1:B2:B3:…:BN. B1 WCDMA_I_IMT_2000 B2 WCDMA_II_PCS_1900 B3 WCDMA_III_1700 B4 WCDMA_IV_1700 B5 WCDMA_V_850 B6 WCDMA_VI_800 B7 WCDMA_VII_2600 B8 WCDMA_VIII_900 B9 WCDMA_IX_1700 B11 WCDMA_XI_1500 B19 WCDMA_XIX_850
<rat_acq_order>	String Type. Use the colon as a separator to specify RAT priority. The

	parameter format is: 5:9:12. 5 – WCDMA 9 – LTE 12 – NR5G

Example

```
AT+CSYSSEL="nr5g_disable"
+CSYSSEL: "nr5g_disable",0

OK
AT+CSYSSEL="nr5g_disable",2
OK
AT+CSYSSEL="nr5g_disable"
+CSYSSEL: "nr5g_disable",2

OK
AT+CSYSSEL="nr5g_band"
+CSYSSEL: "nr5g_band",1:28:41:78

OK
AT+CSYSSEL="nr5g_band",41:78
OK
AT+CSYSSEL="lte_band"
+CSYSSEL:
"lte_band",1:2:3:4:5:7:8:12:13:14:17:18:19:20:25:26:28:29:30:32:34:38:39:40:41:42:43:48:66:7
1

OK
AT+CSYSSEL="lte_band",1:2:3
OK
AT+CSYSSEL="w_band"
+CSYSSEL: "w_band",1:2:3:4:5:6

OK
AT+CSYSSEL="w_band",1:2:3:4:5:6:8
OK
AT+CSYSSEL="rat_acq_order"
+CSYSSEL: "rat_acq_order",12:9:5

OK
AT+CSYSSEL="rat_acq_order",9:12:5
OK
```

//If modem supported NRDC
AT+CSYSSEL=?
+CSYSSEL: "nr5g_disable",(0-2)

+CSYSSEL:
"nr5g_band",1:2:3:5:7:8:12:13:14:18:20:25:26:28:29:30:38:40:41:48:66:71:75:77:78:79

+CSYSSEL:
"nsa_nr5g_band",1:2:3:5:7:8:12:13:14:18:20:25:26:28:29:30:38:40:41:48:66:71:75:77:78:79:257:258:260:261

+CSYSSEL:
"nrdc_nr5g_band",1:2:3:5:7:8:12:13:14:18:20:25:26:28:29:30:38:40:41:48:66:71:75:77:78:79:257:258:260:261

+CSYSSEL:
"lte_band",1:2:3:4:5:7:8:12:13:14:17:18:19:20:25:26:28:29:30:32:34:38:39:40:41:42:43:46:48:66:71

+CSYSSEL: "w_band",1:2:4:5:6:8:19

+CSYSSEL: "rat_acq_order",5:9:12

OK
//If modem supported NRDC
AT+CSYSSEL="nrdc_nr5g_band",3:258

OK
//If modem supported NRDC
AT+CSYSSEL="nrdc_nr5g_band"
+CSYSSEL: "nrdc_nr5g_band",3:258

OK

4.2.20 AT+CCELLCFG Set lte cell configuration

AT+CCELLCFG Set lte cell configuration	
Read Command AT+CCELLCFG?	Response +CCELLCFG: <pci>,<freq> OK

	or **ERROR**
Write Command **AT+CCELLCFG=<enable>[,<pci>,<freq>]**	Response **OK** or **ERROR**

Defined Values

<enable>	0 – Clean cell lock 1 – Set cell lock with pci and freq
<pci>	Physical cell id, unsigned short
<freq>	Frequency, unsigned long int

Example

AT+CCELLCFG=1,255,1850
OK

AT+CCELLCFG=0 //Clean cell lock set
OK
AT+CCELLCFG?
+CCELLCFG: 255,1850

OK

NOTE

The lock command take effect immediately, the clean command take effect after reboot.

4.2.21 AT+C5GCELLCFG Set NR5G cell configuration

AT+C5GCELLCFG Set NR5G cell configuration	
Read Command **AT+C5GCELLCFG?**	Response If lock cell id was set: **+C5GCELLCFG: "pci",<pci>,<freq>,<scs>,<band>** **OK** If arfcn list was set: **+C5GCELLCFG:**

	"arfcn",<arfcn_list_len>,<scs:freq>[,<scs:freq>[…]] **OK** Baseline in SDX55_LE14: **+C5GCELLCFG: 0** **OK** or **OK** or **ERROR**
Write Command **AT+C5GCELLCFG="pci",<pci>,<freq>,<scs>,<band>** **AT+C5GCELLCFG="arfcn",<arfcn_list_len>,<scs:freq>[,<scs:freq>[…]]** **AT+C5GCELLCFG="unlock"**	Response **OK** or **ERROR**

Defined Values

<pci>	Physical cell id, unsigned short
<freq>	Frequency, unsigned long int
<scs>	0x00 –SUB_CARRIER_SPACING_SSB_15KHZ 0x01 –SUB_CARRIER_SPACING_SSB_30KHZ 0x02–SUB_CARRIER_SPACING_SSB_60KHZ 0x03–SUB_CARRIER_SPACING_SSB_120KHZ 0x04–SUB_CARRIER_SPACING_SSB_240KHZ 0x05–SUB_CARRIER_SPACING_SSB_SPARE3 0x06–SUB_CARRIER_SPACING_SSB_SPARE2 0x07–SUB_CARRIER_SPACING_SSB_SPARE1
<band>	NR band
<arfcn_list_len>	An integer value, 1-20.

Example

```
AT+C5GCELLCFG="pci",100,518670,1,41
OK
AT+C5GCELLCFG?
+C5GCELLCFG: "pci",100,518670,1,41

OK
```

```
AT+C5GCELLCFG="arfcn",1,1:518670
OK

AT+C5GCELLCFG?
+C5GCELLCFG: "arfcn",1,1:518670

OK

AT+C5GCELLCFG="unlock"
OK
```

NOTE

This command was not verified test, so suggest customer not use it now until SIMCom test it ok.

4.2.22 AT+CRSSI Getting RSSI for each RX

AT+CRSSI Getting RSSI for each RX	
Read Command AT+CRSSI?	Response **OK** or **ERROR** or **+CRSSI:** **<type>,<RX0_RSSI>,<RX1_RSSI>,<RX2_RSSI>,<RX3_RSSI>** **[<type>,<RX0_RSSI>,<RX1_RSSI>,<RX2_RSSI>,<RX3_RSSI>]** **OK**

Defined Values

<type>	RAT control type: “NR5G”, “LTE”
<RX0_RSSI>	Integer type, value range -40~ -140dbm, if not get value, set default to 0, For PRX.
<RX1_RSSI>	Integer type, value range -40~ -140dbm, if not get value, set default to 0, For DRX1.
<RX2_RSSI>	Integer type, value range -40~ -140dbm, if not get value, set default to 0. For DRX2.
<RX3_RSSI>	Integer type, value range -40~ -140dbm, if not get value, set default to 0. For DRX3.

Example

```
// for NR(SA)
AT+CRSSI?
+CRSSI: NR5G,-79,-75,0,0

OK
// for NR(NSA)
AT+CRSSI?
+CRSSI: LTE,-48,-53,-49,-53
+CRSSI: NR5G,-79,-75,-80,-81

OK
//for LTE
AT+CRSSI?
+CRSSI: LTE,-48,-53,-49,-53

OK
```

5AT Commands According to Call Control

5.1 Overview of AT Commands According to 3GPP Call Control

Command	Description
AT+CVHU	Voice hang up control
AT+CHUP	Hang up call
AT+CBST	Select bearer service type
AT+CRLP	Radio link protocol
AT+CR	Service reporting control
AT+CRC	Cellular result codes
AT+CLCC	List current calls
AT+CEER	Extended error report
AT+CCWA	Call waiting
AT+CHLD	Call related supplementary services
AT+CCFC	Call forwarding number and conditions
AT+CLIP	Calling line identification presentation
AT+CLIR	Calling line identification restriction
AT+COLP	Connected line identification presentation
AT+VTS	DTMF and tone generation
AT+VTD	Tone duration
AT+CSTA	Select type of address
AT+CMOD	Call mode
AT+VMUTE	Speaker mute control
AT+CMUT	Microphone mute control
AT+MORING	Enable or disable report MO ring URC
AT+CLVL	Loudspeaker volume level
AT+CRXVOL	Adjust RX voice output speaker volume
AT+ CTXVOL	Adjust TX voice mic volume
AT+ CTXMICGAIN	Adjust TX voice mic gain
AT+CECH	Inhibit far-end echo
AT+CECDT	Inhibit echo during doubletalk

5.2 Detailed Description of AT Commands According to Call Control

5.2.1 AT+CVHU Voice hang up control

Write command selects whether ATH or "drop DTR" shall cause a voice connection to be disconnected or not. By voice connection is also meant alternating mode calls that are currently in voice mode.

AT+CVHU Voice hang up control	
Test Command AT+CVHU=?	Response **+CVHU:** (range of supported **<mode>s)** **OK**
Read Command AT+CVHU?	Response **+CVHU: <mode>** **OK**
Write Command AT+CVHU=<mode>	Response **OK** or **ERROR**
Execution Command AT+CVHU	Response (Restore the default value) **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 "Drop DTR" ignored but **OK** response given. ATH disconnects. 1 "Drop DTR" and ATH ignored but **OK** response given.

Example

```
AT+CVHU=0
OK
AT+CVHU?
+CVHU: 0

OK
```

5.2.2 AT+CHUP Hang up call

This command is used to cancel voice calls. If there is no call, it will do nothing but **OK** response is given. After running AT+CHUP, multiple "VOICE CALL END: " may be reported which relies on how many calls exist before calling this command.

AT+CHUP Hang up call	
Test Command **AT+CHUP=?**	Response **OK**
Execution Command **AT+CHUP**	Response **OK** **VOICE CALL: END: <time>** **[...** **VOICE CALL: END: <time>]** No call: **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<time>	Voice call connection time. Format HHMMSS (HH: hour, MM: minute, SS: second)

Example

```
AT+CHUP
OK

VOICE CALL: END: 000017
```

5.2.3 AT+CBST Select bearer service type

Write command selects the bearer service <name> with data rate <speed>, and the connection element <ce> to be used when data calls are originated. Values may also be used during mobile terminated data call setup, especially in case of single numbering scheme calls.

AT+CBST Select bearer service type	
Test Command **AT+CBST=?**	Response **+CBST: (**list of supported **<speed>s),(**list of supported **<name>s),(**list of supported **<ce>s)** **OK**
Read Command **AT+CBST?**	Response **+CBST: <speed>,<name>,<ce>** **OK**
Write Command **AT+CBST=<speed>[,<name>[,<ce>]]**	Response **OK** or **ERROR**
Execution Command **AT+CBST**	Response **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<speed>	0 autobauding(automatic selection of the speed; this setting is possible in case of 3.1 kHz modem and non-transparent service) 7 – 9600 bps (V.32) 12 – 9600 bps (V.34) 14 – 14400 bps(V.34) 16 – 28800 bps(V.34) 17 – 33600 bps(V.34) 39 – 9600 bps(V.120) 43 – 14400 bps(V.120) 48 – 28800 bps(V.120) 51 – 56000 bps(V.120) 71 – 9600 bps(V.110) 75 – 14400 bps(V.110) 80 – 28800 bps(V.110 or X.31 flag stuffing) 81 – 38400 bps(V.110 or X.31 flag stuffing) 83 – 56000 bps(V.110 or X.31 flag stuffing) 84 – 64000 bps(X.31 flag stuffing) 116 – 64000 bps(bit transparent) 134 – 64000 bps(multimedia)

<name>	0 Asynchronous modem 1 Synchronous modem 4 data circuit asynchronous (RDI)
<ce>	0 transparent 1 non-transparent
NOTE: If <speed> is set to 116 or 134, it is necessary that <name> is equal to 1 and <ce> is equal to 0.	

Example

```
AT+CBST=0,0,1
OK
AT+CBST?
+CBST: 0,0,1

OK
```

5.2.4 AT+CRLP Radio link protocol

Radio Link Protocol(RLP) parameters used when non-transparent data calls are originated may be altered with write command.

Read command returns current settings for each supported RLP version <verX>. Only RLP parameters applicable to the corresponding <verX> are returned.

Test command returns values supported by the TA as a compound value. If ME/TA supports several RLP versions <verX>, the RLP parameter value ranges for each <verX> are returned in a separate line.

AT+CRLP Radio link protocol	
Test Command AT+CRLP=?	Response **+CRLP:** (list of supported **<iws>s),**(list of supported **<mws>s),**(list of supported **<T1>s),**(list of supported **<N2>s)[,<ver1>[,**(list of supported **<T4>s)]][<CR><LF>** **+CRLP:** (list of supported **<iws>s),**(list of supported **<mws>s),**(list of supported **<T1>s),**(list of supported **<N2>s)[,<ver2>[,**(list of supported **<T4>s)]]** **[...]]** **OK**
Read Command AT+CRLP?	Response **+CRLP: <iws>,<mws>,<T1>,<N2>[,<ver1>[,<T4>]][<CR><LF>** **+CRLP: <iws>,<mws>,<T1>,<N2>[,<ver2>[,<T4>]]**

	[...]] OK
Write Command AT+CRLP=<iws> [,<mws>[,<T1>[,<N2> [,<ver>[,<T4>]]]]]	Response OK or ERROR
Execution Command AT+CRLP	Response OK
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<ver>,<verX>	RLP version number in integer format, and it can be 0, 1 or 2; when version indication is not present it shall equal 1.
<iws>	IWF to MS window size.
<mws>	MS to IWF window size.
<T1>	Acknowledgement timer.
<N2>	Retransmission attempts.
<T4>	Re-sequencing period in integer format.

NOTE:<T1> and <T4> are in units of 10 ms.

Example

```
AT+CRLP?
+CRLP: 61,61,48,6,0
+CRLP: 61,61,48,6,1
+CRLP: 240,240,52,6,2

OK
```

5.2.5 AT+CR Service reporting control

Write command controls whether or not intermediate result code "+CR: <serv>" is returned from the TA to the TE. If enabled, the intermediate result code is transmitted at the point during connect negotiation at which the TA has determined which speed and quality of service will be used, before any error control or data compression reports are transmitted, and before the intermediate result code CONNECT is transmitted.

AT+CR Service reporting control

Test Command **AT+CR=?**	Response **+CR:** (list of supported **<mode>s)** **OK**
Read Command **AT+CR?**	Response **+CR: <mode>** **OK**
Write Command **AT+CR=<mode>**	Response **OK** or **ERROR**
Execution Command **AT+CR**	Response (Restore the default value) **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 disables reporting 1 enables reporting
<serv>	ASYNC asynchronous transparent SYNC synchronous transparent REL ASYNC asynchronous non-transparent REL sync synchronous non-transparent GPRS [<L2P>] GPRS The optional <L2P> proposes a layer 2 protocol to use between the MT and the TE.s

Example

```
AT+CR=1
OK

AT+CR?
+CR: 1

OK
```

5.2.6 AT+CRC Cellular result codes

Write command controls whether or not the extended format of incoming call indication or GPRS network request for PDP context activation is used. When enabled, an incoming call is indicated to the TE with unsolicited result code "+CRING: <type>" instead of the normal RING.
Test command returns values supported by the TA as a compound value.

AT+CRC Cellular result codes	
Test Command **AT+CRC=?**	Response **+CRC: (**list of supported **<mode>s)** **OK**
Read Command **AT+CRC?**	Response **+CRC: <mode>** **OK**
Write Command **AT+CRC=<mode>**	Response **OK** or **ERROR**
Execution Command **AT+CRC**	Response (Restore the default value) **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 – disable extended format 1 – enable extended format
<type>	ASYNC asynchronous transparent SYNC synchronous transparent REL ASYNC asynchronous non-transparent REL SYNC synchronous non-transparent FAX facsimile VOICE normal voice VOICE/XXX voice followed by data(XXX is ASYNC, SYNC, REL ASYNC or REL SYNC) ALT VOICE/XXX alternating voice/data, voice first ALT XXX/VOICE alternating voice/data, data first ALT FAX/VOICE alternating voice/fax, fax first GPRS GPRS network request for PDP context activation

Example

```
AT+CRC=1
OK

AT+CRC?
+CRC: 1

OK
```

5.2.7 AT+CLCC List current calls

This command issued to return list of current calls of ME. If command succeeds but no calls are available, no information response is sent to TE.

AT+CLCC List current calls	
Test Command **AT+CLCC=?**	Response **+CLCC:** (range of supported **<n>s)** **OK**
Read Command **AT+CLCC?**	Response **+CLCC: <n>** **OK**
Write Command **AT+CLCC=<n>**	Response **OK** or **ERROR**
Execution Command **AT+CLCC**	Response **+CLCC:** **<id1>,<dir>,<stat>,<mode>,<mpty>[,<number>,<type>[,<alpha>]][** **<CR><LF>** **+CLCC:** **<id2>,<dir>,<stat>,<mode>,<mpty>[,<number>,<type>[,<alpha>]]** **[...]]** **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	0 Don't report a list of current calls of ME automatically when the current call status changes. 1 Report a list of current calls of ME automatically when the current call status changes.
<idX>	Integer type, call identification number, this number can be used in +CHLD command operations.
<dir>	0 mobile originated (MO) call 1 mobile terminated (MT) call
<stat>	State of the call: 0 active 1 held 2 dialing (MO call) 3 alerting (MO call) 4 incoming (MT call) 5 waiting (MT call) 6 disconnect
<mode>	bearer/teleservice: 0 voice 1 data 2 fax 9 unknown
<mpty>	0 call is not one of multiparty (conference) call parties 1 call is one of multiparty (conference) call parties
<number>	String type phone number in format specified by <type>.
<type>	Type of address octet in integer format; 128 Restricted number type includes unknown type and format 145 International number type 161 national number.The network support for this type is optional 177 network specific number,ISDN format 129 Otherwise
<alpha>	String type alphanumeric representation of <number> corresponding to the entry found in phonebook; used character set should be the one selected with command Select TE Character Set AT+CSCS.

Example

```
ATD10011;
OK

AT+CLCC
```

```
+CLCC: 1,0,0,0,0,"10011",129,"sm"

OK

RING (with incoming call)

AT+CLCC
+CLCC: 1,1,4,0,0,"02152063113",128,"gongsi"

OK
```

5.2.8 AT+CEER Extended error report

Execution command causes the TA to return the information text <report>, which should offer the user of the TA an extended report of the reason for:

1 The failure in the last unsuccessful call setup(originating or answering) or in-call modification.

2 The last call release.

3 The last unsuccessful GPRS attach or unsuccessful PDP context activation.

The last GPRS detach or PDP context deactivation.

AT+CEER Extended error report	
Test Command **AT+CEER=?**	Response **OK**
Execution Command **AT+CEER**	Response **+CEER: <report>** **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<report>	Wrong information which is possibly occurred.

Example

```
AT+CEER
```

+CEER: Invalid/incomplete number

OK

5.2.9 AT+CCWA Call waiting

This command allows control of the Call Waiting supplementary service. Activation, deactivation and status query are supported. When querying the status of a network service (<mode>=2) the response line for 'not active' case (<status>=0) should be returned only if service is not active for any <class>. Parameter <n> is used to disable/enable the presentation of an unsolicited result code +CCWA: <number>,<type>,<class> to the TE when call waiting service is enabled. Command should be abortable when network is interrogated.

AT+CCWA Call waiting	
Test Command **AT+CCWA=?**	Response **+CCWA:** (range of supported **<n>s)** **OK**
Read Command **AT+CCWA?**	Response **+CCWA: <n>** **OK**
Write Command **AT+CCWA=<n>[,<mode>[,<class>]]**	Response When <mode>=2 and command successful: **+CCWA: <status>,<class>[<CR><LF>** **+CCWA: <status>,<class>[...]]** **OK** or **ERROR**
Execution Command **AT+CCWA**	Response (Restore the default value) **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	Sets/shows the result code presentation status in the TA 0 disable 1 enable
<mode>	When <mode> parameter is not given, network is not interrogated:

	0 disable 1 enable 2 query status
<class>	It is a sum of integers each representing a class of information (default 7) 1 voice (telephony) 2 data (refers to all bearer services) 4 fax (facsimile services) 7 voice,data and fax(1+2+4) 8 short message service 16 data circuit sync 32 data circuit async 64 dedicated packet access 128 dedicated PAD access 255 The value 255 covers all classes
<status>	0 not active 1 active
<number>	String type phone number of calling address in format specified by <type>.
<type>	Type of address octet in integer format; 128 Restricted number type includes unknown type and format 145 International number type 129 Otherwise

Example

```
AT+CCWA=?
+CCWA: (0-1)

OK

AT+CCWA?
+CCWA: 0

OK
```

5.2.10 AT+CHLD Call related supplementary services

This command allows the control the following call related services:

1. A call can be temporarily disconnected from the ME but the connection is retained by the network.
2. Multiparty conversation (conference calls).

3. The served subscriber who has two calls (one held and the other either active or alerting) can connect the other parties and release the served subscriber's own connection.
Calls can be put on hold, recovered, released, added to conversation, and transferred. This is based on the GSM/UMTS supplementary services.

AT+CHLD Call related supplementary services	
Test Command **AT+CHLD=?**	Response **+CHLD:** (list of supported **<n>s)** **OK**
Write Command **AT+CHLD=<n>**	Response **OK** or **ERROR**
Execution Command **AT+CHLD** **Default to <n>=2.**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	0 Terminate all held calls; or set User Determined User Busy for a waiting call 1 Terminate all active calls and accept the other call (waiting call or held call) 1X Terminate a specific call X 2 Place all active calls on hold and accept the other call (waiting call or held call) as the active call 2X Place all active calls except call X on hold 3 Add the held call to the active calls 4 Connect two calls and cut off the connection between users and them simultaneously

Example

```
AT+CHLD=?
+CHLD: (0,1,1x,2,2x,3,4)
```

OK

5.2.11 AT+CCFC Call forwarding number and conditions

This command allows control of the call forwarding supplementary service. Registration, erasure, activation, deactivation, and status query are supported.

AT+CCFC Call forwarding number and conditions	
Test Command **AT+CCFC=?**	Response **+CCFC: (**list of supported **<reason>s)** **OK**
Write Command **AT+CCFC=<reason>,<mode>[,<number>[,<type>[,<class>[,<subaddr>[,<satype>[,<time>]]]]]]**	Response When <mode>=2 and command successful: **+CCFC: <status>,<class1>[,<number>,<type>[,<subaddr>,<satype>[,<time>]]][<CR><LF> +CCFC: <status>,<class2>[,<number>,<type>[,<subaddr>,<satype>[,<time>]]][...]]** **OK** When <mode>!=2 and command successful: **OK** or **ERROR** or **+CME ERROR: <err>**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<reason>	0 unconditional 1 mobile busy 2 no reply 3 not reachable 4 all call forwarding 5 all conditional call forwarding
<mode>	0 disable 1 enable

	2 query status 3 registration 4 erasure
<number>	String type phone number of forwarding address in format specified by <type>.
<type>	Type of address octet in integer format: 145 dialing string <number> includes international access code character '+' 129 otherwise
<subaddr>	String type sub address of format specified by <satype>.
<satype>	Type of sub address octet in integer format, default 128.
<classX>	It is a sum of integers each representing a class of information (default 7): 1 voice (telephony) 2 data (refers to all bearer services) 4 fax (facsimile services) 16 data circuit sync 32 – data circuit async 64 dedicated packet access 128 dedicated PAD access 255 The value 255 covers all classes
<time>	1...30 when "no reply" is enabled or queried, this gives the time in seconds to wait before call is forwarded, default value 20.
<status>	0 not active 1 active

Example

```
AT+CCFC=?
+CCFC: (0,1,2,3,4,5)

OK
AT+CCFC=0,2
+CCFC: 0,255

OK
```

5.2.12 AT+CLIP Calling line identification presentation

This command refers to the GSM/UMTS supplementary service CLIP (Calling Line Identification Presentation) that enables a called subscriber to get the calling line identity (CLI) of the calling party when

receiving a mobile terminated call.
Write command enables or disables the presentation of the CLI at the TE. It has no effect on the execution of the supplementary service CLIP in the network.
When the presentation of the CLI at the TE is enabled (and calling subscriber allows), +CLIP: <number>,<type>,,[,[<alpha>][,<CLI validity>]] response is returned after every RING (or +CRING: <type>; refer sub clause "Cellular result codes +CRC") result code sent from TA to TE. It is manufacturer specific if this response is used when normal voice call is answered.

AT+CLIP Calling line identification presentation	
Test Command **AT+CLIP=?**	Response **+CLIP: (**range of supported **<n>s)** **OK**
Read Command **AT+CLIP?**	Response **+CLIP: <n>,<m>** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+CLIP=<n>**	Response **OK** or **ERROR**
Execution Command **AT+CLIP**	Response Set default value(<n>=0): **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	Parameter sets/shows the result code presentation status in the TA: 0 disable 1 enable
<m>	0 CLIP not provisioned 1 CLIP provisioned 2 unknown (e.g. no network, etc.)
<number>	String type phone number of calling address in format specified by <type>
<type>	Type of address octet in integer format;

	128 Restricted number type includes unknown type and format 145 International number type 161 national number.The network support for this type is optional 177 network specific number,ISDN format 129 Otherwise
<alpha>	String type alphanumeric representation of <number> corresponding to the entry found in phone book.
<CLI validity>	0 CLI valid 1 CLI has been withheld by the originator 2 CLI is not available due to interworking problems or limitations of originating network

Example

```
AT+CLIP=1
OK
RING (with incoming call)
+CLIP: "02152063113",128,,,"gongsi",0
```

5.2.13 AT+CLIR Calling line identification restriction

This command refers to CLIR service that allows a calling subscriber to enable or disable the presentation of the CLI to the called party when originating a call.
Write command overrides the CLIR subscription (default is restricted or allowed) when temporary mode is provisioned as a default adjustment for all following outgoing calls. This adjustment can be revoked by using the opposite command.. If this command is used by a subscriber without provision of CLIR in permanent mode the network will act.
Read command gives the default adjustment for all outgoing calls (given in <n>), and also triggers an interrogation of the provision status of the CLIR service (given in <m>).
Test command returns values supported as a compound value.

AT+CLIR Calling line identification restriction	
Test Command **AT+CLIR=?**	Response **+CLIR:** (range of supported **<n>s)** **OK**
Read Command **AT+CLIR?**	Response **+CLIR: <n>,<m>** **OK** or

	ERROR or **+CME ERROR: <err>**
Write Command **AT+CLIR=<n>**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	0 presentation indicator is used according to the subscription of the CLIR service 1 CLIR invocation 2 CLIR suppression
<m>	0 CLIR not provisioned 1 CLIR provisioned in permanent mode 2 unknown (e.g. no network, etc.) 3 CLIR temporary mode presentation restricted 4 CLIR temporary mode presentation allowed

Example

AT+CLIR=?
+CLIR: (0-2)

OK

5.2.14 AT+COLP Connected line identification presentation

This command refers to the GSM/UMTS supplementary service COLP(Connected Line Identification Presentation) that enables a calling subscriber to get the connected line identity (COL) of the called party after setting up a mobile originated call. The command enables or disables the presentation of the COL at the TE. It has no effect on the execution of the supplementary service COLR in the network.
When enabled (and called subscriber allows), +COLP:<number>,<type> [,<subaddr>,<satype> [,<alpha>]] intermediate result code is returned from TA to TE before any +CR Response.It is manufacturer specific if this response is used when normal voice call is established.

When the AT+COLP=1 is set, any data input immediately after the launching of "ATDXXX;" will stop the execution of the ATD command, which may cancel the establishing of the call.

AT+COLP Connected line identification presentation	
Test Command **AT+COLP=?**	Response **+COLP:** (range of supported **<n>s)** **OK**
Read Command **AT+COLP?**	Response **+COLP: <n>,<m>** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+COLP=<n>**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+COLP**	Response Set default value(<n>=0, <m>=0): **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	Parameter sets/shows the result code presentation status in the TA: 0 disable 1 enable
<m>	0 COLP not provisioned 1 COLP provisioned 2 unknown (e.g. no network, etc.)

Example

AT+COLP?
+COLP: 1,0

```
OK
ATD10086;
VOICE CALL: BEGIN

+COLP: "10086",129,,,

OK
```

5.2.15 AT+VTS DTMF and tone generation

This command allows the transmission of DTMF tones and arbitrary tones which cause the Mobile Switching Center (MSC) to transmit tones to a remote subscriber. The command can only be used in voice mode of operation (active voice call).

NOTE: The END event of voice call will terminate the transmission of tones, and as an operator option, the tone may be ceased after a pre-determined time whether or not tone duration has been reached.

When receive DTMF, will report URC +RXDTMF:

AT+VTS DTMF and tone generation	
Test Command **AT+VTS=?**	Response **+VTS:** (list of supported **<dtmf>s)** **OK**
Write Command **AT+VTS=<dtmf> [,<duration>]** **AT+VTS=<dtmf-string>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<dtmf>	A single ASCII character in the set 0-9, *, #, A, B, C, D.
<duration>	Tone duration in 1/10 seconds, from 0 to 255. This is interpreted as a DTMF tone of different duration from that mandated by the AT+VTD command, otherwise, the duration which be set the AT+VTD command will be used for the tone (<duration> is omitted).
<dtmf-string>	A sequence of ASCII character in the set 0-9, *, #, A, B, C, D, and maximal length of the string is 29. The string must be enclosed in double quotes (""), and separated by commas between the ASCII

	characters (e.g. "1,3,5,7,9,*"). Each of the tones with a duration which is set by the AT+VTD command.

Example

```
AT+VTS=1
OK
AT+VTS=1,20
OK
AT+VTS="1,3,5"
OK
AT+VTS=?
+VTS: (0-9,*,#,A,B,C,D)

OK
```

5.2.16 AT+VTD Tone duration

This refers to an integer <n> that defines the length of tones emitted as a result of the AT+VTS command. A value different than zero causes a tone of duration <n>/10 seconds.

AT+VTD Tone duration	
Test Command **AT+VTD=?**	Response **+VTD:** (list of supported **<n>s)** **OK**
Read Command **AT+VTD?**	Response **+VTD: <n>** **OK**
Write Command **AT+VTD=<n>**	Response **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<n>	Tone duration in integer format, from 0 to 255, and 0 is factory value. 0 Tone duration of every single tone is dependent on the

	network. 1…255 one duration of every single tone in 1/10 seconds.

Example

```
AT+VTD=?
+VTD: (0-255)

OK
AT+VTD?
+VTD: 0

OK
AT+VTD=5
OK
```

5.2.17 AT+CSTA Select type of address

Write command is used to select the type of number for further dialing commands (ATD) according to GSM/UMTS specifications.
Read command returns the current type of number.
Test command returns values supported by the Module as a compound value.

AT+CSTA Select type of address	
Test Command **AT+CSTA=?**	Response **+CSTA: (list of supported <type>s)** **OK**
Read Command **AT+CSTA?**	Response **+CSTA: <type>** **OK**
Write Command **AT+CSTA=<type>**	Response **OK** or **ERROR**
Execution Command **AT+CSTA**	Response (Restore the default value) **OK**
Parameter Saving Mode	NO_SAVE
Maximum Response Time	-

Reference	

Defined Values

<type>	Type of address octet in integer format: 145 – when dialing string includes international access code character “+” 161 – national number.The network support for this type is optional 177 – network specific number,ISDN format 129 – otherwise

NOTE

Because the type of address is automatically detected on the dial string of dialing command, command AT+CSTA has really no effect.

Example

```
AT+CSTA?
+CSTA: 129

OK
AT+CSTA=145
OK
```

5.2.18 AT+CMOD Call mode

Write command is used to select the type of number for further dialing commands (ATD) according to GSM/UMTS specifications.
Read command returns the current type of number.
Test command returns values supported by the Module as a compound value.

AT+CMOD Call mode	
Test Command **AT+CMOD=?**	Response **+CMOD:** (list of supported **<mode>s)** **OK**
Read Command **AT+CMOD?**	Response **+CMOD: <mode>** **OK**
Write Command **AT+CMOD=<mode>**	Response **OK** or **ERROR**
Execution Command **AT+CMOD**	Response Set default value: **OK**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 single mode(only supported)

Example

```
AT+CMOD?
+CMOD: 0

OK
AT+CMOD=0
OK
```

NOTE

The value of <mode> shall be set to zero after a successfully completed alternating mode call. It shall be set to zero also after a failed answering. The power-on, factory and user resets shall also set the value to zero. This reduces the possibility that alternating mode calls are originated or answered accidentally.

5.2.19 AT+VMUTE Speaker mute control

This command is used to control the loudspeaker to mute and unmute during a voice call which is connected. If there is not a connected call, write command can't be used.
When all calls are disconnected, the Module sets the subparameter as 0 automatically.

AT+VMUTE Speaker mute control	
Test Command **AT+VMUTE=?**	Response **+VMUTE: (**range of supported **<mode>**s**)** **OK**
Read Command **AT+VMUTE?**	Response **+VMUTE: <mode>** **OK**
Write Command **AT+VMUTE=<mode>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 – mute off 1 – mute on

Example

```
AT+VMUTE?
+VMUTE: 0
```

```
OK
AT+VMUTE=1
OK
```

5.2.20 AT+CMUT Microphone mute control

This command is used to enable and disable the uplink voice muting during a voice call which is connected.
If there is not a connected call, write command can't be used.
When all calls are disconnected, the Module sets the subparameter as 0 automatically.

AT+CMUT Microphone mute control	
Test Command **AT+CMUT=?**	Response **+CMUT:** (range of supported **<mode>**s) **OK**
Read Command **AT+CMUT?**	Response **+CMUT: <mode>** **OK**
Write Command **AT+CMUT=<mode>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	0 – mute off 1 – mute on

Example

```
AT+CMUT?
+CMUT: 0

OK
AT+CMUT=1
OK
```

5.2.21 AT+MORING Enable or disable report MO ring URC

This command is used to enable or disable report MO ring URC.

AT+MORING Enable or disable report MO ring URC	
Test Command **AT+MORING=?**	Response **+MORING: (0-1)** **OK**
Read Command **AT+MORING?**	Response **+MORING: <mode>** **OK**
Write Command **AT+MORING=<mode>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	Enable or disable report MO ring report 0 disable 1 enable

Example

```
AT+MORING?
+MORING: 0

OK
AT+MORING=1
OK
ATD10086;
OK
```

VOICE CALL: RING

VOICE CALL: BEGIN

5.2.22 AT+CLVL Loudspeaker volume level

Write command is used to select the volume of the internal loudspeaker audio output of the device.
Read command returns the volume of the internal loudspeaker audio output of the device.
Test command returns supported values as compound value.

AT+CLVL Loudspeaker volume level	
Test Command **AT+CLVL=?**	Response **+CLVL: (**range of supported **<level>**s) **OK**
Read Command **AT+CLVL?**	Response **+CLVL: <level>** **OK**
Write Command **AT+CLVL=<level>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<level>	0..5 Integer type value which represents loudspeaker volume level. The range is from 0 to 5, and 0 represents the lowest loudspeaker volume level, 5 is default factory value.

Example

AT+CLVL?
+CLVL: 5

OK
AT+CLVL=1
OK

NOTE

<level> is not saved, and it resets default value when restart.

5.2.23 AT+CRXVOL Adjust RX voice output speaker volume

This command is used to adjust digital Volume of output signal after speech decoder, before summation of sidetone and DAC. It modify the RX_VOICE_SPK_GAIN in DSP. This command only be used during call and don't save the parameter after call.

AT+CRXVOL Adjust RX voice output speaker volume	
Test Command AT+CRXVOL=?	Response **+CRXVOL:** (range of supported **<value>**s) **OK**
Read Command AT+CRXVOL?	Response **+CRXVOL: <value>** **OK**
Write Command AT+CRXVOL=<value>	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<value>	Gain value from 0x0000-0xffff, default value is not a fixed value. It varies with different versions.

Example

AT+CRXVOL?
+CRXVOL: 0x3fd9

OK
AT+CRXVOL=0x1234

OK

NOTE

This command only be used during call and don't save the parameter after call.

5.2.24 AT+CTXVOL Adjust TX voice mic volume

This command is used to adjust mic gain. It modify the TX_VOICE_VOL in DSP. This command only be used during call and don't save the parameter after call.

AT+ CTXVOL Adjust TX voice mic volume	
Test Command AT+CTXVOL=?	Response **+CTXVOL: (**range of supported **<value>**s) **OK**
Read Command AT+CTXVOL?	Response **+CTXVOL: <value>** **OK**
Write Command AT+CTXVOL=<value>	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<value>	Gain value from 0x0000-0xffff, default value is not a fixed value. It varies with different versions.

Example

AT+CTXVOL?
+CTXVOL: 0x2d33

OK
AT+CTXVOL=0x1234

OK

NOTE

This command only be used during call and don't save the parameter after call.

5.2.25 AT+CTXMICGAIN Adjust TX voice mic gain

This command is used to adjust mic gain. It modify the TX_VOICE_MIC_GAIN in DSP. This command only be used during call and don't save the parameter after call.

AT+ CTXMICGAIN Adjust TX voice mic gain	
Test Command **AT+CTXMICGAIN=?**	Response **+CTXMICGAIN:** (list of supported **<mode>**s),(range of supported **<value>**s) **OK**
Read Command **AT+CTXMICGAIN?**	Response **+CTXMICGAIN: <mode>,<value>** **OK**
Write Command **AT+CTXMICGAIN =<mode>,<value>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<mode>	mode value from 0-1, default value is not a fixed value. It varies with different versions.
<value>	gain value from 0x0000-0xffff, default value is not a fixed value. It varies with different versions.

Example

AT+CTXMICGAIN?
+CTXMICGAIN: 1,0x2000

```
OK
AT+CTXMICGAIN=1,0x1234
OK
```

NOTE

This command only be used during call and don't save the parameter after call.

5.2.26 AT+CECH Inhibit far-end echo

This command is used to adjust additional muting gain applied in DES during far-end only. It modify the pp_gamma_e_high of SMECNS_V2 MODULE TX in DSP. The bigger the value, the stronger the inhibition .This command only be used during call and don't save the parameter after call.

AT+ CECH Inhibit far-end echo	
Test Command **AT+CECH=?**	Response **+CECH:** (rang of supported **<value>s)** **OK**
Read Command **AT+CECH?**	Response **+CECH: <value>** **OK**
Write Command **AT+CECH=<value>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<value>	gain value from 0x0000-0x7fff, default value is not a fixed value. It varies with different versions.

Example

```
AT+CECH?
```

+CECH: 0x0200

OK
AT+CECH=0x1234
OK

NOTE

This command only be used during call and don't save the parameter after call.

5.2.27 AT+CECDT Inhibit echo during doubletalk

This command is used to adjust additional muting gain applied in DES during doubletalk. It modify the pp_gamma_e_dt of SMECNS_V2 MODULE TX in DSP. The bigger the value, the stronger the inhibition .This command only be used during call and don't save the parameter after call.

AT+ CECDT Inhibit echo during doubletalk	
Test Command **AT+CECDT=?**	Response **+CECDT:** (rang of supported **<value>**s) **OK**
Read Command **AT+CECDT?**	Response **+CECDT: <value>** **OK**
Write Command **AT+CECDT=<value>**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	

Defined Values

<value>	gain value from 0x0000-0x7fff, default value is not a fixed value. It varies with different versions.

Example

```
AT+CECDT?
+CECDT: 0x0100

OK
AT+CECDT=0x1234
OK
```

NOTE

This command only be used during call and don't save the parameter after call.

6 AT Commands for Phonebook

6.1 Overview of AT Commands for Phonebook

Command	Description
AT+CPBS	Select Phonebook memory storage
AT+CPBR	Read Phonebook entries
AT+CPBF	Find Phonebook entries
AT+CPBW	Write Phonebook entry
AT+CNUM	Subscriber number

6.2 Detailed Description of AT Commands for Phonebook

6.2.1 AT+CPBS Select Phonebook memory storage

AT+CPBS Select Phonebook memory storage	
Test Command **AT+CPBS=?**	Response **+CPBS:** (list of supported <**storage**>s) **OK**
Read Command **AT+CPBS?**	Response **+CPBS: <storage>[,<used>,<total>]** **OK**
Write Command **AT+CPBS=<storage>**	Response **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Execution Command	Response (Set default value "SM")

AT+CPBS	OK

Defined Values

<storage>	Values reserved by the present document: "DC" ME dialed calls list Capacity: max. 20 entries AT+CPBW command is not applicable to this storage. "MC" ME missed (unanswered received) calls list Capacity: max. 20 entries AT+CPBW command is not applicable to this storage. "RC" ME received calls list Capacity: max. 20 entries AT+CPBW command is not applicable to this storage. "SM" SIM phonebook Capacity: depending on SIM card "ME" Mobile Equipment phonebook Capacity: max. 25 entries "FD" SIM fixdialling-phonebook Capacity:depending on SIM card "ON" MSISDN list Capacity:depending on SIM card "LD" Last number dialed phonebook Capacity: depending on SIM card AT+CPBW command is not applicable to this storage "EN" Emergency numbers Capacity: depending on SIM card AT+CPBW command is not applicable to this storage.
<used>	Integer type value indicating the number of used locations in selected memory.
<total>	Integer type value indicating the total number of locations in selected memory.

Example

```
AT+CPBS=?
+CPBS: ("SM","DC","FD","LD","MC","ME","RC","EN","ON")

OK
AT+CPBS="SM"
OK
AT+CPBS?
+CPBS: "SM",1,200
```

OK

NOTE

Select the active phonebook storage,i.e.the phonebook storage that all subsequent phonebook commands will be operating on

6.2.2 AT+CPBR Read Phonebook entries

AT+CPBR Read Phonebook entries	
Test Command **AT+CPBR=?**	Response **+CPBR: (<minIndex>-<maxIndex>),[<nlength>],[<tlength>]** **OK** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CPBR=<index1>[,<index2>]**	Response **[+CPBR: <index1>,<number>,<type>,<text>[<CR><LF>** **+CPBR: <index2>,<number>,<type>,<text>[…]]]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<index1>	Integer type value in the range of location numbers of phonebook memory.
<index2>	Integer type value in the range of location numbers of phonebook memory.
<index>	Integer type.the current position number of the Phonebook index.
<minIndex>	Integer type the minimum <index> number.
<maxIndex>	Integer type the maximum <index> number.
<number>	String type, phone number of format <type>, the maximum length is <nlength>.
<type>	Type of phone number octet in integer format, default 145 when dialing string includes international access code character "+"; 128 for restricted number type including unknown type and format, otherwise 129.

<text>	String type field of maximum length <tlength>; often this value is set as name.
<nlength>	Integer type value indicating the maximum length of field <number>
<tlength>	Integer type value indicating the maximum length of field <text>.

Example

```
AT+CPBS?
+CPBS: "SM",2,200

OK
AT+CPBR=1,10
+CPBR: 1,"1234567890",129,"James"
+CPBR: 2,"0987654321",129,"Kevin"

OK
```

NOTE

If the storage is selected as "SM" then the command will return the record in SIM phonebook, the same to others.

6.2.3 AT+CPBF Find Phonebook entries

AT+CPBF Find Phonebook entries	
Test Command **AT+CPBF=?**	Response **+CPBF: [<nlength>],[<tlength>]** **OK**
Write Command **AT+CPBF=[<findtext>]**	Response **[+CPBF: <index1>,<number>,<type>,<text>[<CR><LF>** **+CPBF: <indexN>,<number>,<type>,<text>[…]]]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<findtext>	String type, this value is used to find the record. Character set should be the one selected with command AT+CSCS.
<index>	Integer type.the current position number of the Phonebook index.
<number>	String type, phone number of format <type>, the maximum length is <nlength>.
<type>	Type of phone number octet in integer format, default 145 when dialing string includes international access code character "+"; 128 for restricted number type including unknown type and format,otherwise 129.
<text>	String type field of maximum length <tlength>; often this value is set as name.
<nlength>	Integer type value indicating the maximum length of field <number>
<tlength>	Integer type value indicating the maximum length of field <text>.

Example

```
AT+CPBF="James"
+CPBF: 1,"1234567890",129,"James"

OK
```

NOTE

If <findtext> is null, it will lists all the entries.

6.2.4 AT+CPBW Write Phonebook entry

AT+CPBW Write Phonebook entry	
Test Command **AT+CPBW=?**	Response **+CPBW: (**list of supported **<index>s),[<nlength>],** (list of supported **<type>s),[<tlength>]** **OK** or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**
Write Command **AT+CPBW=[<index>][,<num**	Response **OK**

ber>[,<type>[,<text>]]]	or **ERROR** If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<index>	Integer type values in the range of location numbers of phonebook memory.If <index> is not given,the first free entry will be used. If <index> is given as the only parameter, the phonebook entry specified by <index> is deleted.If record number <index> already exists, it will be overwritten.
<number>	String type, phone number of format <type>, the maximum length is <nlength>.It must be an non-empty string.
<type>	Type of address octet in integer format, The range of value is from 129 to 255. If <number> contains a leading "+" <type> = 145 (international) is used.Supported value are: 145 when dialling string includes international access code character "+" 161 national number.The network support for this type is optional 177 network specific number,ISDN format 129 otherwise NOTE: Other value refer TS 24.008 [8] subclause 10.5.4.7.
<text>	String type field of maximum length <tlength>; character set as specified by command Select TE Character Set AT+CSCS.
<nlength>	Integer type value indicating the maximum length of field <number>.
<tlength>	Integer type value indicating the maximum length of field <text>.

Example

```
AT+CPBW=3,"88888888",129,"John"
OK
AT+CPBW=,"6666666",129,"mary"
OK
AT+CPBW=1
OK
```

NOTE

If the parameters of <type> and <text> are omitted and the first character of <number> is ‘+’, it will specify <type> as 145(129 if the first character isn’t ‘+’) and <text> as NULL.

6.2.5 AT+CNUM Subscriber number

AT+CNUM Subscriber number	
Test Command **AT+CNUM=?**	Response **OK**
Execution Command **AT+CNUM**	Response **[+CNUM: <alpha>,<number>,<type>[<CR><LF> +CNUM: <alpha>, <number>,<type> [...]]]** **OK** or If error is related to ME functionality: **+CME ERROR: <err>**

Defined Values

<alpha>	Optional alphanumeric string associated with <number>, used character set should be the one selected with command Select TE Character Set AT+CSCS.
<number>	String type phone number of format specified by <type>.
<type>	Type of address octet in integer format.see also AT+CPBR <type>

Example

```
AT+CNUM
+CNUM: "","13697252277",129


OK
```

NOTE

If the subscriber has different MSISDN for different services, each MSISDN is returned in a separate line

7 AT Commands for SIM Application Toolkit

7.1 Overview of AT Commands for SIM Application Toolkit

Command	Description
AT+STIN	SAT Indication
AT+STGI	Get SAT information
AT+STGR	SAT respond
AT+STK	STK switch
AT+STKFMT	Set STK pdu format
AT+STENV	Original STK PDU Envelope Command
AT+STSM	Get STK Setup Menu List with PDU Mode

7.2 Detailed Description of AT Commands for SIM Application Toolkit

7.2.1 AT+STIN SAT Indication

AT+STIN SAT Indication	
Test Command **AT+STIN=?**	Response **OK**
Read Command **AT+STIN?**	**+STIN: <cmd_id>** **OK**

Unsolicited Result Codes

<cmd_id>	Proactive Command notification 21 Display text 22 Get inkey

	23 Get input 24 Select item
+STIN: 25	Notification that SIM Application has returned to main menu. If user doesn't do any action in 2 minutes, application will return to main menu automatically.

Defined Values

<cmd_id>	21 Display text 22 Get inkey 23 Get input 24 Select item 25 Set up menu 81 Session end (pdu mode only) 0 None command
<time>	Service time

Example

AT+STIN?
+STIN: 24

OK

NOTE

Every time the SIM Application issues a Proactive Command, via the ME, the TA will receive an indication. This indicates the type of Proactive Command issued.

7.2.2 AT+STGI Get SAT information

AT+STGI Get SAT information	
Test Command **AT+STGI=?**	Response **OK**
Write Command **AT+STGI=<cmd_id>**	Response **(PDU format)** **+STGI: <cmd_id>,<tag>,<pdu_len>,<pdu_value>** **OK**
AT+STGI=<cmd_id>	Response **(NOT PDU format, listed below)** If <cmd_id>=10: **OK** If <cmd_id>=21:

	+STGI: 21,<prio>,<clear_mode>,<text_len>,<text> **OK**
	If <cmd_id>=22: **+STGI: 22,<rsp_format>,<help>,<text_len>,<text>** **OK**
	If <cmd_id>=23: **+STGI:** **23,<rsp_format>,<max_len>,<min_len>,<help>,<show>,<text_len>,<text>** **OK**
	If <cmd_id>=24: **+STGI:** **24,<help>,<softkey>,<present>,<title_len>,<title>,<item_num>** **+STGI: 24,<item_id>,<item_len>,<item_data>** **[…]** **OK**
	If <cmd_id>=25: **+STGI: 25,<help>,<softkey>,<title_len>,<title>,<item_num>** **+STGI: 25,<item_id>,<item_len>,<item_data>** **[…]** **OK**

Defined Values

<cmd_id>	Proactive Command notification 10 Setup call 21 Display text 22 Get inkey 23 Get input 24 Select item 25 Set up menu
<prio>	Priority of display text 0 Normal priority 1 High priority
<clear_mode>	0 Clear after a delay 1 Clear by user
<text_len>	Length of text
<rsp_format>	0 SMS default alphabet 1 YES or NO 2 numerical only 3 UCS2
<help>	0 Help unavailable 1 Help available
<max_len>	Maximum length of input

<min_len>	Minimum length of input
<show>	0 Hide input text 1 Display input text
<softkey>	0 No softkey preferred 1 Softkey preferred
<present>	Menu presentation format available for select item 0 Presentation not specified 1 Data value presentation 2 Navigation presentation
<title_len>	Length of title
<item_num>	Number of items in the menu
<item_id>	Identifier of item
<item_len>	Length of item
<title>	Title in ucs2 format
<item_data>	Content of the item in ucs2 format
<text>	Text in ucs2 format.
<tag>	Not used now.
<pdu_len>	Integer type, pdu string length
<pdu_val>	String type, the pdu string.

Example

```
AT+STGI=25 (NOT PDU format)
+STGI: 25,0,0,10,"795E5DDE884C59295730",15
+STGI: 25,1,8,"8F7B677E95EE5019"
+STGI: 25,2,8,"77ED4FE17FA453D1"
+STGI: 25,3,8,"4F1860E05FEB8BAF"
+STGI: 25,4,8,"4E1A52A17CBE9009"
+STGI: 25,5,8,"8D448D3963A88350"
+STGI: 25,6,8,"81EA52A9670D52A1"
+STGI: 25,7,8,"8F7B677E5F6994C3"
+STGI: 25,8,8,"8BED97F367425FD7"
+STGI: 25,9,10,"97F34E506392884C699C"
+STGI: 25,10,8,"65B095FB59296C14"
+STGI: 25,11,8,"94C358F056FE7247"
+STGI: 25,12,8,"804A59294EA453CB"
+STGI: 25,13,8,"5F005FC34F1195F2"
+STGI: 25,14,8,"751F6D3B5E388BC6"
+STGI: 25,21,12,"00530049004D53614FE1606F"

OK

AT+STGI=24 (PDU format)
```

```
+STGI:
24,0,48,"D02E81030124008202818285098070ED70B963A883508
F0A018053057F574E078C618F0C02809177917777ED6D88606F"

OK
```

7.2.3 AT+STGR SAT respond

AT+STGR SAT respond	
Test Command **AT+STGR=?**	Response **OK**
Write Command **AT+STGR=<cmd_id>[,<data>]**	Response **(NOT PDU format)** **OK**
AT+STGR=<pdu_len>,<pdu_value>	Response **(PDU format)** **OK**

Defined Values

<cmd_id>	Proactive Command notification 21 Display text 22 Get inkey 23 Get input 24 Select item 25 Set up menu 81 Session end 83 Session end by user 84 Go backward
<data>	If <cmd_id>=22: Input a character If <cmd_id>=23: Input a string. If <rsp_format> is YES or NO, input of a character in case of ANSI character set requests one byte, e.g. "Y". If <rsp_format> is numerical only, input the characters in decimal number, e.g. "123" If <rsp_ format> is UCS2, requests a 4 byte string, e.g. "0031" <rsp_ format> refer to the response by AT+STGI=23 If <cmd_id>=24: Input the identifier of the item selected by user If <cmd_id>=25:

	Input the identifier of the item selected by user If <cmd_id>=83: <data> ignore Note: It could return main menu during Proactive Command id is not 22 or 23 If <cmd_id>= 84: <data> ignore
<pdu_len>	Integer type, pdu string length
<pdu_value>	String type, the pdu string.

Example

```
AT+STGR=25,1
OK
+STIN: 24

AT+STGR=30,"810301240002028281830100900101"
OK
```

NOTE

After selected an item, different SIM/USIM cards will report different +STIN: command.

7.2.4 AT+STK STK switch

AT+STK STK switch	
Test Command **AT+STK=?**	Response **+STK:** (list of supported <**value**>s) **OK**
Read Command **AT+STK?**	Response **+STK: <value>** **OK**
Write Command **AT+STK=<value>**	Response **OK** or **ERROR**
Execution Command **AT+STK**	Response **OK**

Defined Values

<value>	0 Disable STK 1 Enable STK

Example

```
AT+STK=1
OK
```

NOTE

Module should reboot to take effective

7.2.5 AT+STKFMT Set STK pdu format

AT+STKFMT Set STK pdu format	
Read Command AT+STKFMT?	Response +STKFMT: <value> OK
Write Command AT+STKFMT=<value>	Response OK or ERROR

Defined Values

<value>	0 Disable STK pdu format, decoded command mode. 1 Enable STK pdu format

Example

```
AT+STKFMT=1
OK
```

NOTE

Module should reboot to take effective

7.2.6 AT+STENV Original STK PDU Envelope Command

AT+STENV Original STK PDU Envelope Command	
Test Command **AT+STENV=?**	Response **OK**
Write Command **AT+STENV=<len>,<pdu>**	Response **OK** or **ERROR**

Defined Values

<len>	Integer type, pdu string length
<pdu>	String type, pdu value

Example

AT+STENV=18,"D30782020181900101"
OK

NOTE

Module should reboot to take effective

7.2.7 AT+STSM Get STK Setup Menu List with PDU Mode

AT+STSM Get STK Setup Menu List with PDU Mode	
Test Command **AT+STSM=?**	Response **OK**
Read Command **AT+STSM?**	Response **+STSM: <cmd_id>,<tag>,<pdu_len>,<pdu_value>** **OK** or **ERROR**

Defined Values

<cmd_id>	Integer type, please refer to AT+STIN
<tag>	Not used now.
<pdu_len>	Integer type, pdu string length
<pdu_value>	String type, the pdu string.

Example

```
AT+STSM?
+STSM:
25,0,120,"D0768103012500820281828507806 5B052BF5
29B8F0A018070ED70B963A883508F06028070AB94C38
F0A03806D41884C77ED4FE18F0A048081EA52A9670D
52A18F0A0580624B673A97F34E508F0606808D854FE1
8F0A07805A314E50753162118F0A0880767E53D8751F6
D3B8F0A09806D596C5F98919053"

OK
```

NOTE

Setup main menu info got first before envelope command sent.

8 AT Commands for GPRS

8.1 Overview of AT Commands for GPRS

Command	Description
AT+CGREG	GPRS network registration status
AT+CGATT	Packet domain attach or detach
AT+CGACT	PDP context activate or deactivate
AT+CGDCONT	Define PDP context
AT+CGDSCONT	Define Secondary PDP Context
AT+CGTFT	Traffic Flow Template
AT+CGQREQ	Quality of service profile (requested)
AT+CGEQREQ	3G quality of service profile (requested)
AT+CGQMIN	Quality of service profile (minimum acceptable)
AT+CGEQMIN	3G quality of service profile (minimum acceptable)
AT+CGDATA	Enter data state
AT+CGPADDR	Show PDP address
AT+CGCLASS	GPRS mobile station class
AT+CGEREP	GPRS event reporting
AT+CGAUTH	Set type of authentication for PDP-IP connections of GPRS

8.2 Detailed Description of AT Commands for GPRS

8.2.1 AT+CGREG GPRS network registration status

This command controls the presentation of an unsolicited result code "+CGREG: <stat>" when <n>=1 and there is a change in the MT's GPRS network registration status.

The read command returns the status of result code presentation and an integer <stat> which shows Whether the network has currently indicated the registration of the MT.

AT+CGREG GPRS network registration status	
Test Command **AT+CGREG=?**	Response **+CGREG: (**list of supported **<n>s)** **OK**
Read Command **AT+CGREG?**	Response **+CGREG: <n>,<stat>[,<lac>,<ci>]** **OK**
Write Command **AT+CGREG=<n>**	Response **OK**
Execution Command Set default value: **AT+CGREG**	Response **OK**

Defined Values

<n>	0 disable network registration unsolicited result code 1 enable network registration unsolicited result code +CGREG: <stat> 2 there is a change in the ME network registration status or a change of the network cell: +CGREG: <stat>[,<lac>,<ci>]
<stat>	0 not registered, ME is not currently searching an operator to register to 1 registered, home network 2 not registered, but ME is currently trying to attach or searching an operator to register to 3 registration denied 4 unknown 5 registered, roaming
<lac>	Two bytes location area code in hexadecimal format (e.g."00C3" equals 193 in decimal).
<ci>	Cell ID in hexadecimal format. GSM : Maximum is two byte WCDMA : Maximum is four byte TDS-CDMA : Maximum is four byte

NOTE

The <lac> not supported in CDMA/HDR mode
The <ci> not supported in CDMA/HDR mode

Example

```
AT+CGREG=?
+CGREG: (0-2)

OK
AT+CGREG?
+CGREG: 0,0

OK
```

8.2.2 AT+CGATT Packet domain attach or detach

The write command is used to attach the MT to, or detach the MT from, the Packet Domain service. The read command returns the current Packet Domain service state.

AT+CGATT Packet domain attach or detach	
Test Command **AT+CGATT=?**	Response **+CGATT:** (list of supported **<state>s)** **OK**
Read Command **AT+CGATT?**	Response **+CGATT: <state>** **OK**
Write Command **AT+CGATT=<state>**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<state>	Indicates the state of Packet Domain attachment: 0 detached 1 attached

Example

```
AT+CGATT?
+CGATT: 0

OK
AT+CGATT=1
OK
```

8.2.3 AT+CGACT PDP context activate or deactivate

The write command is used to activate or deactivate the specified PDP context(s).

AT+CGACT PDP context activate or deactivate	
Test Command **AT+CGACT=?**	Response **+CGACT: (**list of supported **<state>s)** **OK**
Read Command **AT+CGACT?**	Response **+CGACT: [<cid>,<state> [<CR><LF>** **+CGACT: <cid>,<state>** **[...]]]** **OK**
Write Command **AT+CGACT=<state>[,<cid>]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<state>	Indicates the state of PDP context activation: 0 deactivated 1 activated
<cid>	A numeric parameter which specifies a particular PDP context definition (see AT+CGDCONT command). 1…42

Example

```
AT+CGACT=?
+CGACT: (0,1)

OK
AT+CGACT?
+CGACT: 1,1

OK
AT+CGACT=0,1
OK
```


8.2.4 AT+CGDCONT Define PDP context

The set command specifies PDP context parameter values for a PDP context identified by the (local) context identification parameter **<cid>**. The number of PDP contexts that may be in a defined state at the same time is given by the range returned by the test command. A special form of the write command (**AT+CGDCONT**=**<cid>**) causes the values for context **<cid>** to become undefined.

AT+CGDCONT Define PDP context	
Test Command **AT+CGDCONT=?**	Response **+CGDCONT:** (range of supported **<cid>s),<PDP_type>,,,**(list of supported **<d_comp>s),**(list of supported **<h_comp>s),(list of <ipv4_ctrl>s),(list of <emergency_flag>s),,,,,,,,(<ssc_mode>s),,(<pref_access_type>),,,(<always_on_req>)** **OK** or **ERROR**
Read Command **AT+CGDCONT?**	Response **+CGDCONT: [<cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,<ipv4_ctrl>,<emergency_flag>,,,,,,,,[<ssc_mode>],,<pref_access_type>,,,<always_on_req> [<CR><LF>** **+CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>,<ipv4_ctrl>,<emergency_flag>,,,,,,,,[<ssc_mode>],,<pref_access_type>,,,<always_on_req>[...]]]**

	OK or **ERROR**
Write Command **AT+CGDCONT=<cid>[,[<PDP_type>[,<APN>[,<PDP_addr>[,<d_comp>[,<h_comp>[,<ipv4_ctrl>[,<emergency_flag>[,,,,,,,,,<ssc_mode>[,,<pref_access_type>[,,,<always_on_req>]]]]]]]]]]]**	Response **OK** or **ERROR**
Execution Command **AT+CGDCONT**	Response **OK** or **ERROR**

Defined Values

<cid>	(PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. 1…42
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack
<APN>	(Access Point Name) a string parameter which is a logical name that is used to select the GGSN or the external packet data network.
<PDP_addr>	A string parameter that identifies the MT in the address space applicable to the PDP. Read command will continue to return the null string even if an address has been allocated during the PDP startup procedure. The allocated address may be read using command AT+CGPADDR.
<d_comp>	A numeric parameter that controls PDP data compression, this value may depend on platform: 0 off (default if value is omitted) 1 on 2 V.42bis 3 V.44
<h_comp>	A numeric parameter that controls PDP header compression, this

	value may depend on platform: 0 off (default if value is omitted) 1 on 2 RFC1144 3 RFC2507 4 RFC3095
<ipv4_ctrl>	Parameter that controls how the MT/TA requests to get the IPv4 address information: 0 Address Allocation through NAS Signaling 1 on
<emergency_flag>	emergency_flag: 0 off (default if value is omitted) 1 on
<ssc_mode>	ssc mode(if not encode in OTA,ssc_mode is not used in the at command) 0 ssc mode 1(default if value is omitted) 1 ssc mode 2
<pref_access_type>	prefer access type: 0 access unspecified(default if value is omitted) 1 access 3gpp
<always_on_req>	alway on req: 0 off (default if value is omitted) 1 on

Example

```
AT+CGDCONT=?
+CGDCONT: (1-42),"IP",,,(0-3),(0-4),(0-1),(0-1),,,,,,,,,(0-1),,(0-1),,,(0-1)
+CGDCONT: (1-42),"PPP",,,(0-3),(0-4),(0-1),(0-1),,,,,,,,,(0-1),,(0-1),,,(0-1)
+CGDCONT: (1-42),"IPV6",,,(0-3),(0-4),(0-1),(0-1),,,,,,,,,(0-1),,(0-1),,,(0-1)
+CGDCONT: (1-42),"IPV4V6",,,(0-3),(0-4),(0-1),(0-1),,,,,,,,,(0-1),,(0-1),,,(0-1)

OK
AT+CGDCONT?
+CGDCONT: 1,"IPV4V6","","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0,,,,,,,,,"",,,,0
+CGDCONT: 2,"IPV4V6","ims","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0,,,,,,,,,"",,,,0
+CGDCONT: 3,"IPV4V6","sos","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,1,,,,,,,,,"",,,,0

OK
```

8.2.5 AT+CGDSCONT Define Secondary PDP Context

The set command specifies PDP context parameter values for a Secondary PDP context identified by the (local) context identification parameter, **<cid>**. The number of PDP contexts that may be in a defined state at the same time is given by the range returned by the test command. A special form of the set command, AT+CGDSCONT=**<cid>** causes the values for context number **<cid>** to become undefined.

AT+CGDSCONT Define Secondary PDP Context	
Test Command **AT+CGDSCONT=?**	Response **+CGDSCONT: (**range of supported **<cid>s),(list of <p_cid>s for active primary contexts), <PDP_type>,(**list of supported **<d_comp>s),(**list of supported **<h_comp>s)** **OK** or **ERROR**
Read Command **AT+CGDSCONT?**	Response **+CGDSCONT: [<cid>,<p_cid>,<d_comp>,<h_comp>** **[<CR><LF>+CGDSCONT: <cid>,<p_cid>,<d_comp>,<h_comp>** **[...]]]** **OK** or **ERROR**
Write Command **AT+CGDSCONT=<cid>[,<p_cid>[,<d_comp>[,<h_comp>]]]**	Response **OK** or **ERROR**

Defined Values

<cid>	a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command.
<p_cid>	a numeric parameter which specifies a particular PDP context definition which has been specified by use of the +CGDCONT command. The parameter is local to the TE-MT interface. The list of permitted values is returned by the test form of the command.
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol

	PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack
<d_comp>	a numeric parameter that controls PDP data compression (applicable for SNDCPonly) (refer 3GPP TS 44.065 [61]) 0 off 1 on (manufacturer preferred compression) 2 V.42bis data compression 3 V.44bis data compression Other values are reserved.
<h_comp>	a numeric parameter that controls PDP header compression (refer 3GPP TS 44.065 [61] and 3GPP TS 25.323 [62]) 0 off 1 on (manufacturer preferred compression) 2 RFC1144 (applicable for SNDCP only) 3 RFC2507 4 RFC3095 (applicable for PDCP only) Other values are reserved.

NOTE

The <cid>s for network-initiated PDP contexts will have values outside the ranges indicated for the <cid> in the test form of the commands +CGDCONT and +CGDSCONT.

Example

```
AT+CGDSCONT=?
+CGDSCONT: (1-42),(1,2,3),"IP",(0-3),(0-4)
+CGDSCONT: (1-42),(1,2,3),"PPP",(0-3),(0-4)
+CGDSCONT: (1-42),(1,2,3),"IPV6",(0-3),(0-4)
+CGDSCONT: (1-42),(1,2,3),"IPV4V6",(0-3),(0-4)

OK
AT+CGDSCONT?
+CGDSCONT: 2,1,0,0

OK
AT+CGDSCONT=2,1
OK
```

8.2.6 AT+CGTFT Traffic Flow Template

This command allows the TE to specify a Packet Filter - PF for a Traffic Flow Template - TFT that is used in the GGSN in UMTS/GPRS and Packet GW in EPS for routing of packets onto different QoS flows towards the TE. The concept is further described in the 3GPP TS 23.060 [47]. A TFT consists of from one and up to 16 Packet Filters, each identified by a unique <packet filter identifier>. A Packet Filter also has an <evaluation precedence index> that is unique within all TFTs associated with all PDP contexts that are associated with the same PDP address.

AT+CGTFT Traffic Flow Template	
Test Command **AT+CGTFT=?**	Response **+CGTFT: <PDP_type>,**(list of supported **<packet filter identifier>s),**(list of supported **<evaluation precedence index>s),**(list of supported **<source address and subnet mask>s),**(list of supported **<protocol number (ipv4) / next header (ipv6)>s),**(list of supported **<destination port range>s),**(list of supported **<source port range>s),**(list of supported **<ipsec security parameter index (spi)>s),**(list of supported **<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>s),**(list of supported **<flow label (ipv6)>s)** **[<CR><LF>+CGTFT: <PDP_type>,**(list of supported **<packet filter identifier>s),**(list of supported **<evaluation precedence index>s),**(list of supported **<source address and subnet mask>s),**(list of supported **<protocol number (ipv4) / next header (ipv6)>s),**(list of supported **<destination port range>s),**(list of supported **<source port range>s),**(list of supported **<ipsec security parameter index (spi)>s),**(list of supported **<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>s),**(list of supported **<flow label (ipv6)>s)** **[...]]** **OK** or **ERROR**
Read Command **AT+CGTFT?**	Response **+CGTFT: [<cid>,<packet filter identifier>,<evaluation precedence index>,<source address and subnet mask>,<protocol number (ipv4) / next header (ipv6)>,<destination port range>,<source port range>,<ipsec security parameter index (spi)>,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label (ipv6)>** **[<CR><LF>+CGTFT: <cid>,<packet filter identifier>,<evaluation precedence index>,<source address and subnet**

	mask>,<protocol number (ipv4) / next header (ipv6)>,<destination port range>,<source port range>,<ipsec security parameter index (spi)>,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>,<flow label (ipv6)> **[...]]]** **OK** or **ERROR**
Write Command **AT+CGTFT=<cid>[,[<packet filter identifier>,<evaluation precedence index>[,<source address and subnet mask>[,<protocol number (ipv4) / next header (ipv6)>[,<destination port range>[,<source port range>[,<ipsec security parameter index (spi)>[,<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>[,<flow label (ipv6)>]]]]]]]]**	Response **OK** or **ERROR**
Execution Command **AT+CGTFT**	Response **OK** or **ERROR**

Defined Values

<cid>	a numeric parameter which specifies a particular PDP context definition (see the**AT+CGDCONT** and **AT+CGDSCONT** commands).
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack
<packet filter identifier>	a numeric parameter, value range from 1 to 16.
<evaluation precedence	a numeric parameter. The value range is from 0 to 255.

index>	
<source address and subnet mask>	string type The string is given as dot-separated numeric (0-255) parameters on the form: "a1.a2.a3.a4.m1.m2.m3.m4" for IPv4 or "a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16.m1.m2.m3.m4.m5.m6.m7.m8.m9.m10.m11.m12.m13.m14.m15.m16", for IPv6.
<protocol number (ipv4) / next header (ipv6)>	a numeric parameter, value range from 0 to 255.
<destination port range>	string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".
<source port range>	string type. The string is given as dot-separated numeric (0-65535) parameters on the form "f.t".
<ipsec security parameter index (spi)>	numeric value in hexadecimal format. The value range is from 00000000 to FFFFFFFF.
<type of service (tos) (ipv4) and mask / traffic class (ipv6) and mask>	string type. The string is given as dot-separated numeric (0-255) parameters on the form "t.m".
<flow label (ipv6)>	numeric value in hexadecimal format. The value range is from 00000 to FFFFF. Valid for IPv6 only.

Example

```
AT+CGTFT=?
+CGTFT:
"IP",(1-16),(0-255),,(0-255),(0-65535.0-65535),(0-65535.0-65535),(0-FFFFFFFF),(0-255.0-255),(0-FFFFF)
+CGTFT:
"PPP",(1-16),(0-255),,(0-255),(0-65535.0-65535),(0-65535.0-65535),(0-FFFFFFFF),(0-255.0-255),(0-FFFFF)
+CGTFT:
"IPV6",(1-16),(0-255),,(0-255),(0-65535.0-65535),(0-65535.0-65535),(0-FFFFFFFF),(0-255.0-255),(0-FFFFF)
+CGTFT:
"IPV4V6",(1-16),(0-255),,(0-255),(0-65535.0-65535),(0-65535.0-65535),(0-FFFFFFFF),(0-255.0-255),(0-FFFFF)

OK
AT+CGTFT?
+CGTFT: 2,1,0,"74.125.71.99.255.255.255.255",0,0.0,0.0,0,0.0,0

OK
AT+CGTFT=2,1,0,"74.125.71.99.255.255.255.255"
OK
```

8.2.7 AT+CGQREQ Quality of service profile (requested)

This command allows the TE to specify a Quality of Service Profile that is used when the MT sends an Activate PDP Context Request message to the network. A special form of the set command (AT+CGQREQ=<cid>) causes the requested profile for context number <cid> to become undefined.

AT+CGQREQ Quality of service profile (requested)	
Test Command **AT+CGQREQ=?**	Response **+CGQREQ: <PDP_type>,**(list of supported **<precedence>s),**(list of supported **<delay>s),**(list of supported **<reliability>s),**(list of supported **<peak>s),**(list of supported **<mean>s) [<CR><LF> +CGQREQ: <PDP_type>,**(list of supported **<precedence>s),**(list of supported **<delay>s),**(list of supported **<reliability>s),**(list of supported **<peak>s),**(list of supported **<mean>s)** **[…]]** **OK** or **ERROR**
Read Command **AT+CGQREQ?**	Response **+CGQREQ:** **[<cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[<CR><LF>** **+CGQREQ:** **<cid>,<precedence>,<delay>,<reliability.>,<peak>,<mean>[…]]]** **OK** or **ERROR**
Write Command **AT+CGQREQ=<cid>[,<precedence>[,<delay>[,<reliability>[,<peak>[,<mean>]]]]]**	Response **OK** or **ERROR**
Execution Command **AT+CGQREQ**	Response **OK** or **ERROR**

Defined Values

<cid>	A numeric parameter which specifies a particular PDP context

	definition (see AT+CGDCONT command). The range is from 1 to 42.
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack
<precedence>	A numeric parameter which specifies the precedence class: 0 network subscribed value 1 high priority 2 normal priority 3 low priority
<delay>	A numeric parameter which specifies the delay class: 0 network subscribed value 1 delay class 1 2 delay class 2 3 delay class 3 4 delay class 4
<reliability>	A numeric parameter which specifies the reliability class: 0 network subscribed value 1 Non real-time traffic,error-sensitive application that cannot cope with data loss 2 Non real-time traffic,error-sensitive application that can cope with infrequent data loss 3 Non real-time traffic,error-sensitive application that can cope with data loss, GMM/SM,and SMS 4 Real-time traffic,error-sensitive application that can cope with data loss 5 Real-time traffic error non-sensitive application that can cope with data loss
<peak>	A numeric parameter which specifies the peak throughput class: 0 network subscribed value 1 Up to 1000 (8 kbit/s) 2 Up to 2000 (16 kbit/s) 3 Up to 4000 (32 kbit/s) 4 Up to 8000 (64 kbit/s) 5 Up to 16000 (128 kbit/s) 6 Up to 32000 (256 kbit/s) 7 Up to 64000 (512 kbit/s) 8 Up to 128000 (1024 kbit/s) 9 Up to 256000 (2048 kbit/s)
<mean>	A numeric parameter which specifies the mean throughput class: 0 network subscribed value 1 100 (~0.22 bit/s)

	2 200 (~0.44 bit/s) 3 500 (~1.11 bit/s) 4 1000 (~2.2 bit/s) 5 2000 (~4.4 bit/s) 6 5000 (~11.1 bit/s) 7 10000 (~22 bit/s) 8 20000 (~44 bit/s) 9 50000 (~111 bit/s) 10 100000 (~0.22 kbit/s) 11 200000 (~0.44 kbit/s) 12 500000 (~1.11 kbit/s) 13 1000000 (~2.2 kbit/s) 14 2000000 (~4.4 kbit/s) 15 5000000 (~11.1 kbit/s) 16 10000000 (~22 kbit/s) 17 20000000 (~44 kbit/s) 18 50000000 (~111 kbit/s) 31 optimization

Example

```
AT+CGQREQ=?
+CGQREQ: "IP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQREQ: "PPP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQREQ:
"IPV6",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQREQ:
"IPV4V6",(0-3),(0-4),(0-5),(0-9),(0-18,31)

OK
AT+CGQREQ?
+CGQREQ:

OK
```

8.2.8 AT+CGEQREQ 3G quality of service profile (requested)

The test command returns values supported as a compound value.

The read command returns the current settings for each defined context for which a QOS was explicitly specified.

The write command allows the TE to specify a Quality of Service Profile for the context identified by the context identification parameter **<cid>** which is used when the MT sends an Activate PDP Context Request message to the network.

A special form of the write command, **AT+CGEQREQ=<cid>** causes the requested profile for context number **<cid>** to become undefined.

AT+CGEQREQ 3G quality of service profile (requested)	
Test Command AT+CGEQREQ=?	Response **+CGEQREQ: <PDP_type>,(list of supported <Traffic class>s),(list of supported <Maximum bitrate UL>s),(list of supported <Maximum bitrate DL>s),(list of supported <Guaranteed bitrate UL>s,(list of supported <Guaranteed bitrate DL>s),(list of supported <Delivery order>s),(list of supported <Maximum SDU size>s),(list of supported <SDU error ratio>s),(list of supported <Residual bit error Ratio>s),(list of supported <Delivery of erroneous SDUs>s),(list of Supported <Transfer delay>s),(list of supported <Traffic handling priority>s),(list of supported <Source statistics descriptor>s),(list of supported <Signaling indication flag>s)** **OK** or **ERROR**
Read Command AT+CGEQREQ?	Response **+CGEQREQ: [<cid>,<Traffic class>,<Maximum bitrate UL>,<Maximum bitrate DL>,<Guaranteed bitrate UL>,<Guaranteed bitrate DL>,<Delivery order>,<Maximum SDU size>,<SDU error ratio>,<Residual bit error ratio>,<Delivery of erroneous SDUs>,<Transfer delay>,<Traffic handling priority>,<Source statistics descriptor>,<Signaling indication flag>][<CR><LF>+CGEQREQ: <cid>,<Traffic class>,<Maximum bitrate UL>,<Ma ximum bitrate DL>,<Guaranteed bitrate UL>,<Guaranteed bitrate DL>,<Delivery order>,<Maximum SDU size>,<SDU error ratio>,<Residual bit error ratio>,<Delivery of erroneous SDUs>,<Transfer delay>,<Traffic handling priority>,<Source statistics descriptor>,<Signaling indication flag> […]]** **OK** or **ERROR**
Write Command AT+CGEQREQ=<cid>[,<Traf	Response **OK**

fic class>[,<Maximum bitrate UL>[,<Maximum bitrate DL>[,<Guaranteed bitrateUL>[,<Guaranteed bitrate DL>[,<Delivery order>[,<Maximum SDU size>[,<SDU error ratio>[,<Residual bit error ratio>[,<Delivery of erroneous SDUs>[,<Transfer delay>[,<Traffic handling priority>[,<Source statistics descriptor>[,<Signaling indication flag>]]]]]]]]]]]]]	or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+CGEQREQ**	Response **OK** or **ERROR**

Defined Values

<cid>	Parameter specifies a particular PDP context definition.The parameter is also used in other PDP context-related commands. The range is from 1 to 42.
<Traffic class>	0 conversational 1 streaming 2 interactive 3 background 4 subscribed value
<Maximum bitrate UL>	This parameter indicates the maximum number of kbits/s delivered to UMTS(up-link traffic)at a SAP. As an example a bitrate of 32kbit/s would be specified as 32(e.g. AT+CGEQREQ=…,32,…). The range is from 0 to 11520, but for sim82XX the range is from 0 to 5760. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Maximum bitrate DL>	This parameter indicates the maximum number of kbits/s delivered to UMTS(down-link traffic)at a SAP.As an example a bitrate of 32kbit/s would be specified as 32(e.g. AT+CGEQREQ=…,32,…). The range is from 0 to 42200. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Guaranteed bitrate UL>	This parameter indicates the guaranteed number of kbit/s delivered to UMTS(up-link traffic)at a SAP(provided that there is data to deliver).As an example a bitrate of 32kbit/s would be specified as

	32(e.g.AT+CGEQREQ=…,32,…). The range is from 0 to 11520, but for sim82XX the range is from 0 to 5760. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Guaranteed bitrate DL>	This parameter indicates the guaranteed number of kbit/s delivered to UMTS(up-link traffic)at a SAP(provided that there is data to deliver).As an example a bitrate of 32kbit/s would be specified as 32(e.g.AT+CGEQREQ=…,32,…). The range is from 0 to 42200. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Delivery order>	This parameter indicates whether the UMTS bearer shall provide in-sequence SDU delivery or not. 0 no 1 yes 2 subscribed value
<Maximum SDU size>	This parameter indicates the maximum allowed SDU size in octets. The range is from 0 to 1520. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<SDU error ratio>	This parameter indicates the target value for the fraction of SDUs lost or detected as erroneous.SDU error ratio is defined only for conforming traffic.As an example a target SDU error ratio of 5*10-3 would be specified as "5E3"(e.g.AT+CGEQREQ=..,"5E3",…). "0E0" subscribed value "1E2" "7E3" "1E3" "1E4" "1E5" "1E6" "1E1"
<Residual bit error ratio>	This parameter indicates the target value for the undetected bit error ratio in the delivered SDUs. If no error detection is requested,Residual bit error ratio indicates the bit error ratio in the delivered SDUs.As an example a target residual bit error ratio of 5*10-3 would be specified as "5E3"(e.g. AT+CGEQREQ=…,"5E3",..). "0E0" subscribed value "5E2" "1E2" "5E3" "4E3" "1E3" "1E4" "1E5"

	"1E6" "6E8"
<Delivery of erroneous SDUs>	This parameter indicates whether SDUs detected as erroneous shall be delivered or not. 0 no 1 yes 2 no detect 3 subscribed value
<Transfer delay>	This parameter indicates the targeted time between request to transfer an SDU at one SAP to its delivery at the other SAP,in milliseconds. The range is from 0 and 100 to 4000. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Traffic handling priority>	This parameter specifies the relative importance for handling of all SDUs belonging to the UMTS Bearer compared to the SDUs of the other bearers. The range is from 0 to 3. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Source statistics descriptor>	This parameter indicates profile parameter that Source statistics descriptor for requested UMTS QoS The range is from 0 to 1. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Signaling indication flag>	This parameter indicates Signaling flag. The range is from 0 to 1 The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack

Example

```
AT+CGEQREQ=?
+CGEQREQ:
"IP",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3","1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4000),(0-3),(0-1),(0-1)
+CGEQREQ:
"PPP",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3","1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4000),(0-3),(0-1),(0-1)
+CGEQREQ:
```

"IPV6",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3","1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4000),(0-3),(0-1),(0-1)
+CGEQREQ:
"IPV4V6",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3","1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4000),(0-3),(0-1),(0-1)

OK
AT+CGEQREQ?
+CGEQREQ:

OK

8.2.9 AT+CGQMIN Quality of service profile (minimum acceptable)

This command allows the TE to specify a minimum acceptable profile which is checked by the MT against the negotiated profile returned in the Activate PDP Context Accept message.A special form of the set command.**AT+CGQMIN=<cid>** causes the minimum acceptable profile for context number **<cid>** to become undefined.

AT+CGQMIN Quality of service profile (minimum acceptable)	
Test Command **AT+CGQMIN=?**	Response **+CGQMIN: <PDP_type>,**(list of supported **<precedence>s),**(list of supported **<delay>s),**(list of supported **<reliability>s),**(list of supported **<peak>s),**(list of supported **<mean>s) [<CR><LF> +CGQMIN: <PDP_type>,**(list of supported **<precedence>s),**(list of supported **<delay>s),**(list of supported **<reliability>s),**(list of supported **<peak>s),**(list of supported **<mean>s)[…]]** **OK** or **ERROR**
Read Command **AT+CGQMIN?**	Response **+CGQMIN:** **[<cid>,<precedence>,<delay>,<reliability>,<peak>,<mean>[<CR><LF>+CGQMIN:** **<cid>,<precedence>,<delay>,<reliability.>,<peak>,<mean>** **[…]]]** **OK**

	or **ERROR**
Write Command **AT+CGQMIN=<cid>[,<precedence>[,<delay>[,<reliability>[,<peak> [,<mean>]]]]]**	Response **OK** or **ERROR**
Execution Command **AT+CGQMIN**	Response **OK** or **ERROR**

Defined Values

<cid>	A numeric parameter which specifies a particular PDP context definition (see AT+CGDCONT command). The range is from 1 to 42.
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack
<precedence>	A numeric parameter which specifies the precedence class: 0 network subscribed value 1 high priority 2 normal priority 3 low priority
<delay>	A numeric parameter which specifies the delay class: 0 network subscribed value 1 delay class 1 2 delay class 2 3 delay class 3 4 delay class 4
<reliability>	A numeric parameter which specifies the reliability class: 0 network subscribed value 1 Non real-time traffic,error-sensitive application that cannot cope with data loss 2 Non real-time traffic,error-sensitive application that can cope with infrequent data loss 3 Non real-time traffic,error-sensitive application that can cope with data loss, GMM/- SM,and SMS 4 Real-time traffic,error-sensitive application that can cope with data loss 5 Real-time traffic error non-sensitive application that can cope with

	data loss
<peak>	A numeric parameter which specifies the peak throughput class: 0 network subscribed value 1 Up to 1000 (8 kbit/s) 2 Up to 2000 (16 kbit/s) 3 Up to 4000 (32 kbit/s) 4 Up to 8000 (64 kbit/s) 5 Up to 16000 (128 kbit/s) 6 Up to 32000 (256 kbit/s) 7 Up to 64000 (512 kbit/s) 8 Up to 128000 (1024 kbit/s) 9 Up to 256000 (2048 kbit/s)
<mean>	A numeric parameter which specifies the mean throughput class: 0 network subscribed value 1 100 (~0.22 bit/s) 2 200 (~0.44 bit/s) 3 500 (~1.11 bit/s) 4 1000 (~2.2 bit/s) 5 2000 (~4.4 bit/s) 6 5000 (~11.1 bit/s) 7 10000 (~22 bit/s) 8 20000 (~44 bit/s) 9 50000 (~111 bit/s) 10 100000 (~0.22 kbit/s) 11 200000 (~0.44 kbit/s) 12 500000 (~1.11 kbit/s) 13 1000000 (~2.2 kbit/s) 14 2000000 (~4.4 kbit/s) 15 5000000 (~11.1 kbit/s) 16 10000000 (~22 kbit/s) 17 20000000 (~44 kbit/s) 18 50000000 (~111 kbit/s) 31 optimization

Example

```
AT+CGQMIN=?
+CGQMIN: "IP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQMIN: "PPP",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQMIN: "IPV6",(0-3),(0-4),(0-5),(0-9),(0-18,31)
+CGQMIN:
"IPV4V6",(0-3),(0-4),(0-5),(0-9),(0-18,31)

OK
```

AT+CGQMIN? +CGQMIN: OK

8.2.10 AT+CGEQMIN 3G quality of service profile (minimum acceptable)

The test command returns values supported as a compound value.

The read command returns the current settings for each defined context for which a QOS was explicitly specified.

AT+CGEQMIN 3G quality of service profile (minimum acceptable)	
Test Command AT+CGEQMIN=?	Response **+CGEQMIN: <PDP_type>,(**list of supported **<Traffic class>s),(**list of supported **<Maximum bitrate UL>s),(**list of supported **<Maximum bitrate DL>s),(**list of supported **<Guaranteed bitrate UL>s,(**list of supported **<Guaranteed bitrate DL>s),(**list of supported **<Delivery order>s),(**list of supported **<Maximum SDU size>s),(**list of supported **<SDU error ratio>s),(**list of supported **<Residual bit error ratio>s),(**list of supported **<Delivery of erroneous SDUs>s),(**list of supported **<Transfer delay>s),(**list of supported **<Traffic handling priority>s),(**list of supported **<Source statistics descriptor>s),(**list of supported **<Signaling indication flag>s)** **OK** or **ERROR**
Read Command AT+CGEQMIN?	Response **+CGEQMIN: [<cid>,<Traffic class>,<Maximum bitrate UL>,<Ma ximum bitrate DL>,<Guaranteed bitrate UL>,<Guaranteed bitrate DL>,<Delivery order>,<Maximum SDU size>,<SDU error ratio>,<Residual bit error ratio>,<Delivery of erroneous SDUs>,<Transfer delay>,<Traffic handling priority>,<Source statistics descriptor>,<Signaling indication flag>][<CR><LF>+CGEQMIN: <cid>,<Traffic class>,<Maximum bitrate UL>,<Ma ximum bitrate DL>,<Guaranteed bitrate UL>,<Guaranteed bitrate DL>,<Delivery order>,<Maximum SDU size>,<SDU error ratio>,<Residual bit error ratio>,<Delivery of erroneous SDUs>,<Transfer delay>,<Traffic handling priority>,<Source statistics descriptor>,<Signaling indication**

	flag>[…]] **OK** or **ERROR**
Write Command **AT+CGEQMIN=<cid>[,<Traffic class>[,<Maximum bitrate UL>[,<Maximum bitrate DL>[,<Guaranteed bitrate UL>[,<Guaranteed bitrate DL>[,<Delivery order>[,<Maximum SDU size>[,<SDU error ratio>[,<Residual bit error ratio>[,<Delivery of erroneous SDUs>[,<Transfer delay>[,<Traffic handling priority>[,Source statistics descriptor[,Signaling indication flag]]]]]]]]]]]]]]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+CGEQMIN**	Response **OK** or **ERROR**

Defined Values

<cid>	Parameter specifies a particular PDP context definition.The parameter is also used in other PDP context-related commands. The range is from 1 to 42.
<Traffic class>	0 conversational 1 streaming 2 interactive 3 background 4 subscribed value
<Maximum bitrate UL>	This parameter indicates the maximum number of kbits/s delivered to UMTS(up-link traffic)at a SAP.As an example a bitrate of 32kbit/s would be specified as 32(e.g. AT+CGEQMIN=…,32,…). The range is from 0 to 11520, but for sim82XX the range is from 0 to 5760. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Maximum bitrate DL>	This parameter indicates the maximum number of kbits/s delivered to

	UMTS(down-link traffic)at a SAP.As an example a bitrate of 32kbit/s would be specified as 32(e.g. AT+CGEQMIN=…,32,…). The range is from 0 to 42200. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Guaranteed bitrate UL>	This parameter indicates the guaranteed number of kbit/s delivered to UMTS(up-link traffic)at a SAP(provided that there is data to deliver).As an example a bitrate of 32kbit/s would be specified as 32(e.g.AT+CGEQMIN=…,32,…). The range is from 0 to 11520, but for sim82XX the range is from 0 to 5760.The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Guaranteed bitrate DL>	This parameter indicates the guaranteed number of kbit/s delivered to UMTS(down-link traffic)at a SAP(provided that there is data to deliver).As an example a bitrate of 32kbit/s would be specified as 32(e.g.AT+CGEQMIN=…,32,…). The range is from 0 to 42200. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Delivery order>	This parameter indicates whether the UMTS bearer shall provide in-sequence SDU delivery or not. 0 no 1 yes 2 subscribed value
<Maximum SDU size>	This parameter indicates the maximum allowed SDU size inoctets. The range is from 0 to 1520. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<SDU error ratio>	This parameter indicates the target value for the fraction of SDUs lost or detected as erroneous.SDU error ratio is defined only for conforming traffic.As an example a target SDU error ratio of 5*10-3 would be specified as "5E3"(e.g.AT+CGEQMIN=..,"5E3",…). "0E0" subscribed value "1E2" "7E3" "1E3" "1E4" "1E5" "1E6" "1E1"
<Residual bit error ratio>	This parameter indicates the target value for the undetected bit error ratio in the delivered SDUs. If no error detection is requested,Residual bit error ratio indicates the bit error ratio in the delivered SDUs.As an example a target residual bit error ratio of 5*10-3 would be specified as "5E3"(e.g. AT+CGEQMIN=…,"5E3",..). "0E0" subscribed value

	"5E2" "1E2" "5E3" "4E3" "1E3" "1E4" "1E5" "1E6" "6E8"
<Delivery of erroneous SDUs>	This parameter indicates whether SDUs detected as erroneous shall be delivered or not. 0 no 1 yes 2 no detect 3 subscribed value
<Transfer delay>	This parameter indicates the targeted time between request to transfer an SDU at one SAP to its delivery at the other SAP,in milliseconds. The range is from 0,100 to 4000. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Traffic handling priority>	This parameter specifies the relative importance for handling of all SDUs belonging to the UMTS Bearer compared to the SDUs of the other bearers. The range is from 0 to 3. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Source statistics descriptor>	This parameter indicates profile parameter that Source statistics descriptor for requested UMTS QoS The range is from 0 to 1. The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<Signaling indication flag>	This parameter indicates Signaling flag. The range is from 0 to 1 The default value is 0. If the parameter is set to '0' the subscribed value will be requested.
<PDP_type>	(Packet Data Protocol type) a string parameter which specifies the type of packet data protocol. IP Internet Protocol PPP Point to Point Protocol IPV6 Internet Protocol Version 6 IPV4V6 Dual PDN Stack

Example

AT+CGEQMIN=?
+CGEQMIN:
"IP",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3","1

```
E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-40
00),(0-3),(0,1),(0,1)
+CGEQMIN:
"PPP",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3",
"1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4
000),(0-3),(0,1),(0,1)
+CGEQMIN:
"IPV6",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E3",
"1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,100-4
000),(0-3),(0,1),(0,1)
+CGEQMIN:
"IPV4V6",(0-4),(0-11520),(0-42200),(0-11520),(0-42200),(0-2),(0-1520),("0E0","1E1","1E2","7E3","1E
3","1E4","1E5","1E6"),("0E0","5E2","1E2","5E3","4E3","1E3","1E4","1E5","1E6","6E8"),(0-3),(0,10
0-4000),(0-3),(0,1),(0,1)

OK
AT+CGEQMIN?
+CGEQMIN:

OK
```

8.2.11 AT+CGDATA Enter data state

The command causes the MT to perform whatever actions are necessary to establish communication between the TE and the network using one or more Packet Domain PDP types. This may include performing a PS attach and one or more PDP context activations.

AT+CGDATA Enter data state	
Test Command **AT+CGDATA=?**	Response **+CGDATA: (**list of supported **<L2P>s)** **OK** or **ERROR**
Write Command **AT+CGDATA=<L2P>,<cid>**	Response **NO CARRIER** or **CONNECT [<text>]** or **ERROR** or

	+CME ERROR: <err>

Defined Values

<L2P>	A string parameter that indicates the layer 2 protocol to be used between the TE and MT. PPP Point-to-point protocol for a PDP such as IP
<text>	CONNECT result code string; the string formats please refer ATX/AT\V/AT&E command.
<cid>	A numeric parameter which specifies a particular PDP context definition (see AT+CGDCONT command). 1…42

Example

AT+CGDATA=? **+CGDATA: ("PPP")** **OK** AT+CGDATA="PPP",1 **CONNECT 115200**

8.2.12 AT+CGPADDR Show PDP address

The write command returns a list of PDP addresses for the specified context identifiers.

AT+CGPADDR Show PDP address	
Test Command AT+CGPADDR=?	Response **[+CGPADDR: (list of defined <cid>s)]** **OK** or **ERROR**
Write Command AT+CGPADDR=<cid>[,<cid>[,…]]	Response **+CGPADDR: <cid>,<PDP_addr>** **OK** or **OK** or **ERROR**

	or **+CME ERROR: <err>**
Execution Command AT+CGPADDR	Response **[+CGPADDR: <cid>,<PDP_addr>]** **+CGPADDR: <cid>,<PDP_addr>[...]]]** **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<cid>	A numeric parameter which specifies a particular PDP context definition (see AT+CGDCONT command). If no <cid> is specified, the addresses for all defined contexts are returned. 1…42
<PDP_addr>	A string that identifies the MT in the address space applicable to the PDP. The address may be static or dynamic. For a static address, it will be the one set by the AT+CGDCONT command when the context was defined. For a dynamic address it will be the one assigned during the last PDP context activation that used the context definition referred to by <cid>. <PDP_addr> is omitted if none is available.

Example

```
AT+CGPADDR=?
+CGPADDR: (1)

OK
AT+CGPADDR=1
+CGPADDR: 1,10.237.48.122

OK
```

8.2.13 AT+CGCLASS GPRS mobile station class

This command is used to set the MT to operate according to the specified GPRS mobile class.

AT+CGCLASS GPRS mobile station class	
Test Command AT+CGCLASS=?	Response **+CGCLASS: (**list of supported **<class>s)** **OK** or **ERROR**
Read Command AT+CGCLASS?	Response **+CGCLASS: <class>** **OK** or **ERROR**
Write Command AT+CGCLASS=<class>	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command Set default value: AT+CGCLASS	Response **OK** or **ERROR**

Defined Values

<class>	A string parameter which indicates the GPRS mobile class (in descending order of functionality) A class A (highest)

Example

```
AT+CGCLASS=?
+CGCLASS: ("A")

OK
AT+CGCLASS?
+CGCLASS: "A"

OK
```

8.2.14 AT+CGEREP GPRS event reporting

The write command enables or disables sending of unsolicited result codes, "+CGEV" from MT to TE in the case of certain events occurring in the Packet Domain MT or the network. <mode> controls the processing of unsolicited result codes specified within this command. <bfr> controls the effect on buffered codes when <mode> 1 or 2 is entered. If a setting is not supported by the MT, **ERROR**or **+CME ERROR**: is returned.

Read command returns the current <mode> and buffer settings.

Test command returns the modes and buffer settings supported by the MT as compound values.

AT+CGEREP GPRS event reporting	
Test Command **AT+CGEREP=?**	Response **+CGEREP:** (list of supported **<mode>s),(**list of supported **<bfr>s)** **OK** or **ERROR**
Read Command **AT+CGEREP?**	Response **+CGEREP: <mode>,<bfr>** **OK** or **ERROR**
Write Command **AT+CGEREP=<mode>[,<bfr>]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command Set default value: **AT+CGEREP**	Response **OK** or **ERROR**

Defined Values

<mode>	0 – buffer unsolicited result codes in the MT; if MT result code buffer is full, the oldest ones can be discarded. No codes are forwarded to the TE. 1 – discard unsolicited result codes when MT-TE link is reserved (e.g. in on-line data mode); otherwise forward them directly to the TE.

	2 – buffer unsolicited result codes in the MT when MT-TE link is reserved (e.g. in on-line data mode) and flush them to the TE when MT-TE link becomes available; otherwise forward them directly to the TE.
<bfr>	0 – MT buffer of unsolicited result codes defined within this command is cleared when <mode> 1 or 2 is entered. 1 – MT buffer of unsolicited result codes defined within this command is flushed to the TE when <mode> 1 or 2 is entered (OK response shall be given before flushing the codes).

Example

AT+CGEREP=? +CGEREP: (0-2),(0-1) OK AT+CGEREP? +CGEREP: 0,0 OK AT+CGEREP=2 OK	 If set the <mode> to 2, then sending unsolicited result codes, "+CGEV" from MT to TE in the case of certain events occurring in the Packet Domain MT or the network.
AT+CGACT=1,1 OK +CGEV: PDN ACT 1 AT+CGACT=0,1 OK +CGEV: PDN DEACT 1	 When links set to available, "+CGEV: PDN ACT" will be sent; otherwise when link set to deactive, then sending "+CGEV: PDN DEACT".

8.2.15 AT+CGAUTH Set type of authentication for PDP-IP connections of GPRS

This command is used to set type of authentication for PDP-IP connections of GPRS.

AT+CGAUTH	Set type of authentication for PDP-IP connections of GPRS
Test Command AT+CGAUTH=?	Response **+CGAUTH: ,,127,127(for CDMA1x-EvDo only)** **+CGAUTH:** (range of supported **<cid>s),(**list of supported **<auth_type>s),,**

	OK or **ERROR** or **+CME ERROR: <err>**
Read Command **AT+CGAUTH?**	Response **+CGAUTH: <cid>,<auth_type>[,<user>,<passwd>]<CR><LF>** **+CGAUTH: <cid>,<auth_type>[,<user>,<passwd>]<CR><LF>** **…** **OK** or **ERROR** or **+CME ERROR: <err>**
Write Command **AT+CGAUTH=<cid>[,<auth_type>[,<passwd>[,<user>]]]**	Response **OK** or **ERROR** or **+CME ERROR: <err>**
Execution Command **AT+CGAUTH**	Response **OK** or **ERROR** or **+CME ERROR: <err>**

Defined Values

<cid>	Parameter specifies a particular PDP context definition. This is also used in other PDP context-related commands. 1…42
<auth_type>	Indicate the type of authentication to be used for the specified context. If CHAP is selected another parameter <passwd> needs to be specified. If PAP is selected two additional parameters <passwd> and <user> need to specified. 0 none 1 PAP 2 CHAP 3 PAP or CHAP
<passwd>	Parameter specifies the password used for authentication.
<user>	Parameter specifies the user name used for authentication.

Example

```
AT+CGAUTH=?
+CGAUTH: ,,127,127(for CDMA1x-EvDo only)
+CGAUTH: (1-42),(0-3),127,127

OK
AT+CGAUTH=1,1,"123","SIMCOM"
OK
```

8.3 Summary of Unsolicited Result Codes

Unsolicited codes	Description
+CGEV: PDN ACT<cid>	When the PDP in <cid> channel is activated , this unsolicited result code will be reported
+CGEV: REJECT <PDP_type>,<PDP_addr>	A network request for PDP context activation occurred when the MT was unable to report it to the TE with a +CRING unsolicited result code and was automatically rejected.
+CGEV: NW REACT <PDP_type>,<PDP_addr>,[<cid>]	The network has requested a context reactivation. The <cid> that was used to reactivate the context is provided if known to the MT.
+CGEV: NW DEACT <PDP_type>,<PDP_addr>,[<cid>]	The network has forced a context deactivation. The <cid> that was used to activate the context is provided if known to the MT.
+CGEV: ME DEACT <PDP_type>,<PDP_addr>,[<cid>]	The mobile equipment has forced a context deactivation. The <cid> that was used to activate the context is provided if known to the MT.
+CGEV: NW DETACH	The network has forced a Packet Domain detach. This implies that all active contexts have been deactivated. These are not reported separately.
+CGEV: ME DETACH	The mobile equipment has forced a Packet Domain detach. This implies that all active contexts have been deactivated. These are not reported

	separately.
+CGEV: NW CLASS <class>	The network has forced a change of MS class. The highest available class is reported (see AT+CGCLASS).
+CGEV: ME CLASS <class>	The mobile equipment has forced a change of MS class. The highest available class is reported (see AT+CGCLASS).

9 AT Commands for SMS

9.1 Overview of AT Commands for SMS

Command	Description
AT+CSMS	Select message service
AT+CPMS	Preferred message storage
AT+CMGF	Select SMS message format
AT+CSCA	SMS service center address
AT+CSCB	Select cell broadcast message indication
AT+CSMP	Set text mode parameters
AT+CSDH	Show text mode parameters
AT+CNMA	New message acknowledgement to ME/TA
AT+CNMI	New message indications to TE
AT+CGSMS	Select service for MO SMS messages
AT+CMGL	List SMS messages from preferred store
AT+CMGR	Read message
AT+CMGS	Send message
AT+CMSS	Send message from storage
AT+CMGW	Write message to memory
AT+CMGD	Delete message
AT+CMGMT	Change message status
AT+CMVP	Set message valid period
AT+CMGRD	Read and delete message
AT+CMGSEX	Send message
AT+CMSSEX	Send multi messages from storage

9.2 Detailed Description of AT Commands for SMS

9.2.1 AT+CSMS Select message service

This command is used to select messaging service <service>.
Note: This command not support in CDMA/EVDO mode

AT+CSMS Select message service	
Test Command **AT+CSMS=?**	Response a) **+CSMS:** (range of supported **<service>s)** **OK** b)If failed: **ERROR**
Read Command **AT+CSMS?**	Response **+CSMS: <service>,<mt>,<mo>,<bm>** **OK**
Write Command **AT+CSMS=<service>**	Response a) **+CSMS: <mt>,<mo>,<bm>** **OK** b)If failed: **+CMS ERROR: <err>**

Defined Values

<service>	0 SMS at command is compatible with GSM phase 2. 1 SMS at command is compatible with GSM phase 2+.
<mt>	Mobile terminated messages: 0 type not supported. 1 type supported.
<mo>	Mobile originated messages: 0 type not supported. 1 type supported1 SMS at command is compatible with GSM phase 2+.
<bm>	Broadcast type messages: 0 type not supported. 1 type supported.

Example

```
AT+CSMS=0
```

+CSMS: 1,1,1

OK

9.2.2 AT+CPMS Preferred message storage

This command is used to select memory storages <mem1>,<mem2> and <mem3> to be used for reading, writing, etc.

AT+CPMS Preferred message storage	
Test Command **AT+CPMS=?**	Response a) **+CPMS: (**list of supported **<mem1>s),(**list of supported **<mem2>s),(**list of supported **<mem3>s)** **OK** b)If failed: **ERROR**
Read Command **AT+CPMS?**	Response **+CPMS: <mem1>,<used1>,<total1>,<mem2>,<used2>,<total2>,<mem3>,<used3>,<total3>** **OK**
Write Command **AT+CPMS=<mem1>[,<mem2>[,<mem3>]]**	Response a) **+CPMS: <used1>,<total1>,<used2>,<total2>,<used3>,<total3>** **OK** b)If failed: **+CMS ERROR: <err>**

Defined Values

<mem1>	String type, memory from which messages are read and deleted (commands List Messages AT+CMGL, Read Message AT+CMGR and Delete Message AT+CMGD). "ME" and "MT" FLASH message storage "SM" SIM message storage "SR" Status report storage (not used in CDMA/EVDO mode)

<mem2>	String type, memory to which writing and sending operations are made (commands Send Message from Storage AT+CMSS and Write Message to Memory AT+CMGW). "ME" and "MT" FLASH message storage "SM" SIM message storage
<mem3>	String type, memory to which received SMS is preferred to be stored (unless forwarded directly to TE; refer command New Message Indications AT+CNMI). "ME" FLASH message storage "SM" SIM message storage GSM phase 2+.
<usedX>	Integer type, number of messages currently in <memX>.
<totalX>	Integer type, total number of message locations in <memX>.

Example

```
AT+CPMS=?
+CPMS: ("ME","MT","SM","SR"),("ME","MT","SM"),("ME","SM")

OK
AT+CPMS?
+CPMS: "ME", 0,23,"ME", 0,23,"ME", 0,23

OK
AT+CPMS="SM","SM","SM"
+CPMS: 3,50,3,50,3,50

OK
```

9.2.3 AT+CMGF Select SMS message format

This command is used to specify the input and output format of the short messages.

AT+CMGF Select SMS message format	
Test Command **AT+CMGF=?**	Response a) **+CMGF: (**range of supported **<mode>s)** **OK** b)If failed: **ERROR**
Read Command	Response

AT+CMGF?	a) **+CMGF: <mode>** **OK** b)If failed: **ERROR**
Write Command AT+CMGF=<mode>	Response a) **OK** b)If failed: **ERROR**
Execution Command AT+CMGF	Response a)Set default value (<mode>=0): **OK** b)If failed: **ERROR**

Defined Values

<mode>	0 PDU mode 1 Text mode

Example

```
AT+CMGF=1
OK
```

9.2.4 AT+CSCA SMS service center address

This command is used to update the SMSC address, through which mobile originated SMS are transmitted.
Note: This command not support in CDMA/EVDO mode

AT+CSCA SMS service center address	
Test Command AT+CSCA=?	Response **OK**
Read Command AT+CSCA?	Response **+CSCA: <sca>,<tosca>** **OK**
Write Command AT+CSCA=<sca>[,<tosca>]	Response **OK**

Defined Values

<sca>	Service Center Address, value field in string format, BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set (refer to command AT+CSCS), type of address given by <tosca>.
<tosca>	SC address Type-of-Address octet in integer format, when first character of <sca> is + (IRA 43) default is 145, otherwise default is 129.

Example

```
AT+CSCA="+8613012345678"
OK
AT+CSCA?
+CSCA: "+8613012345678",145

OK
```

9.2.5 AT+CSCB Select cell broadcast message indication

The test command returns the supported <mode>s as a compound value.
The read command displays the accepted message types.
Depending on the <mode> parameter, the write command adds or deletes the message types accepted.
Note: This command not support in CDMA/EVDO mode

AT+CSCB Select cell broadcast message indication	
Test Command AT+CSCB=?	Response a) **+CSCB: (**range of supported **<mode>s)** **OK** b)If failed: **ERROR**
Read Command AT+CSCB?	Response a) **+CSCB: <mode>,<mids>,<dcss>** **OK** b)If failed:

	ERROR
Write Command **AT+CSCB=<mode>[,<mids>[,<dcss>]]**	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<mode>	0 message types specified in <mids> and <dcss> are accepted. 1 message types specified in <mids> and <dcss> are not accepted.
<mids>	String type; all different possible combinations of CBM message identifiers.
<dcss>	String type; all different possible combinations of CBM data coding schemes(default is empty string)

Example

```
AT+CSCB=?
+CSCB: (0-1)

OK
```

9.2.6 AT+CSMP Set text mode parameters

This command is used to select values for additional parameters needed when SM is sent to the network or placed in storage when text format message mode is selected.
Note: This command not support in CDMA/EVDO mode

AT+CSMP Set text mode parameters	
Test Command **AT+CSMP=?**	Response **OK**
Read Command **AT+CSMP?**	Response **+CSMP: <fo>,<vp>,<pid>,<dcs>** **OK**
Write Command **AT+CSMP=[<fo>[,<vp>[,<pid>[,<dcs>]]]]**	Response a) **OK**

	b)If failed: **ERROR**

Defined Values

<fo>	Depending on the Command or result code: first octet of GSM 03.40 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default 2) in integer format. SMS status report is supported under text mode if <fo> is set to 49.
<vp>	Depending on SMS-SUBMIT <fo> setting: GSM 03.40,TP-Validity-Period either in integer format (default 167), in time-string format, or if is supported, in enhanced format (hexadecimal coded string with quotes),(<vp> is in range 0... 255).
<pid>	GSM 03.40 TP-Protocol-Identifier in integer format (default 0).
<dcs>	GSM 03.38 SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme in integer format depending on the command or result code.

Example

```
AT+CSMP=17,23,64,244
OK
```

9.2.7 AT+CSDH Show text mode parameters

This command is used to control whether detailed header information is shown in text mode result codes.
Note: This command not support in CDMA/EVDO mode

AT+CSDH Show text mode parameters	
Test Command AT+CSDH=?	Response a) **+CSDH: (**range of supported **<show>s)** **OK** b)If failed: **ERROR**
Read Command AT+CSDH?	Response **+CSDH: <show>** **OK**
Write Command	Response

AT+CSDH=<show>	a) **OK** b)If failed: **ERROR**
Execution Command AT+CSDH	Response a)Set default value (<show>=0): **OK** b)If failed: **ERROR**

Defined Values

<show>	0 do not show header values defined in commands AT+CSCA and AT+CSMP (<sca>,<tosca>,<fo>,<vp>,<pid> and <dcs>) nor <length>,<toda>or<tooa> in +CMT, AT+CMGL, AT+CMGR result codes for SMS-DELIVERs and SMS-SUBMITs in text mode; for SMS-COMMANDs in AT+CMGR result code, do not show <pid>,<mn>,<da>,<toda>,<length>or<data> 1 show the values in result codes

Example

```
AT+CSDH=1
OK
```

9.2.8 AT+CNMA New message acknowledgement to ME/TA

This command is used to confirm successful receipt of a new message (SMS-DELIVER or SMS-STATUSREPORT) routed directly to the TE. If ME does not receive acknowledgement within required time (network timeout), it will send RP-ERROR to the network.
Note: This command not support in CDMA/EVDO mode

AT+CNMA New message acknowledgement to ME/TA	

Test Command
AT+CNMA=?
	Response
	if text mode(AT+CMGF=1):
	OK
	if PDU mode (AT+CMGF=0):
	+CNMA: (list of supported **<n>s)**
	
	OK

Write Command	Response

AT+CNMA=<n>	a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <ERR>**
Execution Command **AT+CNMA**	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<n>	Parameter required only for PDU mode. 0 Command operates similarly as execution command in text mode. 1 Send positive (RP-ACK) acknowledgement to the network. Accepted only in PDU mode. 2 Send negative (RP-ERROR) acknowledgement to the network. Accepted only in PDU mode.

Example

AT+CNMI=1,2,0,0,0
OK
+CMT: "1380022xxxx","","02/04/03,11:06:38+32"<CR><LF>
Testing
(receive new short message)
AT+CNMA(send ACK to the network)
OK
AT+CNMA
+CMS ERROR: 340
(the second time return error, it needs ACK only once)

9.2.9 AT+CNMI New message indications to TE

This command is used to select the procedure how receiving of new messages from the network is indicated to the TE when TE is active, e.g. DTR signal is ON. If TE is inactive (e.g. DTR signal is OFF). If set

<mt> = 3 or<ds> = 1, make sure <mode> = 1, If set <mt>=2, make sure <mode>=1 or 2, otherwise it will return error.

AT+CNMI New message indications to TE	
Test Command **AT+CNMI=?**	Response **+CNMI:** (list of supported **<mode>s),(**list of supported **<mt>s),(**list of supported **<bm>s),(**list of supported **<ds>s),(**list of supported **<bfr>s)**
	OK
Read Command **AT+CNMI?**	Response **+CNMI: <mode>,<mt>,<bm>,<ds>,<bfr>** **OK**
Write Command **AT+CNMI=<mode>[,<mt>[,<bm>[,<ds> [,<bfr>]]]]**	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**
Execution Command **AT+CNMI**	Response Set default value: **OK**

Defined Values

<mode>	0 Buffer unsolicited result codes in the TA. If TA result code buffer is full, indications can be buffered in some other place or the oldest indications may be discarded and replaced with the new received indications. 1 Discard indication and reject new received message unsolicited result codes when TA-TE link is reserved (e.g. in on-line data mode). Otherwise forward them directly to the TE. 2 Buffer unsolicited result codes in the TA when TA-TE link is reserved (e.g. in on-line data mode) and flush them to the TE after reservation. Otherwise forward them directly to the TE.
<mt>	The rules for storing received SMS depend on its data coding scheme, preferred memory storage (AT+CPMS) setting and this value: 0 No SMS-DELIVER indications are routed to the TE. 1 If SMS-DELIVER is stored into ME/TA, indication of the memory location is routed to the TE using unsolicited result code: **+CMTI: <mem3>,<index>.**

	2 SMS-DELIVERs (except class 2 messages and messages in the message waiting indication group (store message)) are routed directly to the TE using unsolicited result code: **+CMT: [<alpha>],<length><CR><LF><pdu>(PDU mode enabled); or** **+CMT:** **<oa>,[<alpha>],<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]** **<CR><LF><data>** (text mode enabled, about parameters in italics, refer command Show Text Mode Parameters AT+CSDH). 3 Class 3 SMS-DELIVERs are routed directly to TE using unsolicited result codes defined in <mt>=2. Messages of other data coding schemes result in indication as defined in <mt>=1.
<bm>	（not used in CDMA/EVDO mode） The rules for storing received CBMs depend on its data coding scheme, the setting of Select CBM Types (AT+CSCB) and this value: 0 No CBM indications are routed to the TE. 2 New CBMs are routed directly to the TE using unsolicited result code: **+CBM: <length><CR><LF><pdu>(PDU mode enabled); or** **+CBM: <sn>,<mid>,<dcs>,<page>,<pages><CR><LF><data> (text mode enabled)**
<ds>	（not used in CDMA/EVDO mode） 0 No SMS-STATUS-REPORTs are routed to the TE. 1 SMS-STATUS-REPORTs are routed to the TE using unsolicited result code: **+CDS: <length><CR><LF><pdu>(PDU mode enabled); or** **+CDS: <fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>** **(text mode enabled)** 2 If SMS-STATUS-REPORT is stored into ME/TA, indication of the memory location is routed to the TE using unsolicited result code: **+CDSI: <mem3>,<index>.**
<bfr>	0 TA buffer of unsolicited result codes defined within this command is flushed to the TE when <mode> 1 to 2 is entered (**OK** response shall be given before flushing the codes). 1 TA buffer of unsolicited result codes defined within this command is cleared when <mode> 1 to 2 is entered.

Example

AT+CNMI=2,1 (unsolicited result codes after received messages.)
OK

9.2.10 AT+CGSMS Select service for MO SMS messages

The write command is used to specify the service or service preference that the MT will use to send MO SMS messages.
The test command is used for requesting information on which services and service preferences can be set by using the AT+CGSMS write command
The read command returns the currently selected service or service preference.
Note: This command not support in CDMA/EVDO mode

AT+CGSMS Select service for MO SMS messages	
Test Command **AT+CGSMS=?**	Response **+CGSMS: (**range of supported **<service>s)** **OK**
Read Command **AT+CGSMS?**	Response **+CGSMS: <service>** **OK**
Write Command **AT+CGSMS=<service>**	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<service>	A numeric parameter which indicates the service or service preference to be used 0 GPRS(value is not really supported and is internally mapped to 2) 1 circuit switched(value is not really supported and is internally mapped to 3) 2 GPRS preferred (use circuit switched if GPRS not available) 3 circuit switched preferred (use GPRS if circuit switched not available)

Example

```
AT+CGSMS?
+CGSMS: 3
```

OK

9.2.11 AT+CMGL List SMS messages from preferred store

This command is used to return messages with status value <stat> from message storage <mem1> to the TE.
If the status of the message is 'received unread', the status in the storage changes to 'received read'.

AT+CMGL List SMS messages from preferred store	
Test Command **AT+CMGL=?**	Response **+CMGL:** (list of supported **<stat>s)** **OK**
Write Command **AT+CMGL=<stat>**	Response a)If text mode (AT+CMGF=1), command successful and SMS-SUBMITs and/or SMS-DELIVERs: **+CMGL: <index>,<stat>,<oa>/<da>,[<alpha>],[<scts>][,<tooa>/<toda>,<fo>,<pid>,<dcs>,[<vp>],<sca>,<tosca>,<length>]<CR><LF><data>[<CR><LF>** **+CMGL: <index>,<stat>,<oa>/<da>,[<alpha>],[<scts>][,<tooa>/<toda>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>[...]]** **OK** b)If text mode (AT+CMGF=1), command successful and SMS-STATUS-REPORTs: **+CMGL: <index>,<stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>[<CR><LF>** **+CMGL: <index>,<stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>[...]]** **OK** c)If text mode (AT+CMGF=1), command successful and SMS-COMMANDs: **+CMGL: <index>,<stat>,<fo>,<ct>[<CR><LF>** **+CMGL: <index>,<stat>,<fo>,<ct>[...]]** **OK**

	d)If text mode (AT+CMGF=1), command successful and CBM storage: **+CMGL: <index>,<stat>,<sn>,<mid>,<page>,<pages>** **<CR><LF><data>[<CR><LF>** **+CMGL: <index>,<stat>,<sn>,<mid>,<page>,<pages>** **<CR><LF><data>[...]]** **OK** e)If PDU mode (AT+CMGF=0) and Command successful: **+CMGL:** **<index>,<stat>,[<alpha>],<length><CR><LF><pdu>[<CR><LF>** **+CMGL: <index>,<stat>,[<alpha>],<length><CR><LF><pdu>** **[…]]** **OK** f)If failed: **+CMS ERROR: <err>**

Defined Values

<stat>	1. Text Mode: "REC UNREAD" received unread message (i.e. new message) "REC READ" received read message "STO UNSENT" stored unsent message "STO SENT" stored sent message "ALL" all messages 2. PDU Mode: 0 received unread message (i.e. new message) 1 received read message 2 stored unsent message 3 stored sent message 4 all messages
<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<oa>	Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>.
<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.
<alpha>	String type alphanumeric representation of <da>or<oa> corresponding to the entry found in MT phonebook; implementation

	of this feature is manufacturer specific; used character set should be the one selected with command Select TE Character Set AT+CSCS.
<scts>	TP-Service-Center-Time-Stamp in time-string format (refer <dt>).
<tooa>	TP-Originating-Address, Type-of-Address octet in integer format. (default refer <toda>).
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<length>	Integer type value indicating in the text mode (AT+CMGF=1) the length of the message body <data> in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. (i.e. the RP layer SMSC address octets are not counted in the length)
<data>	In the case of SMS: TP-User-Data in text mode responses; format: 1. If <dcs> indicates that GSM 7 bit default alphabet is used and <fo> indicates that TP-User-Data-Header-Indication is not set: a. If TE character set other than "HEX": ME/TA converts GSM alphabet into current TE character set. b. If TE character set is "HEX": ME/TA converts each 7-bit character of GSM 7 bit default alphabet into two IRA character long hexadecimal numbers. (e.g. character (GSM 7 bit default alphabet 23) is presented as 17 (IRA 49 and 55)) 2. If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, or<fo> indicates that TP-User-Data-Header-Indication is set: ME/TA converts each 8-bit octet into two IRA character long hexadecimal numbers. (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)) 3. If <dcs> indicates that GSM 7 bit default alphabet is used: a. If TE character set other than "HEX": ME/TA converts GSM alphabet into current TE character set. b. If TE character set is "HEX": ME/TA converts each 7-bit character of the GSM 7 bit default alphabet into two IRA character long hexadecimal numbers. 4. If <dcs> indicates that 8-bit or UCS2 data coding scheme is used: ME/TA converts each 8-bit octet into two IRA character long hexadecimal numbers.
<fo>	Depending on the command or result code: first octet of GSM 03.40 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default 2) in integer format. SMS status report is supported under text mode if <fo> is set to 49.
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format.
<ra>	Recipient Address GSM 03.40 TP-Recipient-Address Address-Value field in string

	format;BCD numbers (or GSM default alphabet characters) are converted to characters of the currently selected TE character set(refer to command AT+CSCS);type of address given by <tora>
<tora>	Type of Recipient Address GSM 04.11 TP-Recipient-Address Type-of-Address octet in integer format (default refer <toda>)
<dt>	Discharge Time GSM 03.40 TP-Discharge-Time in time-string format:"yy/MM/dd,hh:mm:ss+zz",where characters indicate year (two last digits),month,day,hour,minutes,seconds and time zone.
<st>	Status GSM 03.40 TP-Status in integer format 0…255
<ct>	Command Type GSM 03.40 TP-Command-Type in integer format 0…255
<sn>	Serial Number GSM 03.41 CBM Serial Number in integer format
<mid>	Message Identifier GSM 03.41 CBM Message Identifier in integer format
<page>	Page Parameter GSM 03.41 CBM Page Parameter bits 4-7 in integer format
<pages>	Page Parameter GSM 03.41 CBM Page Parameter bits 0-3 in integer format
<pdu>	In the case of SMS: SC address followed by TPDU in hexadecimal format: ME/TA converts each octet of TP data unit into two IRA character long hexadecimal numbers. (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)).

Example

```
AT+CMGL="ALL"
+CMGL: 9,"REC READ","+861310.....","jeck","20/05/20,09:31:00+32",145,0,0,0,"+8613......",145,2
hi
+CMGL: 10,"REC READ","+861310......","leo","20/05/20,09:32:25+32",145,0,0,0,"+8613......",145,4
Fine

OK
```

9.2.12 AT+CMGR Read message

This command is used to return message with location value <index> from message storage <mem1> to the TE.

AT+CMGR Read message	
Test Command **AT+CMGR=?**	Response **OK**
Write Command **AT+CMGR=<index>**	a)If text mode (AT+CMGF=1), command successful and SMS-DELIVER: **+CMGR: <stat>,<oa>,[<alpha>],<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>** **OK** b)If text mode (AT+CMGF=1), command successful and SMS-SUBMIT: **+CMGR: <stat>,<da>,[<alpha>][,<toda>,<fo>,<pid>,<dcs>,[<vp>],<sca>,<tosca>,<length>]<CR><LF><data>** **OK** c)If text mode (AT+CMGF=1), command successful and SMS-STATUS-REPORT: **+CMGR: <stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>** **OK** d)If text mode (AT+CMGF=1), command successful and SMS-COMMAND: **+CMGR: <stat>,<fo>,<ct>[,<pid>,[<mn>],[<da>],[<toda>],<length>]<CR><LF><data>** **OK** e)If text mode (AT+CMGF=1), command successful and CBM storage: **+CMGR: <stat>,<sn>,<mid>,<dcs>,<page>,<pages><CR><LF><data>** **OK** f)If PDU mode (AT+CMGF=0) and Command successful: **+CMGR: <stat>,[<alpha>],<length><CR><LF><pdu>** **OK** g)If failed: **+CMS ERROR: <err>**

Defined Values

<stat>	1. Text Mode: "REC UNREAD" received unread message (i.e. new message) "REC READ" received read message "STO UNSENT" stored unsent message "STO SENT" stored sent message "ALL" all messages 2. PDU Mode: 0 received unread message (i.e. new message) 1 received read message 2 stored unsent message 3 stored sent message 4 all messages
<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<oa>	Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>.
<pid>	Protocol Identifier GSM 03.40 TP-Protocol-Identifier in integer format 0…255
<alpha>	String type alphanumeric representation of <da>or<oa> corresponding to the entry found in MT phonebook; implementation of this feature is manufacturer specific; used character set should be the one selected with command Select TE Character Set AT+CSCS.
<dcs>	Depending on the command or result code: SMS Data Coding Scheme (default 0), or Cell Broadcast Data Coding Scheme in integer format..
<sca>	RP SC address Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tosca>.
<tosca>	RP SC address Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tosca>.
<scts>	TP-Service-Center-Time-Stamp in time-string format (refer <dt>).
<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.

<tooa>	TP-Originating-Address, Type-of-Address octet in integer format. (default refer <toda>).
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<length>	Integer type value indicating in the text mode (AT+CMGF=1) the length of the message body <data> in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. (i.e. the RP layer SMSC address octets are not counted in the length)
<data>	In the case of SMS: TP-User-Data in text mode responses; format: 1. If <dcs> indicates that GSM 7 bit default alphabet is used and <fo> indicates that TP-User-Data-Header-Indication is not set: a. If TE character set other than "HEX": ME/TA converts GSM alphabet into current TE character set. b. If TE character set is "HEX": ME/TA converts each 7-bit character of GSM 7 bit default alphabet into two IRA character long hexadecimal numbers. (e.g. character (GSM 7 bit default alphabet 23) is presented as 17 (IRA 49 and 55)) 2. If <dcs> indicates that 8-bit or UCS2 data coding scheme is used, or<fo> indicates that TP-User-Data-Header-Indication is set: ME/TA converts each 8-bit octet into two IRA character long hexadecimal numbers. (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)) 3. If <dcs> indicates that GSM 7 bit default alphabet is used: a. If TE character set other than "HEX": ME/TA converts GSM alphabet into current TE character set. b. If TE character set is "HEX": ME/TA converts each 7-bit character of the GSM 7 bit default alphabet into two IRA character long hexadecimal numbers. 4. If <dcs> indicates that 8-bit or UCS2 data coding scheme is used: ME/TA converts each 8-bit octet into two IRA character long hexadecimal numbers.
<fo>	Depending on the command or result code: first octet of GSM 03.40 SMS-DELIVER, SMS-SUBMIT (default 17), SMS-STATUS-REPORT, or SMS-COMMAND (default 2) in integer format. SMS status report is supported under text mode if <fo> is set to 49.
<vp>	Depending on SMS-SUBMIT <fo> setting: TP-Validity-Period either in integer format (default 167) or in time-string format (refer <dt>).
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format.
<ra>	Recipient Address GSM 03.40 TP-Recipient-Address Address-Value field in string format;BCD numbers (or GSM default alphabet characters) are

	converted to characters of the currently selected TE character set(refer to command AT+CSCS);type of address given by <tora>
<tora>	Type of Recipient Address GSM 04.11 TP-Recipient-Address Type-of-Address octet in integer format (default refer <toda>)
<dt>	Discharge Time GSM 03.40 TP-Discharge-Time in time-string format:"yy/MM/dd,hh:mm:ss+zz",where characters indicate year (two last digits),month,day,hour,minutes,seconds and time zone.
<st>	Status GSM 03.40 TP-Status in integer format 0…255
<ct>	Command Type GSM 03.40 TP-Command-Type in integer format 0…255
<sn>	Serial Number GSM 03.41 CBM Serial Number in integer format
<mn>	Message Number GSM 03.40 TP-Message-Number in integer format
<mid>	Message Identifier GSM 03.41 CBM Message Identifier in integer format
<page>	Page Parameter GSM 03.41 CBM Page Parameter bits 4-7 in integer format
<pages>	Page Parameter GSM 03.41 CBM Page Parameter bits 0-3 in integer format
<pdu>	In the case of SMS: SC address followed by TPDU in hexadecimal format: ME/TA converts each octet of TP data unit into two IRA character long hexadecimal numbers. (e.g. octet with integer value 42 is presented to TE as two characters 2A (IRA 50 and 65)).

Example

```
AT+CMGR=1
+CMGR: "STO UNSENT","+10011","",145,17,0,0,167,"+8613800100500",145,11
Hello World

OK
```

9.2.13 AT+CMGS Send message

This command is used to send message from a TE to the network (SMS-SUBMIT).

AT+CMGS Send message

Test Command **AT+CMGS=?**	Response **OK**
Write Command If text mode (AT+CMGF=1): **AT+CMGS=<da>[,<toda>]<CR>** **>Text is entered.** **<CTRL-Z/ESC>** If PDU mode(AT+CMGF=0): **AT+CMGS=<length><CR>** **PDU is entered** **<CTRL-Z/ESC>**	Response a)If sending successfully: **+CMGS: <mr>[,<time_stamp>]** **OK** b)If cancel sending: **OK** c)If sending fails: **ERROR** d)If sending fails: **+CMS ERROR: <err>**

Defined Values

<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<length>	integer type value indicating in the text mode (AT+CMGF=1) the length of the message body <data>> (or<cdata>) in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. (i.e. the RP layer SMSC address octets are not counted in the length)
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format.
<time_stamp>	The sending time of the SMS.

Example

AT+CMGS="13012832788"<CR>(TEXT MODE)
> ABCD<ctrl-Z/ESC>
+CMGS: 46

OK

NOTE

In text mode, the maximum length of an SMS depends on the used coding scheme: It is 160 characters if the 7 bit GSM coding scheme is used.

9.2.14 AT+CMSS Send message from storage

This command is used to send message with location value <index> from preferred message storage <mem2> to the network (SMS-SUBMIT or SMS-COMMAND).

AT+CMSS Send message from storage	
Test Command **AT+CMSS=?**	Response **OK**
Write Command **AT+CMSS=<index> [,<da>[,<toda>]]**	Response a) **+CMSS: <mr>[,<time_stamp>]** **OK** b)If failed: **ERROR** c)If sending fails: **+CMS ERROR: <err>**

Defined Values

<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format.
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<time_stamp>	The sending time of the SMS.

Example

```
AT+CMSS=3
+CMSS: 0
```

OK
AT+CMSS=3,"13012345678"
+CMSS: 55

OK

NOTE

In text mode, the maximum length of an SMS depends on the used coding scheme: It is 160 characters if the 7 bit GSM coding scheme is used.

9.2.15 AT+CMGW Write message to memory

This command is used to store message (either SMS-DELIVER or SMS-SUBMIT) to memory storage <mem2>.

AT+CMGW Write message to memory	
Test Command **AT+CMGW=?**	Response **OK**
Write Command If text mode (AT+CMGF=1): **AT+CMGW=<oa>/<da>[,<tooa>/<toda>[,<stat>]]<CR>Text is entered.** **<CTRL-Z/ESC>** If PDU mode(AT+CMGF=0): **AT+CMGW=<length>[,<stat>]<CR>PDU is entered.** **<CTRL-Z/ESC>**	Response a)If write successfully: **+CMGW: <index>** **OK** b)If cancel write: **OK** c)If write fails: **ERROR** d)If write fails: **+CMS ERROR: <err>**

Defined Values

<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<oa>	Originating-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <tooa>.
<tooa>	TP-Originating-Address, Type-of-Address octet in integer format.

	(default refer <toda>).
<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (when first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<length>	Integer type value indicating in the text mode (AT+CMGF=1) the length of the message body <data>> (or<cdata>) in characters; or in PDU mode (AT+CMGF=0), the length of the actual TP data unit in octets. (i.e. the RP layer SMSC address octets are not counted in the length).
<stat>	1. Text Mode: "STO UNSENT" stored unsent message "STO SENT" stored sent message 2. PDU Mode: 2 stored unsent message 3 stored sent message

Example

```
AT+CMGW="13012832788" <CR> (TEXT MODE)
ABCD<ctrl-Z/ESC>
+CMGW: 1

OK
```

NOTE

In text mode, the maximum length of an SMS depends on the used coding scheme: It is 160 characters if the 7 bit GSM coding scheme is used.

9.2.16 AT+CMGD Delete message

This command is used to delete message from preferred message storage <mem1> location <index>. If <delflag> is present and not set to 0 then the ME shall ignore <index> and follow the rules for <delflag> shown below.

AT+CMGD Delete message

Test Command **AT+CMGD=?**	Response **+CMGD: (**list of supported **<index>s)[,(**list of supported **<delflag>s)]** **OK**
Write Command **AT+CMGD=<index>[,<delflag>]**	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<delflag>	0 (or omitted) Delete the message specified in <index>. 1 Delete all read messages from preferred message storage, leaving unread messages and stored mobile originated messages (whether sent or not) untouched. 2 Delete all read messages from preferred message storage and sent mobile originated messages, leaving unread messages and unsent mobile originated messages untouched. 3 Delete all read messages from preferred message storage, sent and unsent mobile originated messages leaving unread messages untouched. 4 Delete all messages from preferred message storage including unread messages.

Example

```
AT+CMGD=1
OK
```

NOTE

If set <delflag>=1, 2, 3 or 4, <index> is omitted, such as AT+CMGD=,1.

9.2.17 AT+CMGMT Change message status

This command is used to change the message status. If the status is unread, it will be changed read. Other statuses don't change.
Note: This command not support in CDMA/EVDO mode

AT+CMGMT Change message status	
Test Command AT+CMGMT=?	Response **OK**
Write Command AT+CMGMT=<index>	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.

Example

```
AT+CMGMT=1
OK
```

9.2.18 AT+CMVP Set message valid period

This command is used to set valid period for sending short message.
Note: This command not support in CDMA/EVDO mode

AT+CMVP Set message valid period	
Test Command AT+CMVP=?	Response **+CMVP:** (range of supported **<vp>s**) **OK**
Read Command AT+CMVP?	Response **+CMVP: <vp>**

	OK
Write Command AT+CMVP=<vp>	Response a) **OK** b)If failed: **ERROR** c)If failed: **+CMS ERROR: <err>**

Defined Values

<vp>	Validity period value: 0 to 143 (<vp>+1) x 5 minutes (up to 12 hours) 144 to 167 12 hours + (<vp>-143) x 30 minutes 168 to 196 (<vp>-166) x 1 day 197 to 255 (<vp>-192) x 1 week

Example

```
AT+CMVP=167
OK
```

9.2.19 AT+CMGRD Read and delete message

This command is used to read message, and delete the message at the same time. It integrate AT+CMGR and AT+CMGD, but it doesn't change the message status.
Note: This command not support in CDMA/EVDO mode

AT+CMGRD Read and delete message	
Test Command AT+CMGRD=?	Response **OK**
Write Command AT+CMGRD=<index>	Response a)If text mode(AT+CMGF=1),command successful and SMS-DE-LIVER: **+CMGRD: <stat>,<oa>,[<alpha>],<scts>[,<tooa>,<fo>,<pid>,<dcs>,<sca>,<tosca>,<length>]<CR><LF><data>** **OK** b)If text mode(AT+CMGF=1),command successful and SMS-SU-BMIT:

	+CMGRD: **<stat>,<da>,[<alpha>][,<toda>,<fo>,<pid>,<dcs>,[<vp>],** **<sca>,<tosca>,<length>]<CR><LF><data>** **OK** c)If text mode(AT+CMGF=1),command successful and SMS-STA-TUS- REPORT: **+CMGRD: <stat>,<fo>,<mr>,[<ra>],[<tora>],<scts>,<dt>,<st>** **OK** d)If text mode(AT+CMGF=1),command successful and SMS-CO-MMAND: **+CMGRD:** **<stat>,<fo>,<ct>[,<pid>,[<mn>],[<da>],[<toda>],<length><CR><LF><data>]** **OK** e)If text mode(AT+CMGF=1),command successful and CBM storage: **+CMGRD:** **<stat>,<sn>,<mid>,<dcs>,<page>,<pages><CR><LF><data>** **OK** f)**If PDU mode(AT+CMGF=0) and command successful:** **+CMGRD: <stat>,[<alpha>],<length><CR><LF><pdu>** **OK** g)If failed: **ERROR** h)If failed: **+CMS ERROR: <err>**

Defined Values

Refer to command AT+CMGR.

Example

```
AT+CMGRD=6
+CMGRD: "REC READ","+8613917787249","","06/07/10,12:09:38+32",145,4,0,0,"+8613800210500",145,5
HELLO

OK
```

9.2.20 AT+CMGSEX Send message

This command is used to send message from a TE to the network (SMS-SUBMIT).
Note: This command not support in CDMA/EVDO mode

AT+CMGSEX Send message	
Test Command **AT+CMGSEX=?**	Response **OK**
Write Command If text mode (AT+CMGF=1): **AT+CMGSEX=<da>[,<toda>][,<mr>,<msg_seg>,<msg_total>]<CR>Text is entered. <CTRL-Z/ESC>**	Response a)If sending successfully: **+CMGSEX: <mr>** **OK** b)If cancel sending: **OK** c)If sending fails: **ERROR** d)If sending fails: **+CMS ERROR: <err>**

Defined Values

<da>	Destination-Address, Address-Value field in string format; BCD numbers (or GSM 7 bit default alphabet characters) are converted to characters of the currently selected TE character set, type of address given by <toda>.
<toda>	TP-Destination-Address, Type-of-Address octet in integer format. (When first character of <da> is + (IRA 43) default is 145, otherwise default is 129). The range of value is from 128 to 255.
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format. The maximum length is 255.
<msg_seg>	The segment number for long sms
<msg_total>	The total number of the segments for long sms. Its range is from 2 to 255.

Example

AT+CMGSEX="13012832788", 190, 1, 2<CR>(TEXT MODE)
> ABCD<ctrl-Z/ESC>

```
+CMGSEX: 190

OK
AT+CMGSEX="13012832788", 190, 2, 2<CR>(TEXT MODE)
> ABCD<ctrl-Z/ESC>
+CMGSEX: 191

OK
```

NOTE

In text mode, the maximum length of an SMS depends on the used coding scheme: For single SMS, it is 160 characters if the 7 bit GSM coding scheme is used; For multiple long sms, it is 153 characters if the 7 bit GSM coding scheme is used.

9.2.21 AT+CMSSEX Send multi messages from storage

This command is used to send messages with location value <index1>,<index2>,<index3>··· from preferred message storage <mem2> to the network (SMS-SUBMIT or SMS-COMMAND).The max count of index is 13 one time. Set AT+CNMI parameter <ds> equal to 0.
Note: This command not support in CDMA/EVDO mode

AT+CMSSEX Send multi messages from storage	
Test Command **AT+CMSSEX=?**	Response **OK**
Write Command **AT+CMSSEX=<index>[,<index>[,…]]**	Response a) **+CMSSEX: <mr>[,<mr>[,…]]** **OK** b)If failed: **ERROR** c)If sending fails: **[+CMSSEX: <mr>[,<mr>[,…]]]** **+CMS ERROR: <err>** d)If sending fails: **[+CMSSEX: <mr>[,<mr>[,…]]]** **ERROR**

Defined Values

<index>	Integer type; value in the range of location numbers supported by the associated memory and start with zero.
<mr>	Message Reference GSM 03.40 TP-Message-Reference in integer format.

Example

```
AT+CMSSEX=0,1
+CMSSEX: 239,240

OK
```

NOTE

In text mode, the maximum length of an SMS depends on the used coding scheme: It is 160 characters if the 7 bit GSM coding scheme is used.

10 AT Commands for GPS

10.1 Overview of AT Commands for GPS

Command	Description
AT+CGPS	Start/Stop GPS session
AT+CGPSINFO	Get GPS fixed position information
AT+CGPSCOLD	Cold start GPS
AT+CGPSHOT	Hot start GPS
AT+CGPSURL	Set AGPS default server URL
AT+CGPSSSL	Set AGPS transport security
AT+CGPSAUTO	Start GPS automatic
AT+CGPSNMEA	Configure NMEA sentence type
AT+CGPSMD	Configure AGPS MO method
AT+CGPSFTM	Start GPS test mode
AT+CGPSDEL	Delete the GPS information
AT+CGPSXE	Enable/Disable GPS XTRA function
AT+CGPSXD	Download XTRA assistant file
AT+CGPSXDAUTO	Download XTRA assistant file automatically
AT+CGPSPMD	Configure positioning mode
AT+CGPSMSB	Configure based mode switch to standalone
AT+CGPSHOR	Configure positioning desired accuracy
AT+CGNSSINFO	Get GNSS fixed position information
AT+CGNSSMODE	Configure GNSS support mode

10.2 Detailed Description of AT Commands for GPS

10.2.1 AT+CGPS Start/Stop GPS session

AT+CGPS Start/Stop GPS session

Test Command **AT+CGPS=?**	Response **+CGPS:** (scope of<**on/off**>),(scope of<**mode**>) **OK**
Read Command **AT+CGPS?**	Response **+CGPS: <on/off>,<mode>** **OK**
Write Command **AT+CGPS=<on/off>[,<mode>]**	Response **OK** If UE-assisted mode, when fixed will report indication: **+CAGPSINFO:<lat>,<lon>,<alt>,<date>,<time>,100,<horUncEllipseSemiMajor>** If <off>, it will report indication: **+CGPS: 0**

Defined Values

<on/off>	0 stop GPS session 1 start GPS session
<mode>	Ignore - standalone mode 1 standalone mode 2 UE-based mode 3 UE-assisted mode
<lat>	Latitude of current position. Unit is in 10^8 degree
<lon>	Longitude of current position. Unit is in 10^8 degree
<date>	UTC Date. Output format is ddmmyy
<time>	UTC Time. Output format is hhmmss.s
<alt>	MSL Altitude. Unit is meters.
100	fixed value
<horUncEllipseSemiMajor>	horizontal elliptical accuracy semi-major axis

Example

```
AT+CGPS?
+CGPS: 0,1

OK
AT+CGPS=0
OK
```

+CGPS: 0

10.2.2 AT+CGPSINFO Get GPS fixed position information

AT+CGPSINFO Get GPS fixed position information	
Test Command **AT+CGPSINFO=?**	Response **+CGPSINFO:** (scope of<**time**>) **OK**
Read Command **AT+CGPSINFO?**	Response **+CGPSINFO: <time>** **OK**
Write Command **AT+CGPSINFO=<time>**	Response **+CGPSINFO:** **[<lat>],[<N/S>],[<lon>],[<E/W>],[<date>],[<UTCtime>],[<alt>],[<speed>],[<course>]** **OK** If <time>=0, it will report indication: **OK**
Execution Command **AT+CGPSINFO**	Response **+CGPSINFO:** **[<lat>],[<N/S>],[<lon>],[<E/W>],[<date>],[<UTCtime>],[<alt>],[<speed>],[<course>]** **OK**

Defined Values

<lat>	Latitude of current position. Output format is ddmm.mmmmmm
<N/S>	N/S Indicator, N=north or S=south
<lon>	Longitude of current position. Output format is dddmm.mmmmmm
<E/W>	E/W Indicator, E=east or W=west
<date>	Date. Output format is ddmmyy
<UTC time>	UTC Time. Output format is hhmmss.s
<alt>	MSL Altitude. Unit is meters.
<speed>	Speed Over Ground. Unit is knots.

<course>	Course. Degrees.
<time>	The range is 0-255, unit is second, after set <time> will report the GPS information every the seconds.

Example

```
AT+CGPSINFO=?
+CGPSINFO: (0-255)

OK
AT+CGPSINFO?
+CGPSINFO: 0

OK
AT+CGPSINFO
+CGPSINFO: 3113.343286,N,12121.234064,E,250311,072809.3,44.1,0.0,0.0

OK
```

10.2.3 AT+CGPSCOLD Cold Start GPS

AT+CGPSCOLD Cold Start GPS	
Test Command **AT+CGPSCOLD=?**	Response **OK**
Execution Command **AT+CGPSCOLD**	Response **OK**

Example

```
AT+CGPSCOLD=?
OK
AT+CGPSCOLD
OK
```

NOTE

Before using this command, ensure GPS is off.

10.2.4 AT+CGPSHOT Hot Start GPS

AT+CGPSHOT Hot Start GPS	
Test Command AT+CGPSHOT=?	Response **OK**
Execution Command AT+CGPSHOT	Response **OK**

Example

```
AT+CGPSHOT=?
OK
AT+CGPSHOT
OK
```

NOTE

Before using this command, ensure GPS is off.

10.2.5 AT+CGPSURL Set AGPS default server URL

AT+CGPSURL Set AGPS default server URL	
Test Command AT+CGPSURL=?	Response **OK**
Read Command AT+CGPSURL?	Response **+CGPSURL: <URL>** **OK** Or **ERROR**
Write Command AT+CGPSURL=<URL>	Response **OK** Or **ERROR**

Defined Values

<URL>	AGPS default server URL. It needs double quotation marks. NOTE: Max length of URL is 128.

Example

```
AT+CGPSURL="123.123.123.123:8888"
OK
AT+CGPSURL?
+CGPSURL: "123.123.123.123:8888"

OK
```

NOTE

It will take effect only after restarting.

10.2.6 AT+CGPSSSL Set AGPS transport security

AT+CGPSSSL Set AGPS transport security	
Test Command **AT+CGPSSSL=?**	Response **+CGPSSSL:** (list of supported <**SSL**>) **OK**
Read Command **AT+CGPSSSL?**	Response **+CGPSSSL: <SSL>** **OK**
Write Command **AT+CGPSSSL=<SSL>**	Response **OK** Or **ERROR**

Defined Values

<SSL>	0 don't use certificate 1 use certificate

Example

```
AT+CGPSSSL=0
OK
```

NOTE

This command is used to select transport security, used certificate or not. The certificate gets from local carrier. If the AGPS server doesn't need certificate, execute AT+CGPSSSL=0.

10.2.7 AT+CGPSAUTO Start GPS automatic

AT+CGPSAUTO Start GPS automatic	
Test Command **AT+CGPSAUTO=?**	Response **+CGPSAUTO:** (list of supported **<auto>**) **OK**
Read Command **AT+CGPSAUTO?**	Response **+CGPSAUTO: <auto>** **OK**
Write Command **AT+CGPSAUTO=<auto>**	Response **OK** Or **ERROR**

Defined Values

<auto>	0 Non-automatic 1 automatic

Example

```
AT+CGPSAUTO=1
OK
```

NOTE

If GPS start automatically, its operation mode is standalone mode.. It will take effect only after restarting.

10.2.8 AT+CGPSNMEA Configure NMEA sentence type

AT+CGPSNMEA Configure NMEA sentence type	
Test Command **AT+CGPSNMEA=?**	Response **+CGPSNMEA:** (scope of <**nmea**>)

	OK
Read Command **AT+CGPSNMEA?**	Response **+CGPSNMEA: <nmea>** **OK** Or **ERROR**
Write Command **AT+CGPSNMEA=<nmea>**	Response **OK** Or **ERROR**

Defined Values

<nmea>	Range – 0 to 2147483647(0 to 2143191039 on SDX65) Each bit enables an NMEA sentence output as follows: Bit 0 – GPGGA (Fix data) Bit 1 – GPRMC (recommended minimum data) Bit 2 – GPGSV (GPS SVs in view) Bit 3 – GPGSA (GPS SV dop and active SV info) Bit 4 – GPVTG (Speed and heading info) Bit 5 – PQXFI Bit 6 – PSTIS (proprietary sentence at beginning of each sess) Bit 7 – GLGSV (Glonass SV in view info) Bit 8 – GNGSA (Dop and Active SV info iff Glonass SVs are used) Bit 9 – GNGNS (new GGA message for GNSS) Bit 10 – GARMC (GAL recommended minimum data) Bit 11 – GAGSV (GAL SVs in view) Bit 12 – GAGSA (GAL SV dop and active SV info) Bit 13 – GAVTG (GAL Speed and heading info) Bit 14 – GAGGA (GAL Fix data) Bit 15 – PQGSA (QZSS Enable PQGSA,Depreciated on SDX65) Bit 16 – PQGSV (QZSS Enable PQGSV,Depreciated on SDX65) Bit 17 – DEBUG (NMEA debugging enable) Bit 18 – GPDTM (new DTM message for GAL) Bit 19 – GNGGA (Fix data) Bit 20 – GNRMC (recommended minimum data) Bit 21 – GNVTG (GNSS: Speed and heading info) Bit 22 – GAGNS (GNS message for GAL,Depreciated on SDX65) Bit 23 – GBGGA (BDS: Fix data) Bit 24 – GBGSA (BDS SV dop and active SV info) Bit 25 – GBGSV (BDS SV in view info) Bit 26 – GBRMC (BDS: recommended minimum data)

	Bit 27 – GBVTG (BDS: Speed and heading info) Bit 28 – GQGSV (QZSS SV in view info) Bit 29 – GIGSV (NAVIC SV in view info) Bit 30 – GNDTM (GNSS Datum Message) Set the desired NMEA sentence bit(s). If multiple NMEA sentence formats are desired, "OR" the desired bits together.

Example

```
AT+CGPSNMEA=200191
OK
```

10.2.9 AT+CGPSMD Configure AGPS MO method

AT+CGPSMD Configure AGPS MO method	
Test Command **AT+CGPSMD=?**	Response **+CGPSMD:** (scope of<**method**>) **OK**
Read Command **AT+CGPSMD?**	Response **+CGPSMD: <method>** **OK**
Write Command **AT+CGPSMD=<method>**	Response **OK**

Defined Values

<method>	0 Control plane 1 User plane

10.2.10 AT+CGPSFTM Start GPS test mode

AT+CGPSFTM Start GPS test mode	
Test Command **AT+CGPSFTM=?**	Response **OK**

Read Command AT+CGPSFTM?	Response +CGPSFTM: <on/off> OK
Write Command AT+CGPSFTM=<on/off>	Response OK Or ERROR

Defined Values

<on/off>	0 Close test mode 1 Start test mode
<SV>	Satellite ID number
<CNo>	Satellite CNo value. Floating value.
URC format	$GPGSV[,<SV>,<CNo>][...] $GLGSV[,<SV>,<CNo>][...] $GBGSV[,<SV>,<CNo>][...] $GAGSV[,<SV>,<CNo>][...] $GQGSV[,<SV>,<CNo>][...] $GIGSV[,<SV>,<CNo>][...]

Example

```
AT+CGPSFTM=1
OK

$GPGSV,3,17.1,6,26.1,9,18.0,17,25.6,19,23.9,28,22.4,2,0.0,4,0.0,5
,0.0,12,0.0

$GBGSV,230,23.5,229,27.4,220,26.9

$GQGSV,193,22.8,195,24.7
```

NOTE

URC sentence will report every 1 second.

10.2.11 AT+CGPSDEL Delete the GPS information

AT+CGPSDEL Delete the GPS information

Test Command **AT+CGPSDEL=?**	Response **OK**
Execution Command **AT+CGPSDEL**	Response **OK**

Example

```
AT+CGPSDEL=?
OK
AT+CGPSDEL
OK
```

NOTE

This command must be executed after GPS stopped

10.2.12 AT+CGPSXE Enable/Disable GPS XTRA function

AT+CGPSXE Enable/Disable GPS XTRA function	
Test Command **AT+CGPSXE=?**	Response **+CGPSXE:** (list of supported **<on/off>**) **OK**
Read Command **AT+CGPSXE?**	Response **+CGPSXE: <on/off>** **OK**
Write Command **AT+CGPSXE=<on/off>**	Response **OK** Or **ERROR**

Defined Values

<on/off>	0 Disable GPS XTRA 1 Enable GPS XTRA

Example

```
AT+CGPSXE=?
+CGPSXE: (0,1)
```

```
OK
AT+CGPSXE=0
OK
```

NOTE

XTRA function must download the assistant file from network by HTTP, so the APN must be set by AT+CGDCONT command.
It will take effect only after restarting

10.2.13 AT+CGPSXD Download XTRA assistant file

AT+CGPSXD Download XTRA assistant file	
Test Command AT+CGPSXD=?	Response **+CGPSXD:** (list of supported **<server>**) **OK**
Read Command AT+CGPSXD?	Response **+CGPSXD: <server>** **OK**
Write Command AT+CGPSXD=<server>	Response **OK** **+CGPSXD: <resp>** Or **ERROR**

Defined Values

<server>	0 XTRA primary server (precedence) 1 XTRA secondary server 2 XTRA tertiary server
<resp>	refer to Unsolicited XTRA download Codes

Example

```
AT+CGPSXD=?
+CGPSXD: (0-2)
```

```
OK
AT+CGPSXD=0
OK

+CGPSXD: 0
```

10.2.14 AT+CGPSXDAUTO Download XTRA assistant file automatically

AT+CGPSXDAUTO	Download XTRA assistant file automatically
Test Command AT+CGPSXDAUTO=?	Response **+CGPSXDAUTO:** (list of supported **<on/off>**) **OK**
Read Command AT+CGPSXDAUTO?	Response **+CGPSXDAUTO: <on/off>** **OK**
Write Command AT+CGPSXDAUTO=<on/off>	Response **OK** Or **ERROR**

Defined Values

<on/off>	0 disable download automatically 1 enable download automatically

Example

```
AT+CGPSXDAUTO=?
+CGPSXDAUTO: (0,1)

OK
AT+CGPSXDAUTO=0
OK
```

10.2.15 AT+CGPSPMD Configure positioning mode

AT+CGPSPMD Configure positioning mode	
Test Command **AT+CGPSPMD=?**	Response **+CGPSPMD:** (scope of <**mode**>) **OK**
Read Command **AT+CGPSPMD?**	Response **+CGPSPMD: <mode>** **OK**
Write Command **AT+CGPSPMD=<mode>**	Response **OK** Or **ERROR**

Defined Values

<mode>	Default - 65407. But for SIM8200/8260 the default value is 775. Range - 1 to 65407 Each bit enables a supported positioning mode as follows: Bit 0 – Standalone Bit 1 – UP MS-based Bit 2 – UP MS-assisted Bit 3 – CP MS-based (2G) Bit 4 – CP MS-assisted (2G) Bit 5 – CP UE-based (3G) Bit 6 – CP UE-assisted (3G) Bit 7 – NOT USED Bit 8 – UP MS-based (4G) Bit 9 – UP MS-assisted(4G) Bit 10 – CP MS-based (4G) Bit 11 – CP MS-assisted (4G) Set the desired mode sentence bit(s). If multiple modes are desired, "OR" the desired bits together. Example, support standalone, UP MS-based and UP MS-assisted, set Binary value 0000 0111, is 7.

Example

```
AT+CGPSPMD=127
OK
```

NOTE

Need to restart the module after setting the mode.

10.2.16 AT+CGPSMSB Configure based mode switch to standalone

AT+CGPSMSB Configure based mode switch to standalone	
Test Command AT+CGPSMSB=?	Response **+CGPSMSB:** (scope of <**mode**>) **OK**
Read Command AT+CGPSMSB?	Response **+CGPSMSB: <mode>** **OK**
Write Command AT+CGPSMSB=<mode>	Response **OK** Or **ERROR**

Defined Values

<mode>	0 Don't switch to standalone mode automatically 1 Switch to standalone mode automatically

Example

```
AT+CGPSMSB=0
OK
```

NOTE

This command take effect next start gps.

10.2.17 AT+CGPSHOR Configure positioning desired accuracy

AT+CGPSHOR Configure positioning desired accuracy

Test Command **AT+CGPSHOR=?**	Response **+CGPSHOR:** (scope of <**acc**>) **OK**
Read Command **AT+CGPSHOR?**	Response **+CGPSHOR: <acc>** **OK**
Write Command **AT+CGPSHOR=<acc>**	Response **OK** Or **ERROR**

Defined Values

<acc>	Range – 0 to 1800000 Default value is 50

Example

```
AT+CGPSHOR=50
OK
```

NOTE

This command take effect next start gps.

10.2.18 AT+CGNSSINFO Get GNSS fixed position information

AT+CGNSSINFO Get GNSS fixed position information	
Test Command **AT+CGNSSINFO=?**	Response **+CGNSSINFO:** (scope of <**time**>) **OK**
Read Command **AT+CGNSSINFO?**	Response **+CGNSSINFO: <time>** **OK**
Write Command **AT+CGNSSINFO=<time>**	Response **+CGNSSINFO:** **[<mode>],[<GPS-SVs>],[<GLONASS-SVs>],[<BEIDOU-SVs>],[<lat**

	>],[<N/S>],[<log>],[<E/W>],[<date>],[<UTC-time>],[<alt>],[<speed>],[<course>],[<PDOP>],[HDOP],[VDOP] **OK** or If <off>, it will report indication: **OK**(if <time>=0)
Execution Command AT+CGNSSINFO	Response **+CGNSSINFO: [<mode>],[<GPS-SVs>],[<GLONASS-SVs>],[<BEIDOU-SVs>],[<lat>],[<N/S>],[<log>],[<E/W>],[<date>],[<UTC-time>],[<alt>],[<speed>],[<course>],[<PDOP>],[<HDOP>],[<VDOP>]** **OK**

Defined Values

<mode>	Fix mode 2=2D fix 3=3D fix
<GPS-SVs>	GPS satellite valid numbers
<GLONASS-SVs>	GLONASS satellite valid numbers
<BEIDOU-SVs>	BEIDOU satellite valid numbers
<lat>	Latitude of current position. Output format is ddmm.mmmmmm
<N/S>	N/S Indicator, N=north or S=south
<log>	Longitude of current position. Output format is dddmm.mmmmmm
<E/W>	E/W Indicator, E=east or W=west
<date>	Date. Output format is ddmmyy
<UTC-time>	UTC Time. Output format is hhmmss.s
<alt>	MSL Altitude. Unit is meters.
<speed>	Speed Over Ground. Unit is knots.
<course>	Course. Degrees.
<PDOP>	Position Dilution Of Precision.
<HDOP>	Horizontal Dilution Of Precision.
<VDOP>	Vertical Dilution Of Precision.
<time>	The range is 0-255, unit is second, after set <time> will report the GNSS information every the seconds.

Example

```
AT+CGNSSINFO=?
+CGNSSINFO: (0-255)

OK
```

```
AT+CGNSSINFO?
+CGNSSINFO: 0

OK
AT+CGNSSINFO
+CGNSSINFO:
2,09,05,00,3113.330650,N,12121.262554,E,131117,091918.0,32.9,
0.0,255.0,1.1,0.8,0.7

OK
AT+CGNSSINFO(if not fix,will report null)
+CGNSSINFO: ,,,,,,,,,,,,,,,

OK
```

10.2.19 AT+CGNSSMODE Configure GNSS support mode

AT+CGNSSMODE Configure GNSS support mode	
Test Command **AT+CGNSSMODE=?**	Response **+CGNSSMODE:** (scope of **<gnss_mode>**),(scope of **<dpo_mode>**) **OK**
Read Command **AT+CGNSSMODE?**	Response **+CGNSSMODE: <gnss_mode>,<dpo_mode>** **OK**
Write Command **AT+CGNSSMODE=<gnss_mode>[,<dpo_mode>]**	Response **OK** Or **ERROR**

Defined Values

<gnss_mode>	Range – 0 to 31 Bit0: GLONASS Bit1: BEIDOU Bit2: GALILEO Bit3: QZSS Bit4: NAVIC
<dpo_mode>	1: enable DPO 0: disable DPO

Example

```
AT+CGNSSMODE=15,1
OK
```

NOTE

Module should reboot to take effective.

10.2.20 Unsolicited XTRA download Codes

Code of <err>

Code of <err>	Description
0	Assistant file download successfully
1	Assistant file doesn’t exist
2	Assistant file check error
225	Memory error
227	Network error（When GPS is enabled, the time will be updated first, then the Xtra data will be downloaded, and network errors will be reported

11 AT Commands for Hardware

11.1 Overview of AT Commands for Hardware

Command	Description
AT+IPREX	Set UART local baud rate permanently
AT+CFGRI	Indicate RI when using URC
AT+CSCLK	Control UART sleep
AT+CMUX	Enable the multiplexer over the UART
AT+CGFUNC	Enable/disable the function for the special GPIO
AT+CGDRT	Set the direction of specified GPIO
AT+CGSETV	Set the value of specified GPIO
AT+CGGETV	Get the value of specified GPIO
AT+CPCIEMODE	Get or set the mode of PCIE

11.2 Detailed Description of AT Commands for Hardware

11.2.1 AT+IPREX Set UART local baud rate permanently

AT+IPREX Set UART local baud rate permanently	
Test Command **AT+IPREX=?**	Response **+IPREX:** (list of supported <**speed**>s) **OK**
Read Command **AT+IPREX?**	Response **+IPREX: <speed>** **OK** or **ERROR**

Write Command AT+IPREX=<speed>	Response **OK** or **ERROR**
Execution Command AT+IPREX	Response a)Restore the default value: **OK** b)If failed: **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<speed>	300,600,1200,2400,4800,9600,19200,38400,57600,115200,230400,460800,921600,3000000,3200000,3686400

Example

```
AT+IPREX?
+IPREX: 115200

OK
AT+IPREX=?
+IPREX:
(300,600,1200,2400,4800,9600,19200,38400,57600,115200,230400,460800,921600,3000000,3200000,
3686400)

OK
AT+IPREX=115200
OK
```

11.2.2 AT+CFGRI Indicate RI when using URC

AT+CFGRI Indicate RI when using URC	
Test Command AT+CFGRI=?	Response **+CFGRI:** (range of supported <**status**>s),(range of supported <**URC time**>s),(range of supported <**SMS time**>s) **OK**

Read Command **AT+CFGRI?**	Response **+CFGRI: <status>,<URC time>,<SMS time>** **OK** or **ERROR**
Write Command **AT+CFGRI=<status>[,<URC time>,<SMS time>]**	Response **OK** or **ERROR**
Execution Command **AT+CFGRI**	Response a)Restore the default value: **OK** b)If failed: **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	-

Defined Values

<status>	0 off (default) 1 on
<URC time>	Integer type. Which is number of milliseconds to assert RI pin. The parameter range is 10 to 6000. 60 (Default)
<SMS time>	Integer type. Which is number of milliseconds to assert RI pin. The parameter range is 20 to 6000. 120 (Default)

Example

```
AT+CFGRI?
+CFGRI: 0,60,120

OK
AT+CFGRI=?
+CFGRI: (0-1),(10-6000),(20-6000)

OK
AT+CFGRI=1
OK
AT+CFGRI
OK
```

AT+CFGRI? **+CFGRI: 1,1000,1000** **OK**	In other M.2 modules,the default value of command is changed.

11.2.3 AT+CSCLK Control UART sleep

AT+CSCLK Control UART sleep	
Test Command **AT+CSCLK=?**	Response **+CSCLK:** (range of supported <**status**>s) **OK**
Read Command **AT+CSCLK?**	Response **+CSCLK: <status>** **OK** or **ERROR**
Write Command **AT+CSCLK=<status>**	Response **OK** or **ERROR**
Execution Command **AT+CSCLK**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<status>	0 off (Default) 1 on

Example

```
AT+CSCLK?
+CSCLK: 0

OK
```

```
AT+CSCLK=?
+CSCLK: (0-1)

OK
AT+CSCLK=1
OK
AT+CSCLK
OK
```

11.2.4 AT+CMUX Enable the multiplexer over the UART

AT+CMUX Enable the multiplexer over the UART	
Test Command **AT+CMUX=?**	Response **+CMUX:** (range of supported <**mode**>s),(range of supported <**subset**>s),(range of supported <**port_speed**>s),(range of supported <**N1**>s),(range of supported <**T1**>s),(range of supported <**N2**>s),(range of supported <**T2**>s) **OK** or **ERROR**
Read Command **AT+CMUX?**	Response **+CMUX: <mode>,<subset>,<port_speed>,<N1>,<T1>,<N2>,<T2>** **OK** or **ERROR**
Write Command **AT+CMUX=<mode>[,<subset>[,<port_speed>[,<N1>[,<T1>[,<N2>[,<T2>]]]]]]**	Response **OK** or **ERROR**
Parameter Saving Mode	NO_SAVE
Max Response Time	-
Reference	-

Defined Values

<mode>	0 basic mode (default)
<subset>	0 UIH frame type (default)
<port_speed>	1-8 CMUX always use current bitrate, the value only use for compatible.

	1 9600 bit/s 2 19200 bit/s 3 38400 bit/s 4 57600 bit/s 5 115200 bit/s 6 230400 bits/s 7 460800 bits/s 8 921600 bits/s The default value is 5(115200 bit/s).
<N1>	Integer type. Max frame size in bytes in Information field. The parameter range is 1 to 1500. 118 bytes (Default)
<T1>	Time UE waits for an acknowledgement before resorting to other action. (Don't Support Setting Currently)
<N2>	The max re-tries. (Don't Support Setting Currently)
<T2>	Integer type. The time in ms mux control channel waits before re-transmitting a command. The parameter range is 2 to 1000. The default value is 600,the mean is 6000 ms (Default)

Example

```
AT+CMUX?
+CMUX: 0,0,5,118,0,0,600

OK
AT+CMUX=?
+CMUX: (0),(0),(1-8),(1-1500),(0),(0),(2-1000)

OK
```

11.2.5 AT+CGFUNC Enable/disable the function for the special GPIO

AT+CGFUNC Enable /disable the function for the special GPIO	
Test Command AT+CGFUNC=?	Response **+CGFUNC:** (list of supported <**gpio**>s),(rang of supported

	<function>s) **OK**
Write Command **AT+CGFUNC=<gpio>**	Response **+CGFUNC: <gpio>,<function>** **OK** or **ERROR**
Write Command **AT+CGFUNC=<gpio>,<function>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<gpio>	The value is GPIO ID, Different hardware versions have the different values.
<function>	GPIO function 0 function 1 1 function 2

Example

```
AT+CGFUNC=?                                2 – GPIO88
+CGFUNC: (2),(0-1)

OK
AT+CGFUNC=2
+CGFUNC: 2,0

OK
AT+CGFUNC=2,1
OK
```

11.2.6 AT+CGDRT Set the direction of specified GPIO

AT+CGDRT Set the direction of specified GPIO

Test Command **AT+CGDRT=?**	Response **+CGDRT:** (list of supported <**gpio**>s),(list of supported <**gpio_io**>s) **OK**
Write Command **AT+CGDRT=<gpio>**	Response **+CGDRT: <gpio>,<gpio_io>** **OK** or **ERROR**
Write Command **AT+CGDRT=<gpio>,<gpio_io>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<gpio>	The value is GPIO ID, Different hardware versions have the different values.
<gpio_io>	0 in 1 out

Example

AT+CGDRT=?
+CGDRT: (1,2,3,4,5,6,8,9,11,12,13,14,15,16,18,19,20,21),(0,1)

OK
AT+CGDRT=1
+CGDRT: 1,0

OK
AT+CGDRT=1,1
OK

1 – PMX_GPIO13
2 – GPIO88 14 – GPIO106
3 – GPIO4 15 – GPIO82
4 – GPIO5 16 – GPIO83
5 – GPIO6 18 – GPIO107
6 – GPIO7 19 – GPIO31
8 – GPIO84 20 – GPIO102
9 – GPIO85 21 – GPIO105
11 – GPIO92
12 – GPIO96
13 – GPIO97

11.2.7 AT+CGSETV Set the value of specified GPIO

AT+CGSETV Set the value of specified GPIO	
Test Command **AT+CGSETV=?**	Response **+CGSETV:** (list of supported <**gpio**>s),(rang of supported <**value**>s) **OK**
Write Command **AT+CGSETV=<gpio>,<value>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<gpio>	Different hardware versions have the different values.
<value>	0 low level 1 high level

Example

```
AT+CGSETV=?
+CGSETV: (1,13),(0-1)

OK
AT+CGSETV=13,1
OK
```

NOTE

Only if the GPIO be set to output through AT+CGDRT, value can be set through AT+CGSETV.

11.2.8 AT+CGGETV Get the value of specified GPIO

AT+CGGETV Get the value of specified GPIO	
Test Command AT+CGGETV=?	Response **+CGGETV:** (list of supported <**gpio**>s) **OK**
Write Command AT+CGGETV=<gpio>	Response **+CGGETV: <gpio>,<values>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<gpio>	Different hardware versions have the different values.
<value>	0 low level 1 high level

Example

```
AT+CGGETV=?
+CGGETV:
(1,2,3,4,5,6,8,9,11,12,13,14,15,16,18,19,20,21)

OK
AT+CGGETV=1
+CGGETV: 1,0

OK
```

NOTE

+CGFUNC +CGDRT +CGSETV +CGGETV not suitable for SIM8260 and SIM8200 M2 Version,and will return ERROR.

11.2.9 AT+CPCIEMODE Get or set the mode of PCIE

This command is used to set or get PCIE mode, which supports M2 and LGA boards, when the AT setting is successful, the module will be restarted.

AT+CPCIEMODE Get or set the mode of PCIE	
Test Command **AT+CPCIEMODE=?**	Response **+CPCIEMODE:** (list of supported <**mode**>s) **OK**
Read Command **AT+CPCIEMODE?**	Response **+CPCIEMODE: <mode>** **OK** or **ERROR**
Write Command **AT+CPCIEMODE=<mode>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<mode>	EP EP mode HOST HOST mode

Example

```
AT+CPCIEMODE=?
+CPCIEMODE: (EP,HOST)

OK
AT+CPCIEMODE=EP
OK
```

12 Hardware Related Commands

12.1 Overview of Hardware Related Commands

Command	Description
AT+CVALARM	Set overvoltage and undervoltage alarm
AT+CADC	Read the value of ADC
AT+CADC2	Read the value of ADC2
AT+CMTE	Set the power action when over the critical temperature
AT+CPMVT	Set the power action when overvoltage and undervoltage
AT+CDELTA	Set module reboot to recovery mode
AT+CBC	Read the voltage value of the power supply
AT+CPMUTEMP	Read the temperature of the modules
AT+CUSBCFG	Set usbid,adb mode,edl mode and bootloader mode
AT+CCPUTEMP	Read the temperature of CPU different zones
AT+CCUART	Open or close the main uart
AT+CWIIC	I2C write command
AT+CRIIC	I2C read command

12.2 Detailed Description of AT Commands for Hardware

12.2.1 AT+CVALARM Set overvoltage and undervoltage alarm

This command is used to open or close the low voltage alarm function.

AT+CVALARM Set overvoltage and undervoltage alarm	
Test Command **AT+CVALARM=?**	Response **+CVALARM:** (list of supported <**enable**>s),(range of supported <**low_voltage**>s),(range of supported <**high_voltage**>s)

	OK
Read Command AT+CVALARM?	Response **+CVALARM: <enable>,<low_voltage>,<high_voltage>** **OK** or **ERROR**
Write Command AT+CVALARM=<enable>[,<low_voltage>[,<high_voltage>]]	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<enable>	0 close alarm(Default) 1 open alarm
<low_voltage>	Integer type. The undervoltage value of the module. The parameter range is 3300mV to 4000mV depending on hardware design. 3300mV(Default)
<high_voltage>	Integer type. The overvoltage value of the module. The parameter range is 4001mV to 4300mV depending on hardware design. 4300mV(Default)

Example

```
AT+CVALARM=?
+CVALARM: (0,1),(3300-4000),(4001-4300)

OK
AT+CVALARM?
+CVALARM: 1,3400,4300

OK
AT+CVALARM=1,3400,4300
OK
```

NOTE

If voltage < <low voltage>, it will report "**UNDER-VOLTAGE WARNNING**" every 10s. If voltage > <high voltage>, it will report "**OVER-VOLTAGE WARNNING**" every 10s.

12.2.2 AT+CADC Read the value of ADC

This command is used to read the ADC value from modem. ME supports 2 types of value, which are raw type and voltage type

AT+CADC Read the value of ADC	
Test Command AT+CADC=?	Response **+CADC:** (list of supported <**type**>s) **OK**
Write Command AT+CADC=<type>	Response **+CADC: <value>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<type>	0 raw data type 2 voltage type in mV
<value>	Integer type.

Example

```
AT+CADC=?
+CADC: (0,2)

OK
AT+CADC=0
+CADC: 187

OK
```

NOTE

+CADC is not suitable for M2 Version,and will return ERROR.

12.2.3 AT+CADC2 Read the value of ADC2

This command is used to read the ADC2 value from modem. ME supports 2 types of value, which are raw type and voltage type

AT+CADC2 Read the value of ADC2	
Test Command AT+CADC2=?	Response **+CADC2:** (list of supported <**type**>s) **OK**
Write Command AT+CADC2=<type>	Response **+CADC2: <value>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<type>	0 raw data type 2 voltage type in mV
<value>	Integer type.

Example

```
AT+CADC2=?
+CADC2: (0,2)

OK
AT+CADC2=0
+CADC2: 187

OK
```

NOTE

+CADC2 is not suitable for M2 Version,and will return ERROR.

12.2.4 AT+CMTE Set the power action when over the critical temperature

This command is used to control the module whether power off when temperature upon the critical temperature.

AT+CMTE Set the power action when over the critical temperature	
Test Command **AT+CMTE=?**	Response **+CMTE:** (list of supported <**power_off**>s) **OK**
Read Command **AT+CMTE?**	Response **+CMTE: <power_off>** **OK** or **ERROR**
Write Command **AT+CMTE=<power_off>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<power_off>	0 no action(Default) 1 power off

Example

AT+CMTE=?
+CMTE: (0,1)

OK

```
AT+CMTE?
+CMTE: 1

OK
AT+CMTE=0
OK
```

12.2.5 AT+CPMVT Set the power action when overvoltage and undervoltage

This command is used to open or close the power off action when undervoltage and overvoltage.

AT+CPMVT Set the power action when overvoltage and undervoltage	
Test Command **AT+CPMVT=?**	Response **+CPMVT:** (list of supported <**power_off**>s),(list of supported <**low_voltage**>s),(list of supported <**high_voltage**>s) **OK**
Read Command **AT+CPMVT?**	Response **+CPMVT: <power_off>,<low_voltage>,<high_voltage>** **OK** or **ERROR**
Write Command **AT+CPMVT=<power_off>[,<low_voltage>[,<high_voltage>]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<power_off>	0 no action(Default) 1 power off
<low_voltage>	Integer type. The undervoltage value of the module. The parameter range is 3200mV to 4000mV depending on hardware design. 3200mV(Default)
<high_voltage>	Integer type. The overvoltage value of the module. The parameter range is 4001mV to 4300mV depending on hardware

	design. 4300mV(Default)

Example

```
AT+CPMVT=?
+CPMVT: (0,1),(3200-4000),(4001-4300)

OK
AT+CPMVT?
+CPMVT: 1,3400,4300

OK
AT+CPMVT=1
OK
```

NOTE

If voltage < <low voltage>, it will report "**UNDER-VOLTAGE WARNNING POWER DOWN**" and power off the module. If voltage > <high voltage>, it will report "**OVER-VOLTAGE WARNNING POWER DOWN**" and power off the module.

12.2.6 AT+CDELTA Set module reboot to recovery mode

AT+CDELTA Set module reboot to recovery mode	
Execution Command **AT+CDELTA**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Example

```
AT+CDELTA
OK
```

NOTE

This command will write a flag to the module and restart, Check the flag during the next boot and enter

recovery mode to prepare for the firmware upgrade.

12.2.7 AT+CBC Read the voltage value of the power supply

AT+CBC Read the voltage value of the power supply	
Execution Command AT+CBC	Response **+CBC: <value>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<value>	The voltage value, such as 3.8V

Example

```
AT+CBC
+CBC: 3.657V

OK
```

12.2.8 AT+CPMUTEMP Read the temperature of the modules

AT+CPMUTEMP Read the temperature of the module	
Execution Command AT+CPMUTEMP	Response **+CPMUTEMP: <temp>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<temp>	Char type

Example

```
AT+CPMUTEMP
+CPMUTEMP: 28

OK
```

12.2.9 AT+CUSBCFG Set usbid,adb mode,edl mode and bootloader mode

AT+CUSBCFG Set usbid,adb mode,edl mode and bootloader mode	
Test Command **AT+CUSBCFG=?**	Response **BOOTLDR** **BOOTEDL** **USBADB:** (list of supported **<adb_state>**s) **USBID: <vendor_id>,**(list of supported **<product_id>**s) **OK**
Read Command **AT+CUSBCFG?**	Response **USBADB: <adb_state>** **USBID: <vendor_id>,<product_id>** **OK** or **ERROR**
Write Command /* Switch to fastboot port */ **AT+CUSBCFG=bootldr**	No Responses
/* Switch to 9008 port */ **AT+CUSBCFG=bootedl**	No Responses
/* Open/close adb port */ **AT+CUSBCFG=usbadb,<adb_state>**	Response **OK** or **ERROR**
/* Modify usb pid/vidt */ **AT+CUSBCFG=usbid,<vendor_id>,<product_id>**	Response **OK** or

	ERROR
/* Query usb mode */ **AT+CUSBCFG=USBMODE**	Response **USBMODE: <usb_state>** **OK** or **ERROR**
Execution Command **AT+CUSBCFG**	Response **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<vendor_id>	1E0E
<product_id>	9001 9011 901E 9003 902B 9B2A
<adb_state>	0 Close ADB port 1 Open ADB port
<usb_state>	USB3.0 USB2.0

Example

```
AT+CUSBCFG=?
BOOTLDR
BOOTEDL
USBADB: (0,1)
USBID: 1E0E,(9001,9011,901E,9003,902B,9B2A)
DIAG: (0,1)

OK
AT+CUSBCFG?
USBADB: 0
USBID: 0X1E0E,0X9001

OK
AT+CUSBCFG=bootldr
```

(NO RESPONSES)
AT+CUSBCFG=bootedl
(NO RESPONSES)
AT+CUSBCFG=usbadb,1

OK
AT+CUSBCFG=usbid,1e0e,9011

OK
AT+CUSBCFG=usbmode
USBMODE: USB3.0

OK

NOTE

1.When default composition is 9011 , you may need to execute AT+NETACT=1 to turn on network. After AT+CUSBCFG=usbadb,1/0 not need to excute AT+CRESET
2. AT+CUSBCFG=bootldr and AT+CUSBCFG=bootedl no response,when use those commands,the device directly enters the corresponding mode.
3. For AT+CUSBCFG=? Command that the sim8202 M2 JE project only has DIAG: (0,1) and the sim8260 project does not have DIAG: (0,1).
4. 902B UAC functionality - When connected to the 902B, an audio device will be enumerated on the computer.

12.2.10 AT+CCPUTEMP Read the temperature of CPU different zones

AT+CCPUTEMP Read the temperature of CPU different zones	
Test Command **AT+CCPUTEMP=?**	Response **+CCPUTEMP: (0-6)** **OK**
Write Command **AT+CCPUTEMP=<zone>**	Response **+CCPUTEMP: <temp>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<temp>	int type
<zone>	0-6

Example

```
AT+CCPUTEMP=2
+CCPUTEMP: 28

OK
```

12.2.11 AT+CCUART Open or close the main uart

AT+CCUART	Open or close the main uart
Test Command AT+CCUART=?	Response **+CCUART: (0-1)** **OK**
Write Command AT+CCUART=<on/off>	Response **OK** or **ERROR**
Read Command AT+CCUART?	Response **+CCUART: <on/off>** **OK** or **ERROR**
Execution Command AT+CCUART	Response **OK**
Parameter Saving Mode	AUTO_SAVE_REBOOT
Max Response Time	-
Reference	-

Defined Values

<on/off>	On: 1 Off: 0

Example

```
AT+CCUART=1
OK
AT+CCUART?
+CCUART: 0

OK
```

NOTE

1.The M2 hardware and open SDK main uart default is close.

12.2.12 AT+CWIIC I2C write command

AT+CWIIC	I2C write command
Test Command **AT+CWIIC=?**	Response **+CWIIC: <addr>,<reg_addr>,<data>,<len>** **OK**
Write Command **AT+CWIIC=<addr>,<reg_addr>,<data>,<len>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<addr>	i2c bus add of the test i2c device, hex value , don't contain 0x, it is 1a for nau8810 and 1b for rt5616
<reg_addr>	i2c device register adder, hex value,don't contain 0x;
<data>	data be writed to register, hex value,don't contain 0x
<len>	the data len (could be always 3);

Example

AT+CWIIC=1a,3a,a,3 **OK**	Write the value of the nau8810' 0x3a register to 0xa

12.2.13 AT+CRIIC I2C read command

AT+CRIIC	I2C read command
Test Command **AT+CRIIC=?**	Response **+CRIIC: <addr>,<reg_addr>,<len>** **OK**
Write Command **AT+CRIIC=<addr>,<reg_addr>,<len>**	Response **+CRIIC: <value>** **OK** or

	ERROR
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<addr>	i2c bus add of the test i2c device, hex value, don't contain 0x, it is 1a for nau8810 and 1b for rt5616;
<reg_addr>	i2c device register adder, hex value,don't contain 0x;
<len>	the data len (could be always 2);
<value>	The data in the register, hex value.

Example

AT+CRIIC=1a,40,2 **+CRIIC: 0xca** **OK**	Read the value of the nau8810' id register 0x40

13 AT Commands for UIM hot swap

13.1 Overview of AT Commands for UIM hot swap

Command	Description
AT+UIMHOTSWAPON	Set UIM hot swap function on or off
AT+UIMHOTSWAPLEVEL	Set UIM card detection level

13.2 Detailed Description of AT Commands for UIM hot swap

13.2.1 AT+UIMHOTSWAPON Set UIM hot swap function on or off

AT+UIMHOTSWAPON Set UIM hot swap function on or off	
Read Command **AT+UIMHOTSWAPON?**	Response **+UIMHOTSWAPON: <on_off>,<slot>** **+UIMHOTSWAPON: <on_off>,<slot>** **OK**
Write Command **AT+UIMHOTSWAPON=<onoff>[,<slot>]**	Response **OK** or **ERROR**
Parameter Saving Mode	AUTO_SAVE_REBOOT
Max Response Time	
Reference	

Defined Values

<on_off>	0 The UIM hot swap function is disabled 1 The UIM hot swap function is enabled
<slot>	Integer type. It indicates the slot number

	1 Slot 1 2 Slot 2

Example

```
AT+UIMHOTSWAPON?
+UIMHOTSWAPON: 0,1
+UIMHOTSWAPON: 0,2

OK
AT+UIMHOTSWAPON=1,2
OK
AT+UIMHOTSWAPON?
+UIMHOTSWAPON: 0,1
+UIMHOTSWAPON: 1,2

OK
```

NOTE

If the second parameter <slot> is omitted in write command, the default slot value is 1.
Module reset to take effect

13.2.2 AT+UIMHOTSWAPLEVEL Set UIM card detection level

AT+UIMHOTSWAPLEVEL	Set UIM card detection level
Read Command **AT+UIMHOTSWAPLEVEL?**	Response **+UIMHOTSWAPLEVEL: <level>,<slot>** **+UIMHOTSWAPLEVEL: <level>,<slot>** **OK**
Write Command **AT+UIMHOTSWAPLEVEL=<level>[,<slot>]**	Response **OK** or **ERROR**
Parameter Saving Mode	AUTO_SAVE_REBOOT
Max Response Time	
Reference	

Defined Values

<level>	0 ACTIVE LOW 1 ACTIVE HIGH
<slot>	Integer type. It indicates the slot number 1 Slot 1 2 Slot 2

Example

```
AT+UIMHOTSWAPLEVEL?
+UIMHOTSWAPLEVEL: 0,1
+UIMHOTSWAPLEVEL: 0,2

OK
AT+UIMHOTSWAPLEVEL=1,2
OK
AT+UIMHOTSWAPLEVEL?
+UIMHOTSWAPLEVEL: 0,1
+UIMHOTSWAPLEVEL: 1,2

OK
```

NOTE

If the second parameter <slot> is omitted in write command, the default slot value is 1.
Module reset to take effect
UIM card detection level depends on the SIM card holder, usually it's a "normal open kind" one.
The default value 1

14 AT Commands for File System

The file system is used to store files in a hierarchical (tree) structure, and there are some definitions and conventions to use the Module.

Local storage space is mapped to "C:", "D:" for TF card, "E:" for multimedia, "F:" for cache.

NOTE

General rules for naming (both directories and files):
The length of actual fully qualified names of directories and files can not exceed 254.
Directory and file names can not include the following characters: \ : * ? " <> | , ;
Between directory name and file/directory name, use character "/" as list separator, so it can not appear in directory name or file name.
The first character of names must be a letter or a numeral or underline, and the last character can not be period "." and oblique "/".

14.1 Overview of AT Commands for File System

Command	Description
AT+FSCD	Select directory as current directory
AT+FSMKDIR	Make new directory in current directory
AT+FSRMDIR	Delete directory in current directory
AT+FSLS	List directories/files in current directory
AT+FSDEL	Delete file in current directory
AT+FSRENAME	Rename file in current directory
AT+FSATTRI	Request file attributes
AT+FSMEM	Check the size of available memory
AT+FSLOCA	Select storage place
AT+FSCOPY	Copy an appointed file
AT+CFTRANRX	Transfer a file to EFS
AT+CFTRANTX	Transfer a file from EFS to host

14.2 Detailed Description of AT Commands for File System

14.2.1 AT+FSCD Select directory as current directory

This command is used to select a directory. The Module supports absolute path and relative path. Read Command will return current directory without double quotation marks. Support "C:", "D:", "E:", "F:".

AT+FSCD Select directory as current directory	
Test Command AT+FSCD=?	Response **OK**
Read Command AT+FSCD?	Response **+FSCD: <curr_path>** **OK**
Write Command AT+FSCD=<path>	Response **+FSCD: <curr_path>** **OK** or **ERROR**

Defined Values

<path>	String without double quotes, directory for selection.
<curr_path>	String without double quotes, current directory.

NOTE

If <**path**> is "..", it will go back to previous level of directory.

Example

```
AT+FSCD=C:
+FSCD: C:/

OK
AT+FSCD=C:/
+FSCD: C:/
```

```
OK
AT+FSCD?
+FSCD: C:/

OK
AT+FSCD=..
+FSCD: C:/

OK
AT+FSCD=D:
+FSCD: D:/

OK
AT+FSCD?
+FSCD: D:/

OK
```

14.2.2 AT+FSMKDIR Make new directory in current directory

This command is used to create a new directory in current directory. Support "C:", "D:", "E:", "F:".

AT+FSMKDIR Make new directory in current directory	
Test Command AT+FSMKDIR=?	Response **OK**
Write Command AT+FSMKDIR=<dir>	Response **OK** or **ERROR**

Defined Values

<dir>	String without double quotes, directory name which does not already exist in current directory.

Example

```
AT+FSMKDIR=SIMTech
OK
AT+FSCD?
```

```
+FSCD: E:/

OK
AT+FSLS
+FSLS: SUBDIRECTORIES:
Audio
SIMTech

OK
```

14.2.3 AT+FSRMDIR Delete directory in current directory

This command is used to delete existing directory in current directory. Support "C:", "D:", "E:", "F:".

AT+FSRMDIR Delete directory in current directory	
Test Command AT+FSRMDIR=?	Response OK
Write Command AT+FSRMDIR=<dir>	Response OK or ERROR

Defined Values

<dir>	String without double quotes.

Example

```
AT+FSRMDIR=SIMTech
OK
AT+FSCD?
+FSCD: E:/

OK
AT+FSLS
+FSLS: SUBDIRECTORIES:
Audio

OK
```

14.2.4 AT+FSLS List directories/files in current directory

This command is used to list information of directories and/or files in current directory. Support "C:", "D:", "E:", "F:".

AT+FSLS List directories/files in current directory	
Test Command **AT+FSLS=?**	Response **+FSLS: (**list of supported **<type>)** **OK**
Read Command **AT+FSLS?**	Response **+FSLS: SUBDIRECTORIES:<dir_num>,FILES:<file_num>** **OK**
Write Command **AT+FSLS=<type>**	Response **[+FSLS: SUBDIRECTORIES:** **<list of subdirectories>** **<CR><LF>]** **[+FSLS: FILES:** **<list of files>** **<CR><LF>]** **OK**
Execution Command **AT+FSLS**	**[+FSLS: SUBDIRECTORIES:** **<list of subdirectories>** **<CR><LF>]** **[+FSLS: FILES:** **<list of files>** **<CR><LF>]** **OK**

Defined Values

<dir_num>	Integer type, the number of subdirectories in current directory.
<file_num>	Integer type, the number of files in current directory.
<type>	0 list both subdirectories and files 1 list subdirectories only 2 list files only

Example

AT+FSLS?
+FSLS: SUBDIRECTORIES:2,FILES:2

OK
AT+FSLS
+FSLS: SUBDIRECTORIES:
FirstDir
SecondDir

+FSLS: FILES:
image_0.jpg
image_1.jpg

OK
AT+FSLS=2
+FSLS: FILES:
image_0.jpg
image_1.jpg

OK

14.2.5 AT+FSDEL Delete file in current directory

This command is used to delete a file in current directory. Before do that, it needs to use AT+FSCD select the father directory as current directory. Support "C:", "D:", "E:", "F:".

AT+FSDEL Delete file in current directory	
Test Command **AT+FSDEL=?**	Response **OK**
Write Command **AT+FSDEL=<filename>**	Response **OK** or **ERROR**

Defined Values

<filename>	String with or without double quotes, file name which is relative and already existing. If **<filename>** is *.*, it means delete all files in current directory. If the file path contains non-ASCII characters, the filename parameter should contain a prefix of {non-ascii} and the quotation mark.

Example

```
AT+FSDEL=image_0.jpg
OK
```

14.2.6 AT+FSRENAME Rename file in current directory

This command is used to rename a file in current directory. Support "C:", "D:", "E:", "F:".

AT+FSRENAME Rename file in current directory	
Test Command **AT+FSRENAME=?**	Response **OK**
Write Command **AT+FSRENAME=<old_name>,<new_name>**	Response **OK** or **ERROR**

Defined Values

<old_name>	String with or without double quotes, file name which is existed in current directory. If the file path contains non-ASCII characters, the file path parameter should contain a prefix of {non-ascii} and the quotation mark.
<new_name>	New name of specified file, string with or without double quotes. If the file path contains non-ASCII characters, the file path parameter should contain a prefix of {non-ascii} and the quotation mark.

Example

```
AT+FSRENAME=image_0.jpg,image_1.jpg
OK
AT+FSRENAME="my test.jpg",{non-ascii}"E6B58BE8AF95E99984E4BBB62E6A7067"
OK
```

14.2.7 AT+FSATTRI Request file attributes

This command is used to request the attributes of file which exists in current directory. Support "C:", "D:", "E:", "F:".

AT+FSATTRI Request file attributes

Test Command **AT+FSATTRI=?**	Response **OK**
Write Command **AT+FSATTRI=<filename>**	Response **+FSATTRI: <file_size>,<create_date>** **OK** or **ERROR**

Defined Values

<filename>	String with or without double quotes, file name which is in current directory. If the file path contains non-ASCII characters, the file path parameter should contain a prefix of {non-ascii} and the quotation mark.
<file_size>	The size of specified file, and the unit is in Byte.
<create_date>	Create date and time of specified file, the format is YYYY/MM/DD HH:MM:SS Week. Week Mon, Tue, Wed, Thu, Fri, Sat, Sun

Example

```
AT+FSATTRI=image_0.jpg
+FSATTRI: 8604,2008/04/28 10:24:46 Tue

OK
AT+FSATTRI={non-ascii}"E6B58BE8AF95E99984E4BBB62E6A7067"
+FSATTRI: 6296,2012/01/06 00:00:00 Sun

OK
```

14.2.8 AT+FSMEM Check the size of available memory

This command is used to check the size of available memory. The response will list total size and used size of local storage space if present and mounted. Support "C:", "D:", "E:", "F:".

AT+FSMEM Check the size of available memory	
Test Command **AT+FSMEM=?**	Response **OK**
Execution Command **AT+FSMEM**	Response **+FSMEM: <loctype>:(<total>,<used>)**

	OK

Defined Values

<loctype>	Support "C:", "D:", "E:", "F:".
<total>	The total size of local storage space.The unit of storage space size is in Byte.
<used>	The used size of local storage space.The unit of storage space size is in Byte.

Example

AT+FSMEM
+FSMEM: C:(11348480,2201600)

OK

14.2.9 AT+FSLOCA Select storage place

This command is used to set the storage place for media files. Support "C:".

AT+FSLOCA Select storage place	
Test Command **AT+FSLOCA=?**	Response **+FSLOCA: (**list of supported **<loca>s)** **OK**
Read Command **AT+FSLOCA?**	**+FSLOCA: <loca>** **OK**
Write Command **AT+FSLOCA=<loca>**	Response **OK** or **ERROR**

Defined Values

<loca>	0 store media files to local storage space (namely "C:/")

Example

```
AT+FSLOCA=0
OK
AT+FSLOCA?
+FSLOCA: 0

OK
```

14.2.10 AT+FSCOPY Copy an appointed file

This command is used to copy an appointed file on C:/ to an appointed directory on C:/, the new file name should give in parameter. Support "C:","D:", "E:", "F:", but copying from "C:" to "D:", "E:", "F:" or from "D:", "E:", "F:" to "C:" is not supported.

AT+FSCOPY Copy an appointed file	
Test Command **AT+FSCOPY=?**	Response **OK**
Write Command **AT+FSCOPY=<file1>,<file2>[,<sync_mode>]**	Response **Sync mode** **+FSCOPY: <percent><CR><LF>** **[+FSCOPY: <percent><CR><LF>]** **OK** **Async mode** **OK** **+FSCOPY: <percent><CR><LF>** **[+FSCOPY: <percent><CR><LF>]** **+FSCOPY: END<CR><LF>** Or **When error, shows one of the following errors and ERROR** **SD CARD NOT PLUGGED IN** **FILE IS EXISTING** **FILE NOT EXISTING** **DIRECTORY IS EXISTED** **DIRECTORY NOT EXISTED** **FORBID CREATE DIRECTORY UNDER \"C:/\"** **FORBID DELETE DIRECTORY** **INVALID PATH NAME** **INVALID FILE NAME** **NO ENOUGH MEMORY** **FILE CREATE ERROR** **READ FILE ERROR**

	WRITE FILE ERROR ERROR

Defined Values

<file1>	The sources file name or the whole path name with sources file name. If the file path contains non-ASCII characters, the file path parameter should contain a prefix of {non-ascii} and the quotation mark.
<file2>	The destination file name or the whole path name with destination file name. If the file path contains non-ASCII characters, the file path parameter should contain a prefix of {non-ascii} and the quotation mark.
<percent>	The percent of copy done. The range is 0.0 to 100.0
<sync_mode>	The execution mode of the command: 0 synchronous mode. Default 0 1 asynchronous mode.

NOTE

1. The **<file1>** and **<file2>** should give the whole path and name, if only given file name, it will refer to current path (**AT+FSCD**) and check the file's validity.
2. If **<file2>** is a whole path and name, make sure the directory exists, make sure that the file name does not exist or the file name is not the same name as the sub folder name, otherwise return error.
3. **<percent>** report refer to the copy file size. The big file maybe report many times, and little file report less.
4. If **<sync_mode>** is 1, the command will return **OK** immediately, and report final result with **+FSCOPY: END**.

Example

```
AT+FSCD?
+FSCD: C:/

OK
AT+FSCOPY=C:/TESTFILE,COPYFILE   (Copy file TESTFILE on C:/ to C:/COPYFILE)
+FSCOPY: 1.0

+FSCOPY: 100.0

OK
AT+FSCOPY="my test.jpg",{non-ascii}"E6B58BE8AF95E99984E4BBB62E6A7067"
```

```
+FSCOPY: 1.0

+FSCOPY: 100.0

OK
```

14.2.11 AT+CFTRANRX Transfer a file to EFS

This command is used to transfer a file to EFS.Support SDcard.

AT+CFTRANRX Transfer a file to EFS	
Test Command **AT+CFTRANRX=?**	Response **+CFTRANRX: [{non-ascii}]"FILEPATH"** **OK**
Write Command **AT+CFTRANRX="<filepath>",<len>**	Response **>** **OK** or **>** **ERROR** or **ERROR**

Defined Values

<filepath>	The path of the file on EFS.
<len>	The length of the file data to send. The range is from 1 to 2147483647.

NOTE

The **<filepath>** must be a full path with the directory path.

Example

AT+CFTRANRX="C:/MyDir/t1.txt",10
>*<input data here>*
OK
AT+CFTRANRX="D:/MyDir/t1.txt",10
>*<input data here>*

OK

14.2.12 AT+CFTRANTX Transfer a file from EFS to host

This command is used to transfer a file from EFS to host. Before using this command, the AT+CATR must be used to set the correct port used. Support SDcard.

AT+CFTRANTX Transfer a file from EFS to host	
Test Command **AT+CFTRANTX=?**	Response **+CFTRANTX: [{non-ascii}]"FILEPATH"** **OK**
Write Command **AT+CFTRANTX ="<filepath>"[,<location>,<size>]**	Response **[+CFTRANTX: DATA,<len>** **…** **+CFTRANTX: DATA,<len>]** **+CFTRANTX: 0** **OK** or **ERROR**

Defined Values

<filepath>	The path of the file on EFS.
<len>	The length of the following file data to output.
<location>	The beginning of the file data to output.
<size>	The length of the file data to output.

NOTE

The **<filepath>** must be a full path with the directory path.

Example

AT+CFTRANTX="C:/MyDir/t1.txt"
+CFTRANTX: DATA,11
Testcontent

+CFTRANTX: 0

```
OK
AT+CFTRANTX="D:/MyDir/t1.txt",1,4
+CFTRANTX: DATA,4
estc

+CFTRANTX: 0

OK
```

15 AT Commands for AUDIO

15.1 Overview of AT Commands for AUDIO

Command	Description
AT+CREC	Record wav audio file
AT+CRECAMR	Record amr audio file
AT+CCMXPLAY	Play audio file
AT+CCMXSTOP	Stop playing audio file

15.2 Detailed Description of AT Commands for AUDIO

15.2.1 AT+CREC Record wav audio file

AT+CREC Record wav audio file	
Read Command **AT+CREC?**	Response **+CREC: <status>** **OK**
Write Command **AT+CREC=<record_path>,<filename>**	Response **+CREC: 1** **OK** or **ERROR**
Write Command **AT+CREC=<mode>**	Response **+CREC: 0** **OK** **+RECSTATE: crec stop**

	or **+CREC: 0** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<status>	Indicate whether the recording is going on. 0 – free, not recording 1 – busy, recording
<record_path>	Source of recorded sound 1 – local path 2 – remote path 3 – local and remote sound mixing
<filename>	The location and name of wav file.
<mode>	Stop recording wav audio file 0 – stop

NOTE

<filename>,The file should be put into the "E:/". Maximum filename length is 90 bytes. (including "")
<record_path>,Only during the call, <record_path> can be set to 2 or 3

Example

```
AT+CREC=1,"e:/rec.wav"
+CREC: 1

OK

AT+CREC=0
+CREC: 0

OK

+RECSTATE: crec stop
```

15.2.2 AT+CRECAMR Record amr audio file

AT+CRECAMR Record amr audio file	
Read Command **AT+CRECAMR?**	Response **+CRECAMR: <status>** **OK**
Write Command **AT+CRECAMR=<record_path>,<filename>**	Response **+CRECAMR: <status>** **OK** or **ERROR**
Write Command **AT+CRECAMR=<mode>**	Response **+CRECAMR: <status>** **OK** **+RECSTATE: crecamr stop** or **+CRECAMR: <status>** **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<status>	Indicate whether the recording is going on. 0 – free, not recording 1 – busy, recording
<record_path>	Source of recorded sound 1 – local path 2 – remote path
<filename>	The location and name of amr file.
<mode>	Stop recording wav audio file 0 – stop

NOTE

<filename>,The wav audio file should be located at "E:/". Maximum filename length is 90 bytes.

(including ""). Support audio file format mp3, aac, amr, wav.
<play_path>,Only during the call, <play_path>can be set to 1 successfully.Only 8k 16bit wav audio and amr audio can be played to remote at present.Only wav audio and amr audio can be played to local during the call,other format don't care.
<repeat>,This parameter is reserved,not used at present, you can input this parameter or not. (0-255)

Example

AT+CRECAMR=1,"e:/rec.amr"
+CRECAMR: 1

OK

AT+CRECAMR=0
+CRECAMR: 0

OK

+RECSTATE: crecamr stop

15.2.3 AT+CCMXPLAY Play audio file

AT+CCMXPLAY Play audio file	
Test Command **AT+CCMXPLAY=?**	Response **+CCMXPLAY: (0-1),(0-255)** **OK**
Read Command **AT+CCMXPLAY?**	Response **+CCMXPLAY: <status>** **OK**
Write Command **AT+CCMXPLAY=<filename>[,<play_path>][,<repeat>]**	Response **+CCMXPLAY:** **OK** **+AUDIOSTATE: audio play** **+AUDIOSTATE: audio play stop** or

	ERROR or **+CCMXPLAY:** **OK** **+AUDIOSTATE: audio play** **+AUDIOSTATE: audio play error** or **ERROR** **+CCMXPLAY: not enough space to play**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<play_path>	Play to local or to remote. Default 0 0 – local 1 – remote
<repeat>	How much times can be played. Default 0
<filename>	The location and name of wav file.
<status>	Indicate playing thread status. Default value is 0 0 – idle 1 – busy, playing

NOTE

<filename>,The wav audio file should be located at "E:/". Maximum filename length is 90 bytes. (including ""). Support audio file format mp3, aac, amr, wav.
<play_path>,Only during the call, <play_path>can be set to 1 successfully.Only 8k 16bit wav audio and amr audio can be played to remote at present.
<repeat>,This parameter is reserved,not used at present, you can input this parameter or not. (0-255)

Example

```
AT+CCMXPLAY=?
+CCMXPLAY: (0-1),(0-255)

OK

AT+CCMXPLAY="E:/rec.mp3",0,0
```

```
+CCMXPLAY:

OK

+AUDIOSTATE: audio play

+AUDIOSTATE: audio play stop
```

15.2.4 AT+CCMXSTOP Stop playing audio file

AT+CCMXSTOP Stop playing audio file	
Test Command AT+CCMXSTOP=?	Response **OK**
Execution Command AT+CCMXSTOP	Response **+CCMXSTOP:** **OK** **+AUDIOSTATE: audio play stop** or **+CCMXSTOP:** **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

-	-

NOTE

"**+AUDIOSTATE: audio play stop**" can be returned only if audio is playing.

Example

```
AT+CCMXSTOP
+CCMXSTOP:

OK
```

+AUDIOSTATE: audio play stop

16 AT Commands for TTS

16.1 Overview of AT Commands for TTS

Command	Description
AT+CDTAM	TTS play path ,local or remote
AT+CTTS	TTS operation ,play or stop
AT+CTTSPARAM	TTS parameters ,set or get

16.2 Detailed Description of AT Commands for TTS

16.2.1 AT+CDTAM TTS play path ,local or remote

AT+CDTAM TTS play path ,local or remote	
Test Command **AT+CDTAM=?**	Response **+CDTAM: (0-1)** **OK**
Read Command **AT+CDTAM?**	Response **+CDTAM: <status>** **OK**
Write Command **AT+CDTAM=<mode>**	Response **+CDTAM:** **OK** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<status>	Indicate play path, play TTS to local or play to remote. 0 – local path 1 – remote path
<mode>	Set TTS play path, local or remote. Default value is 0. 0 – local path 1 – remote path

Example

```
AT+CDTAM=1
+CDTAM:

OK
```

16.2.2 AT+CTTS TTS operation ,play or stop

AT+CTTS TTS operation ,play or stop	
Test Command **AT+CTTS=?**	Response **OK**
Read Command **AT+CTTS?**	Response **+CTTS: <status>** **OK**
Write Command **AT+CTTS=<mode>[,<text>]**	Response If **<mode>**is 0, then**<text>** is not required. When TTS is playing, return: **+CTTS:0** **OK** If **<mode>**is 0, then **<text>** is not required. When TTS is not playing, return: **OK** If **<mode>**is 1 or 2, then **<text>** is must be required. return: **OK** **+CTTS:0**

	or **ERROR** or **OK** **+CTTS:1**
Write Command **AT+CTTS=<mode>[,<text>][,<filename>]**	Response If **<mode>**is 3 or 4, then **<text>** and **<filename>** are must be required. return: **OK** **+CTTS:0** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<status>	Indicate playing thread status. Default value is 0 0 NO_WORKING 1 PLAY_WAV_WORKING 2 AMR_WORKING 3 MP3_WORKING 4 AAC_WORKING 5 WAV_WORKING 6 TTS_WORKING 8 CREC_WORKING
<mode>	Stop or play TTS. 0 stop TTS 1 **<text>** is in UCS2 coding format, Start to synth and play 2 **<text>** is in ASCII coding format for English,Chinese text is in GBK coding format. Start to synth and play 3 **<text>** is in ASCII coding format for English,Chinese text is in GBK coding format. Start to synth and play, and save pcm data as wav file. 4 **<text>** is in UCS2 coding format . Start to synth and play, and save pcm data as wav file.
<filename>	Location and filename for wav file

Example

```
AT+CTTS=1,"6B228FCE4F7F75288BED97F3540862107CFB7EDF"
```

OK

+CTTS:0

AT+CTTS=3,"欢迎使用语音合成系统","E:/tts.wav"
OK

+CTTS:0

AT+CTTS=0
OK

16.2.3 AT+CTTSPARAM TTS Parameters ,set or get

AT+CTTSPARAM TTS Parameters ,set or get	
Test Command AT+CTTSPARAM=?	Response **+CTTSPARAM: (0-2),(0-3),(0-3),(0-2),(0-2)** **OK**
Read Command AT+CTTSPARAM?	Response **+CTTSPARAM: <volume>,<sysvolume>,<digitmode>,<pitch>,<speed>** **OK**
Write Command AT+CTTSPARAM=<volume>[,<sysvolume>[,<digitmode>[,<pitch>[,<speed>]]]]	Response **OK** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<volume>	TTS Speech Volume, default:2. 0 – the mix volume 1 – the normal volume 2 – the max volume
<sysvolume>	The module system volume,default:3. 0 – the mix system volume 1 – the small system volume

	2 – the normal system volume 3 – the max system volume
<digitmode>	The digit read mode, default:0 0 – auto read digit based on number rule first. 1 – auto read digit bases on telegram rule first. 2 – read digit based on telegram rule. 3 – read digit based on number rule.
<pitch>	The voice tone, default:1 0 – the mix voice tone. 1 – the normal voice tone. 2 – the max voice tone.
<speed>	The voice speed, default:1 0 – the mix speed 1 – the normal speed 2 – the max speed

NOTE

<sysvolume>,It takes no effect to set**<sysvolume>**,reserved at present

Example

```
AT+CTTSPARAM=2,3,0,1,1
OK
```

17 AT Commands for MiFi

17.1 Overview of AT Commands for MiFi

Command	Description
AT+CWMAP	Open/Close MiFi
AT+CWSSID	SSID setting
AT+CWBCAST	Broadcast setting
AT+CWAUTH	Authentication type, encrypt mode and password setting
AT+CWMOCH	80211 mode and channel setting
AT+CWISO	Client isolation setting
AT+CWDHCP	Get the current DHCP configuration
AT+CWNAT	NAT type setting
AT+CWCLICNT	Get client number connected to the MiFi
AT+CWRSTD	Restore to default setting
AT+CWMAPCFG	MiFi configuration setting
AT+CWMACADDR	Get MAC address
AT+CWNETCNCT	Query the connection to the network
AT+CWSTASCAN	Scan WIFI network
AT+CWSTACFG	STA mode configuration setting
AT+CWSTAIP	Get STA mode IP address
AT+CWETHMODE	Ethernet mode configuration setting

17.2 Detailed Description of AT Commands for MiFi

17.2.1 AT+CWMAP Open/Close MiFi

AT+CWMAP Open/Close MiFi	
Test Command AT+CWMAP=?	Response **+CWMAP: (0-1),(0-1)**

	OK
Read Command AT+CWMAP?	Response +CWMAP: <cur_mifi_state>,<reboot_mifi_state> OK
Write Command AT+CWMAP=<cur_mifi_state>[,<reboot_mifi_state>]	Response OK or ERROR
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<cur_mifi_state>	0 – Current MiFi off 1 – Current MiFi on
<reboot_mifi_state>	0 – MiFi off after reboot the module 1 – MiFi on after reboot the module
	The default value of cur_mifi_state and reboot_mifi_state are both 1

Example

```
AT+CWMAP?
+CWMAP: 1,1

OK

AT+CWMAP=0
OK

AT+CWMAP?
+CWMAP: 0,1

OK

AT+CWMAP=0,0
OK

AT+CWMAP?
+CWMAP: 0,0
```

```
OK


AT+CWMAP=1
OK

AT+CWMAP?
+CWMAP: 1,0

OK


AT+CWMAP=0,1
OK

AT+CWMAP?
+CWMAP: 0,1

OK

AT+CWMAP=1
OK

AT+CWMAP?
+CWMAP: 1,1

OK
```

17.2.2 AT+CWSSID SSID setting

AT+CWSSID SSID setting	
Read Command **AT+CWSSID?**	Response **+CWSSID: <ssid>** **OK**
Write Command **AT+CWSSID=<ssid>**	Response **OK** or **ERROR**
Parameter Saving Mode	-

Max Response Time	-
Reference	-

Defined Values

<ssid>	Ssid string 1. The max length of <ssid> is 32 bytes when the <ssid> include only ASCII characters. 2. The max length of <ssid> is 20 bytes when <ssid> include only Chinese (One Chinese characters is 2 bytes, so the max Chinese count is 10). 3. The max length of <ssid> is 22 bytes when <ssid> include ASCII and Chinese characters (One Chinese character is 2 bytes, one ASCII character is 1 byte). The default ssid is SIM8X00AP.

Example

```
AT+CWSSID?
+CWSSID: "SIM8X00AP"

OK

AT+CWSSID="SIM8200"
OK
```

17.2.3 AT+CWBCAST Broadcast setting

AT+CWBCAST Broadcast setting	
Test Command **AT+CWBCAST=?**	Response **+CWBCAST: (0-1)** **OK**
Read Command **AT+CWBCAST?**	Response **+CWBCAST: <broadcast>** **OK**
Write Command **AT+CWBCAST=<broadcast>**	Response **OK** or

	ERROR
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<broadcast>	Broadcast parameter, default: 1. 0 – disabled 1 – enabled

Example

```
AT+CWBCAST?
+CWBCAST: 1

OK

AT+CWBCAST=0
OK
```

17.2.4 AT+CWAUTH Authentication setting

AT+CWAUTH Authentication type, encrypt mode and password setting	
Read Command **AT+CWAUTH?**	Response **+CWAUTH: <auth>,<encrypt>[,<password>]** **OK**
Write Command **AT+CWAUTH=<auth>,<encrypt>[,<password>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<auth>	Authentication type parameter, default:4. 0 - auto

	1 - open 2 - share 3 - wpa 4 - wpa2 5 - wpa/wpa2 6 - wpa3 7 - wpa2/wpa3
<encrypt>	Encrypt mode, default:3 0 - null 1 - WEP 2 - TKIP 3 - AES 4 - TKIP-AES
<password>	password string, the length is 5 or between 8 to 64. The char in the password is only allow the ASCII 's decimal code between 32 to 126. The parameter need to meet the following conditions: 1. If (auth = 0 or auth = 1) then (encrypt = 0 or encrypt = 1) 2. If (auth =2) then (encrypt = 1) 3. If (auth >=3) then (encrypt >=2) 4. If(encrypt = 0) then (password is null) 5. If(encrypt = 1) then 1) password can't be set null 2) password format: (5 ASCII characters) or (10 hexadecimal numbers) or(13 ASCII characters) or(26 hexadecimal numbers) 6. if(encrypt >= 2) then 1) password can't be set null 2) password format: (8~63 ASCII characters or 64 hexadecimal numbers)

Example

```
AT+CWAUTH?
+CWAUTH: 4,3,"1234567890"

OK

AT+CWAUTH=5,4,"abcd1234"
OK
```

17.2.5 AT+CWMOCH 80211 mode,channel and bandwidth setting

AT+CWMOCH 80211 mode,channel and bandwidth setting	
Read Command **AT+CWMOCH?**	Response **+CWMOCH: <mode>,<channel>[,<bandwidth>]** **OK**
Write Command **AT+CWMOCH=<mode>,<channel>[,<bandwidth>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<mode>	80211 mode parameter, default：4. 1 a/n 5G mode 2 b 2.4G mode 3 b/g 2.4G mode 4 b/g/n 2.4G mode 5 ac/n 5G mode 6 ax WIFI6/6E mode
<channel>	Channel parameter, default: 0 0 auto select 1~13 2.4G mode channel number 36/40/44/48/149/153/157/161/165 5G mode channel number 191/195/199/203/207/211/215/219/223/227/231/235/239/243/247/251/255/259/263/267/271/275/279/283/287/291/295/299/303/307/311/315/319/323/327/331/335/339/343/347/351/355/359/363/367/371/375/379/383/387/391/395/399/403/407/411/415/419/423 WIFI 6E 6G mode channel number When <mode> is 1/5 <channel> can be set with 5G mode channel number,or set 0 with auto select. When <mode> is 2/3/4, <channel> range is 0~11 When <mode> is 6, <channel> can be set with 2.4G/5G/6G mode channel number,or set 0 with auto select.
<bandwidth>	MiFi band width, it is an optional parameter. 0 20MHz band width vht_oper_chwidth is 0 and vht_capab is HT20

1 40MHz band width
vht_oper_chwidth is 0 and vht_capab is HT40+

2 40MHz band width
vht_oper_chwidth is 0 and vht_capab is HT40-

3 80MHz band width
vht_oper_chwidth is 1 and vht_capab is HT40+

4 80MHz band width
vht_oper_chwidth is 1 and vht_capab is HT40-

5 160MHz band width
vht_oper_chwidth is 2 and vht_capab is HT40+

6 160MHz band width
vht_oper_chwidth is 2 and vht_capab is HT40-

When <bandwidth> is not set, the band width will be set automatically, but if it is set, please fellow the following rules:
(1) When <mode> is set as 2/3/4, or <mode> is 6 and <channel> is set 1-11, <bandwidth> can be only set 0~2
(2) When <channel> is set 1~4, <bandwidth> can't be set to 2, and when <channel> is set 8~11, <bandwidth> can't be set to 1
(3) When <bandwidth> is set, 5G channels can be set 36/40/44/48/52/56/60/ 64/100/104/108/112/116/120/124/128/132/136/140/144/149/153/ 157/ 161/165
As 52/56/60/ 64/100/104/108/112/116/120/124/128/132/136/140/144 are radar detections, they are not recommended to use.
(4)When <channel> is set 36/100/149, <bandwidth> can't be set to 2/4
(5)When <channel> is set 64/144/165, <bandwidth> can't be set to 1/3
(6)When <channel> is set 36/44/149/157, and we want to get 80M Hz band with, the <bandwidth> option must be set to 3
(7)When <channel> is set 40/48/153/161, and we want to get 80M Hz band with, the <bandwidth> option must be set to 4.
(8)For 5G mode,When <bandwidth> is set to 5, <mode> must be 6, and <channel> must be 36.
(9)The <bandwidth> 5 and 6 is only supported in W82.
(10) For WIFI 6E 6G mode, <bandwidth> must be set. And <auth> in +CWAUTH command must be set with 6(wpa3) firstly.
If <channel> is set to 191, <bandwidth> can't be set to 2/4/6.

	If <bandwidth> is set to 1/2, <channel> can't be set to 423. If <bandwidth> is set to 3/4, <channel> can't be set to 415/419/423. If <bandwidth> is set to 5/6, <channel> can't be set to 415/419/423.

Example

```
AT+CWMOCH?
+CWMOCH: 4,6

OK

AT+CWMOCH=4,0
OK
```

17.2.6 AT+CWISO Client isolation setting

AT+CWISO Client isolation setting	
Test Command **AT+CWISO=?**	Response **+CWISO: (0-1)** **OK**
Read Command **AT+CWISO?**	Response **+CWISO: <isolation>** **OK**
Write Command **AT+CWISO=<isolation>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<isolation>	Isolation parameter, default: 0 0 – close 1 – open

Example

```
AT+CWISO?
+CWISO: 0

OK

AT+CWISO=1
OK
```

17.2.7 AT+CWDHCP Get the current DHCP configuration

AT+CWDHCP Get the current DHCP configuration	
Read Command **AT+CWDHCP?**	Response **+CWDHCP:** **<host_ip>,<range_start_ip>,<range_end_ip>,<leasetime>** **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<host_ip>	The AP IP address
<range_start_ip>	The start IP of the IP range that assigned to the client
<range_end_ip>	The end IP of the IP range that assigned to the client
<leasetime>	The lease time

Example

```
AT+CWDHCP?
+CWDHCP:
"192.168.225.1","192.168.225.20","192.168.225.60",12h

OK
```

17.2.8 AT+CWNAT NAT type setting

AT+CWNAT NAT type setting	
Test Command **AT+CWNAT=?**	Response **+CWNAT: (0-1)[,"IP Address"]** **OK**
Read Command **AT+CWNAT?**	Response **+CWNAT: <type>,"<IP Address>"** **OK**
Write Command **AT+CWNAT=<type>[,"<IP Address>"]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<type>	NAT type parameter, default:0. 0 Symmetric 1 Cone
<IP Address>	String type, indicates the IP address corresponding to the domain name.

Example

```
AT+CWNAT?
+CWNAT: 0 ,"0.0.0.0"

OK
AT+CWNAT=1
OK
AT+CWNAT=1, "192.168.225.16"
OK
```

17.2.9 AT+CWCLICNT Get client number connected to the MiFi

AT+CWCLICNT Get client number connected to the MiFi

Read Command AT+CWCLICNT?	Response **+CWCLICNT: <cnt>** **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<cnt>	the connected client count, range is from 0 to 32.

Example

```
AT+CWCLICNT?
+CWCLICNT: 1

OK
```

17.2.10 AT+CWRSTD Restore to default setting

AT+CWRSTD Restore to default setting	
Execution Command AT+CWRSTD	Response **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-The module will reboot after restore

Defined Values

Example

```
AT+CWRSTD
OK
```

17.2.11 AT+CWMAPCFG MiFi configuration setting

AT+CWMAPCFG MiFi configuration setting	
Test Command **AT+CWMAPCFG=?**	Response **+CWMAPCFG: ("enablessid2","configselect"),(0-2)** **OK**
Read Command **AT+CWMAPCFG?**	Response **+CWMAPCFG: <enablessid2_value>,<configselect_value>** **OK**
Write Command **AT+CWMAPCFG=<option>,<value>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<enablessid2_value>	0 – AP mode 1 – AP-AP mode 2 – STA-AP mode
<configselect_value>	Current AP ID: 0,1,2
<option>	"enablessid2" set MiFi mode "configselect" set the current AP ID

<value>	The value of the options
	
	If (option="enablessid2")
	0 AP mode
	1 AP-AP mode
	2 STA-AP mode
	
	If (option="configselect")
	Current AP ID (0 or 1 or 2) to be set.
	
	When current AP ID is 0, the AT+CWSSID/AT+CWBCAST/AT+CWAUTH/AT+CWMOCH/AT+CWISO/AT+CWDHCP/AT+CWCLICNT/AT+CWMACADDR will modify the first AP's settings;

	When current AP ID is 1, the AT+CWSSID/AT+CWBCAST/AT+CWAUTH/AT+CWMOCH/AT+CWISO/ AT+CWDHCP/AT+CWCLICNT/AT+CWMACADDR will modify the second AP's settings; When current AP ID is 2, the AT+CWSSID/AT+CWBCAST/AT+CWAUTH/AT+CWMOCH/AT+CWISO/ AT+CWDHCP/AT+CWCLICNT/AT+CWMACADDR will modify the third AP's settings, the AT+CWSTAIP/AT+CWSTASCAN/AT+CWSTACFG will modify the STA's settings.

NOTE

- It can't set the configselect value to 1 when enablessid2 is 0.
- The configselect value will be changed due to enablessid2.

enablessid2	configselect
0	0
1	0 or 1
2	2

Example

```
AT+CWMAPCFG=?
+CWMAPCFG: ("enablessid2","configselect"),(0-2)

OK
AT+CWMAPCFG?
+CWMAPCFG: 0,0

OK
AT+CWMAPCFG="enablessid2",1
OK
AT+CWMAPCFG="configselect",1
OK
AT+CWMAPCFG="enablessid2",0
OK
```

17.2.12 AT+CWMACADDR Get MAC address

AT+CWMACADDR Get MAC address

Read Command **AT+CWMACADDR?**	Response **OK** **+CWMACADDR:** **<number>,<mac_addr>** **[... ...]** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<number>	0 – host mac addr 1– client mac addr ... client mac addr
<mac_addr>	Device mac address

Example

```
AT+CWMACADDR?
OK

+CWMACADDR:
0,02:03:7F:93:62:62
1,cc:29:f5:e0:ef:f3
```

17.2.13 AT+CWNETCNCT Query the connection to the network

AT+CWNETCNCT Query the connection to the network	
Read Command **AT+CWNETCNCT?**	Response **+CWNETCNCT: <flag>** **OK**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<flag>	0 disconnect 1 connect

Example

AT+CWNETCNCT?
+CWNETCNCT: 1

OK

17.2.14 AT+CWSTASCAN Scan WIFI network

AT+CWSTASCAN Scan WIFI network	
Read Command **AT+CWSTASCAN?**	Response **+CWSTASCAN: <flag_show_signal>** **OK**
Write Command **AT+CWSTASCAN=<flag_show_signal>**	Response **OK** or **ERROR**
Execute Command **AT+CWSTASCAN**	**OK** **[+CWSTASCAN: <bssid>,<ssid>[,<signal>]** **[… …]]** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<flag_show_signal>	0 –Don't show the signal level. It's the default value 1 –Show the signal level.
<bssid>	The MAC address of external wireless network.
<ssid>	The SSID name of external wireless network.

<signal>	The signal level of external wireless network.

Example

```
AT+CWSTASCAN
OK

+CWSTASCAN:
4c:e6:76:49:2a:48,simtest
AT+CWSTASCAN=1
OK
AT+CWSTASCAN?
+CWSTASCAN: 1

OK
AT+CWSTASCAN
OK

+CWSTASCAN:
f4:83:cd:d8:24:c8,TP-LINK_24C8,-52
80:89:17:10:e6:23,TP-LINK_SW2,-58
14:2d:27:24:98:61,Public,-58
bc:46:99:38:e2:ca,TP-LINK_E2CA,-64
0c:72:d9:49:25:8b,nubia-WD670-258B,-92
50:2b:73:c0:aa:d9,Tenda_C0AAD9,-68
```

17.2.15 AT+CWSTACFG STA mode configuration setting

AT+CWSTACFG STA mode configuration setting	
Read Command **AT+CWSTACFG?**	Response **OK** **+CWSTACFG: <ssid>[,<security>,<proto>,<psk>]**
Write Command **AT+CWSTACFG=<ssid>[,<security>,<proto>,<psk>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<ssid>	The SSID name of external wireless network. Refer to +CWSSID command for details.
<security>	The key management of external wireless network 0/6 is supported only now. It is only supported in W82. 0 - NONE 6 - WPA3
<proto>	Reserved value.
<psk>	The password of external wireless network. The length of <psk> string is 8-63.

Example

```
AT+CWSTACFG="simtest",0,1,"1234567890"
OK
AT+CWSTACFG?
OK

+CWSTACFG: "simtest",,,"1234567890"
AT+CWSTACFG="simtest",,,"1234567890"
OK
AT+CWSTACFG?
OK

+CWSTACFG: "simtest",,,"1234567890"
```

17.2.16 AT+CWSTAIP Get STA mode IP address

AT+CWSTAIP Get STA mode IP address	
Read Command **AT+CWSTAIP?**	Response **[+CWSTAIP: <ip address>]** **OK**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<ip address>	The station IP address

Example

```
AT+CWSTAIP?
+CWSTAIP: 192.168.11.27

OK
```

17.2.17 AT+CWETHMODE Ethernet mode configuration setting

AT+CWETHMODE Ethernet mode configuration setting	
Read Command **AT+CWETHMODE?**	Response **+CWETHMODE: <mode>** **OK**
Write Command **AT+CWETHMODE=<mode>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Maximum Response Time	-
Reference	-

Defined Values

<mode>	The Ethernet mode. 0 LAN router 1 WAN router 2 eth0 WAN router,eth1 LAN router,when there are two routers

Example

```
AT+CWETHMODE=0
OK
```

```
AT+CWETHMODE?
+CWETHMODE: 0

OK
AT+CWETHMODE=1
OK

AT+CWETHMODE?
+CWETHMODE: 1

OK
```

18 Gadgets on the AP side

18.1 Overview of AT Commands

Command	Description
AT+NETACT	Open/Close Datacall link
AT+CAPNET	Activate PDP context

18.2 Detailed Description of AT Commands

18.2.1 AT+NETACT Open/Close Datacall link

AT+NETACT Open/Close Datacall link	
Test Command **AT+NETACT=?**	Response **+NETACT: (0-1)** **OK**
Read Command **AT+NETACT?**	Response **+NETACT: <state>** **OK**
Write Command **AT+NETACT=<state>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<state>	0 – disable 1 – enable
	The default value of state is 0

Example

```
AT+CGDCONT?
+CGDCONT: 1,"IP","ctlte","0.0.0.0",0,0,0,0
+CGDCONT: 2,"IPV4V6","ims","0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0",0,0,0,0

OK

AT+CGDCONT=6,"IPV4V6","3gnet"
OK

AT+NETACT?
+NETACT: 0

OK

AT+NETACT=1
OK

AT+NETACT?
+NETACT: 1

OK

AT+NETACT=0
OK

AT+NETACT?
+NETACT: 0

OK
```

18.2.2 AT+CAPNET Activate PDP context

AT+CAPNET Activate PDP context	
Test Command AT+CAPNET=?	Response +CAPNET: (range of supported <cid>), (range of <on_off>) OK

Read Command **AT+CAPNET?**	Response **+CAPNET:** <**cid**>, <**on_off**> … … **OK**
Write Command **AT+CAPNET=<cid>,<on_off>**	Response **OK** If the PDP context has not been activated or deactivated, response: **+CAPNET:** <**cid**>, <**on_off**> other: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<**cid**>	(PDP Context Identifier) a numeric parameter which specifies a particular PDP context definition. The parameter is local to the TE-MT interface and is used in other PDP context-related commands. The range of permitted values (minimum value = 1) is returned by the test form of the command. The following range is for reference only, please refer to actual results. 1…6
<**on_off**>	Integer type, which indicates the state of PDP context activation. 0 – network close (deactivated) 1 – network open(activated)

Example

AT+CAPNET=?
+CAPNET: (1-6),(0-1)

OK
AT+CAPNET?
+CAPNET: 1,0
+CAPNET: 2,0
+CAPNET: 3,0
+CAPNET: 4,0
+CAPNET: 5,0
+CAPNET: 6,0

OK

```
AT+CAPNET=6,1
OK
+CAPNET: 6,1
AT+CAPNET?
+CAPNET: 1,0
+CAPNET: 2,0
+CAPNET: 3,0
+CAPNET: 4,0
+CAPNET: 5,0
+CAPNET: 6,1

OK
```

NOTE

- AT+CAPNET=<cid>,<on_off> will return immediately,so you need to listen the +CAPNET URC message or use AT+CAPNET? to query the status.

19 AT Command for QCMAP

19.1 Overview of AT Commands for QCMAP

Command	Description
AT+CQCMAP="WWAN"	Query the default QCMAP dialed network IP address

AT+CQCMAP="DMZ"	Query/Configure DMZ Function for Default QCMAP Dialing
AT+CQCMAP="LAN"	Query/Lock Default LAN Interface Single IP Address
AT+CQCMAP="LANIP"	Query/Default Modify LAN Interface DHCP Address Pool
AT+CQCMAP="VLAN"	Query/Configure VLAN
AT+CQCMAP="MPDN_rule"	Query/Modify QCMAP Multiplex Dialing Rules
AT+CQCMAP="IPPT_NAT"	Query/configure QCMAP dial-up IPPT NAT working mode
AT+CQCMAP="connect"	Starting/disconnecting QCMAP dialing
AT+CQCMAP="auto_connect"	Query/Modify QCMAP Automatic Dialing Configuration
AT+CQCMAP="MPDN_Status"	Query/Modify QCMAP Multiway Dialing Status
AT+CQCMAP="SFE"	Query/configure SFE software acceleration function
AT+CQCMAP="domain"	Query/configure gateway domain names for LAN/VLAN
AT+CQCMAP="DHCPV6DNS"	Query/Configure IPv6 DNS for QCMAP Dialing
AT+CQCMAP="BACKHAUL_PREF"	Query/Configure Backhaul Network Priority

19.2 Detailed Description of AT Commands for QCMAP

19.2.1 AT+CQCMAP="WWAN" Query the default QCMAP dialed network IP address

AT+CQCMAP="WWAN"	**Query the default QCMAP dialed network IP address**
Write Command **AT+CQCMAP="WWAN"**	Response **+CQCMAP: "WWAN",<status>,<profileID>,<IP_family>,<IP_address>** **+CQCMAP: "WWAN",<status>,<profileID>,<IP_family>,<IP_address>** **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<status>	0 disconnect 1 connect
<profileID>	Profile ID range:1~16
<IP_family>	IPV4 IPV6
<IP_address>	String type default QCMAP dialed network IP address IPv4 disconnected address:0.0.0.0 IPv6 disconnected address:0:0:0:0:0:0:0:0

Example

```
AT+CQCMAP="WWAN"
+CQCMAP: "WWAN",0,1,IPV4,0.0.0.0
+CQCMAP: "WWAN",0,1,IPV6,0:0:0:0:0:0:0:0

OK
```

19.2.2 AT+CQCMAP="DMZ" Query/Configure DMZ Function for Default QCMAP Dialing

AT+CQCMAP="DMZ"	Query/Configure DMZ Function for Default QCMAP Dialing
Write Command **AT+CQCMAP="DMZ"**	Response **+CQCMAP: "DMZ",<enable>,<IP_family>[,<IP_address>]** **+CQCMAP: "DMZ",<enable>,<IP_family>[,<IP_address>]** **OK**
Write Command **AT+CQCMAP="DMZ"[,<enable>,<IP_family>[,<IP_address>]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<enable>	Default QCMAP Dialing Status 0 disconnect 1 connect
<IP_family>	IP type 4 IPv4 6 IPv6
<IP_address>	String type

Example

```
AT+CQCMAP="DMZ"
```

```
+CQCMAP: "DMZ",0,4
+CQCMAP: "DMZ",0,6

OK
AT+CQCMAP="DMZ",1,4,192.168.225.50
OK
AT+CQCMAP="DMZ",0,4
OK
```

19.2.3 AT+CQCMAP="LAN" Query/Lock Default LAN Interface Single IP Address

AT+CQCMAP="LAN"	Query/Lock Default LAN Interface Single IP Address
Write Command **AT+CQCMAP="LAN"**	Response **+CQCMAP: "LAN",<IP_address>** **OK**
Write Command **AT+CQCMAP="LAN"[,<IP_address>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<IP_address>	String type

NOTE

<IP_address>must belong to the network segment of the current default LAN interface。Default LAN IP_address is 192.168.225.x。

Example

```
AT+CQCMAP="LAN",192.168.225.50
OK
AT+CQCMAP="LAN"
```

```
+CQCMAP: "LAN",192.168.225.50

OK
```

19.2.4 AT+CQCMAP="LANIP" Query/Default Modify LAN Interface DHCP Address Pool

AT+CQCMAP="LANIP"	Query/Default Modify LAN Interface DHCP Address Pool
Write Command **AT+CQCMAP="LANIP"**	Response **+CQCMAP: "LANIP",<LAN_IP_start_address>,<LAN_IP_end_address>,<GW_IP_address>** **OK**
Write Command **AT+CQCMAP="LANIP"[,<LAN_IP_start_address>,<LAN_IP_end_address>,<GW_IP_address>[,<effect>]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Defined Values

<LAN_IP_start_address>	String type Start address
<LAN_IP_end_address>	String type End address
<GW_IP_address>	String type Gateway address
<effect>	Int type effective 0 /NULL Effective after the module restart 1 effect immediately

Example

```
AT+CQCMAP="LANIP"
```

```
+CQCMAP: "LANIP",192.168.225.40,192.168.225.60,192.168.225.1

OK
AT+CQCMAP="LANIP",192.168.111.20,192.168.111.60,192.168.111.1,1
OK
AT+CQCMAP="LANIP",192.168.111.20,192.168.111.60,192.168.111.1
OK
```

19.2.5 AT+CQCMAP="VLAN" Query/Configure VLAN

AT+CQCMAP="VLAN"	Query/Configure VLAN
Write Command **AT+CQCMAP="VLAN"**	Response **+CQCMAP: "VLAN",0** **+CQCMAP: "VLAN",<VLAN_ID>,<VLAN_type>** **OK**
Write Command **AT+CQCMAP="VLAN"[,<VLAN_ID>,<enable>[,<VLAN_type>]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<VLAN_ID>	VLAN ID range:0, 2~255.
<enable>	Enable or disable the Vlan specified by vlan_id "enable" "disable"
<VLAN_type>	<enable>="enable" 1 ETH 2 ECM 3 RNDIS 11 ETH, the VLAN data acceleration is not enabled 12 ECM, the VLAN data acceleration is not enabled 13 RNDIS,The VLAN data acceleration is not enabled

NOTE

1. <VLAN_type>=1/2/3，the module will be reboot
2. When the VLAN is enabled and disabled in other cases,the module will not reboot

Example

```
AT+CQCMAP="VLAN"
+CQCMAP:"VLAN",0
+CQCMAP:"VLAN",2,1
+CQCMAP:"VLAN",3,1

OK
AT+CQCMAP="VLAN",4,"enable",1
OK
AT+CQCMAP="VLAN",4,"disable"
OK
```

19.2.6 AT+CQCMAP="MPDN_rule" Query/Modify QCMAP Multiplex Dialing Rules

AT+CQCMAP="MPDN_rule"	Query/Modify QCMAP Multiplex Dialing Rules
Write Command **AT+CQCMAP="MPDN_rule"**	Response **+CQCMAP: "MPDN_rule",<rule_num>,<profile_id>,<VLAN_ID>,<IPPT_mode>,<auto_connect>** **...** **OK**
Write Command **AT+CQCMAP="MPDN_rule"[,<rule_num>[,<profile_id>,<VLAN_ID>,<IPPT_mode>,<auto_connect>[,<IPPT_info>]]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<rule_num>	QCMAP Multiplex Dialing rules ID Range: 0~3
<profileID>	APN Profile ID. Range: 1~16
<VLAN_ID>	VLAN ID Range:0, 2~255. When the VLAN_ID is 0,is default LAN,is not VLAN
<IPPT_mode>	IPPT mode(IP Pass through). 0 disable IPPT 1 enable IPPT(ETH) 2 enable IPPT(WIFI) 3 enable IPPT(USB-ECM/RNDIS)
<auto_connect>	1 enable 0 disable
<IPPT_info>	<IPPT_mode>=1 or 2, the value is MAC address of the peer NIC bound in IPPT mode <IPPT_mode>=3, the value is the name of the peer host bound to the IPPT mode <IPPT_mode>=0, NULL Ps:Enable IPPT 1.IPPT NAT mode is WithNAT(AT+CQCMAP=”IPPT_NAT”,1) must have <IPPT_info>,LAN can get public address，other get private network address .The module will convert the network addresses for all the LAN devices. 2.<IPPT_mode>=1,and<IPPT_info>="FF:FF:FF:FF:FF:FF",module always deliver the public address to the latest connected Ethernet LAN device 3.IPPT NAT mode is WithoutNAT(AT+CQCMAP=”IPPT_NAT”,0) If not<IPPT_info>,the first LAN can get public address,the module does not translate the network address of the LAN device, and other devices cannot obtain any IP address. If <IPPT_info> is specified, the specified LAN device obtains a public IP address, and the module does not translate the IP address of the LAN device. Therefore, other devices do not obtain any IP address

NOTE

1. If only the default LAN interface is required to connect to the network and the VLAN and multiple dial-up are not required, set <rule_num>=0,and set <VLAN_ID> to 0.
2. The implementation of QCMAP dial-up is to bind WAN interfaces based on different APN dials to LAN/VLAN interfaces, and implement NAT configuration between the corresponding WAN and LAN/VLAN. In this way, devices connected to different LAN/VLAN interfaces can access different

networks through corresponding WAN interfaces.
3. If the VLAN interface (<VLAN_ID> is not 0) is required during the QCMAP dial-up rule configuration, run the AT+QCMAP="VLAN" command to set up the corresponding VLAN interface.
4. The IPPT mode(IP Pass through) enables the Public IP address (Public IP address) assigned by the carrier to be transparent A function that is transmitted to a LAN device.
5. By default, if the IPPT mode needs to be enabled for QCMAP dialing using the USB (ECM/RNDIS) port, set this parameter
Set <IPPT_mode> = 3 and set the host name of the LAN device in <IPPT_info>. Because most of the time, USB virtual the MAC address of an Ethernet interface (ECM/RNDIS) is not fixed. But in actual use, you can also set <IPPT_mode>= 1, and set the MAC address of the LAN USB device in <IPPT_info> to implement the IPPT function.
6. WLAN interfaces do not support VLAN. WLAN belongs to VLAN0. In practice, if the public IP address needs to be assigned to a device connected to the WLAN, set <IPPT_mode> to 2 and <VLAN_ID> to 0.
7.<rule_num>=0 is default QCMAP Dialing.
8. The default QCMAP dial-up is bound to the physical LAN interface (VLAN0) by default. If you change the LAN/VLAN bound to the default QCMAP dial-up rule, the VLAN is bound to the VLAN Interface, the module will automatically restart.
AT+QCMAP="MPDN_rule",0,1,2,0,1;if AT+QCMAP="MPDN_rule",0 After disabling the default QCMAP dialing rule, The LAN/VLAN interface bound to the QCMAP dialing rule is automatically switched from the VLAN interface (<VLAN_ID>=2) to the physical LAN Interface (<VLAN_ID>=0) and the module restarts automatically.
9. The network is accessed internally through a network connection created by the default QCMAP dialing rules. That is, if <rule_num>=0 by default
If no dial-up is performed, there is no network in the module.

Example

```
AT+CQCMAP="MPDN_rule"
+QCMAP: "MPDN_rule",0,1,0,0,1
+QCMAP: "MPDN_rule",1,0,0,0,0
+QCMAP: "MPDN_rule",2,0,0,0,0
+QCMAP: "MPDN_rule",3,0,0,0,0

OK
AT+CQCMAP="MPDN_rule",0,1,0,0,1
OK
AT+CQCMAP="MPDN_rule",1,5,2,0,1
OK
AT+CQCMAP="MPDN_rule"
+QCMAP: "MPDN_rule",0,1,0,0,1
+QCMAP: "MPDN_rule",1,5,2,0,1
+QCMAP: "MPDN_rule",2,0,0,0,0
+QCMAP: "MPDN_rule",3,0,0,0,0
```

```
OK
AT+CQCMAP="MPDN_rule",1
OK
AT+CQCMAP="MPDN_rule"
+QCMAP: "MPDN_rule",0,1,0,0,1
+QCMAP: "MPDN_rule",1,0,0,0,0
+QCMAP: "MPDN_rule",2,0,0,0,0
+QCMAP: "MPDN_rule",3,0,0,0,0

OK
```

19.2.7 AT+CQCMAP="IPPT_NAT" Query/Configure QCMAP dial-up IPPT NAT working mode

AT+CQCMAP="IPPT_NAT" Query/Configure QCMAP dial-up IPPT NAT working mode	
Write Command **AT+CQCMAP="IPPT_NAT"**	Response **+CQCMAP: "IPPT_NAT",<IPPT_NAT>** **OK**
Write Command **AT+CQCMAP="IPPT_NAT"[,<IPPT_NAT>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<IPPT_NAT>	QCMAP IPPT mode。 0 WithoutNAT mode 1 WIthNAT mode

NOTE

1. When the QCMAP dial-up IPPT working mode is changed, all QCMAP dial-up connections are disconnected. If the QCMAP dialing rule is set to automatic connection, the QCMAP dialing will be

automatically reconnected after the IPPT working mode is changed. If automatic connection is disabled, the QCMAP dial needs to be manually dialed again (AT+QCMAP="connect") after the IPPT working mode is changed.
2. If the dialing mode is changed from WithNAT to WithoutNAT, the IPPT mode of all previously configured dialing rules is automatically changed to WithoutNAT mode. If the WithoutNAT mode is changed to WithNAT mode, the IPPT mode that <IPPT_info> is omitted in all the previously configured dialing rules is automatically invalid and needs to be reconfigured

Example

```
AT+CQCMAP="IPPT_NAT"
+CQCMAP: "IPPT_NAT",0

OK
AT+CQCMAP="IPPT_NAT",1
OK
```

19.2.8 AT+CQCMAP="connect" Starting/Disconnecting QCMAP dialing

AT+CQCMAP="connect"	Starting/Disconnecting QCMAP dialing
Write Command **AT+CQCMAP="connect",<rule_num>,<connect>**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<rule_num>	QCMAP rule ID. Range:0~3
<connect>	QCMAP DIALING 0 disconnect 1 connect

NOTE

If <auto_connect>=1 of the dialing rule is specified (see AT+QCMAP="MPDN_rule"), automatic dialing will be enabled for this dial. You cannot manually start or disconnect the dial using AT+CQCMAP="connect". If you need to manually control QCMAP dialing through AT+CQCMAP="connect", you must disable automatic dialing in the configuration of AT+CQCMAP="MPDN_rule".

Example

AT+CQCMAP="connect",0,1
OK
AT+CQCMAP="connect",0,0
OK

19.2.9 AT+CQCMAP="auto_connect" Query/Modify QCMAP Automatic Dialing Configuration

AT+CQCMAP="auto_connect" Query/Modify qcamp Automatic Dialing Configuration	
Write Command **AT+CQCMAP="auto_connect"**	Response **+CQCMAP: "auto_connect",<rule_num>,<auto_connect>,<profileID>** **OK**
Write Command **AT+CQCMAP="auto_connect"[,<rule_num>[,<auto_connect>[,<profileID>]]]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<rule_num>	QCMAP rule ID.Range:0~3
<auto_connect>	QCMAP auto connect status. 0 disable 1 enable
<profileID>	APN Profile ID Range:1~16.

NOTE

When changing <auto_connect> of the specified QCMAP dialing rule, ensure that the QCMAP dialing rule has passed AT+CQCMAP="MPDN_rule" configured and enabled.

Example

```
AT+CQCMAP="auto_connect"
+CQCMAP: "auto_connect",0,1,1
+CQCMAP: "auto_connect",1,0,0
+CQCMAP: "auto_connect",2,0,0
+CQCMAP: "auto_connect",3,0,0

OK
AT+CQCMAP="auto_connect",0
+CQCMAP: "auto_connect",0,1

OK
AT+CQCMAP="auto_connect",1,1
OK
AT+CQCMAP="auto_connect",2,1,6
OK
```

19.2.10 AT+CQCMAP="MPDN_status" Query/Modify QCMAP Multiway Dialing Status

AT+CQCMAP="MPDN_status"	Query/Modify qcmap Multiway Dialing Status
Write Command **AT+CQCMAP="MPDN_status"**	Response **+CQCMAP: "MPDN_status",<rule_num>,<profileID>,<IPPT_status>,<connect_status>** ... **OK**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<rule_num>	QCMAP rule id Range:0~3
<profileID>	QCMAP APN Profile ID Range1~16。
<IPPT_status>	IPPT mode 0 disable 1 enable
<connect_status>	QCMAP status 0 disconnect 1 connect

Example

```
AT+CQCMAP="MPDN_status"
+CQCMAP: "MPDN_status",0,1,0,1
+CQCMAP: "MPDN_status",1,2,0,0
+CQCMAP: "MPDN_status",2,3,0,0
+CQCMAP: "MPDN_status",3,0,0,0

OK
```

19.2.11 AT+CQCMAP="SFE" Query/Configure SFE software acceleration function

AT+CQCMAP="SFE"	Query/Configure SFE software acceleration function
Write Command **AT+CQCMAP="SFE"**	Response **+CQCMAP: "SFE",<status>** **OK**
Write Command **AT+CQCMAP="SFE"[,<status>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<status>	SFE status "enable" "disable"

NOTE

The SFE feature provides limited performance optimization only if the module does not support hardware acceleration (IPA); If the module supports hardware acceleration (IPA), this feature is not available.

Example

```
AT+CQCMAP="SFE"
+CQCMAP: "SFE","disable"

OK
AT+CQCMAP="SFE","enable"
OK
```

19.2.12 AT+CQCMAP="domain" Query/Configure gateway domain names for LAN/VLAN

AT+CQCMAP="domain" Query/Configure gateway domain names for LAN/VLAN	
Write Command **AT+CQCMAP="domain"**	Response **+CQCMAP: "domain",<domain_name>** **OK**
Write Command **AT+CQCMAP="domain"[,<domain_name>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<domain_name>	LAN/VLAN gateway domain names.Example: "simcom.com"

Example

```
AT+CQCMAP="domain"
```

```
+CQCMAP: "domain","simcom.mobileap.com"

OK
AT+CQCMAP="domain","simcom.mobileap.com"
OK
```

19.2.13 AT+CQCMAP="DHCPV6DNS" Query/Configure IPv6 DNS for QCMAP Dialing

AT+CQCMAP="DHCPV6DNS" Query/Configure IPv6 DNS for QCMAP Dialing	
Write Command **AT+CQCMAP="DHCPV6DNS"**	Response **+CQCMAP: "DHCPV6DNS",<status>** **OK**
Write Command **AT+CQCMAP="DHCPV6DNS"[,<status>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<status>	DHCPV6DNS status "enable" "disable"

Example

```
AT+CQCMAP="DHCPV6DNS"
+CQCMAP: "DHCPV6DNS","enable"

OK
AT+CQCMAP="DHCPV6DNS","enable"
OK
```

19.2.14 AT+CQCMAP="BACKHAUL_PREF"Query/Configure Backhaul Network Priority

AT+CQCMAP="BACKHAUL_PREF"	Query/Configure Backhaul Network Priority
Write Command **AT+CQCMAP="BACKHAUL_PREF"**	Response **+CQCMAP:"BACKHAUL_PREF"[,<first priority>,<second priority>,<third priority>,<fourth priority>,<fifth priority>]** **OK**
Write Command **AT+CQCMAP="BACKHAUL_PREF"[,<first priority>,<second priority>,<third priority>,<fourth priority>,<fifth priority>]**	Response **OK** or **ERROR**
Parameter Saving Mode	-
Max Response Time	-
Reference	-

Default Values

<priority>	Backhaul Network Priority 1 WWAN 2 USB Cradle 3 WLAN 4 Ethernet 5 BT-WAN

Example

```
AT+CQCMAP="BACKHAUL_PREF"
+CQCMAP: "BACKHAUL_PREF",5,4,2,3,1

OK
AT+CQCMAP="BACKHAUL_PREF",5,2,1,3,4
OK
```